Image Analysis

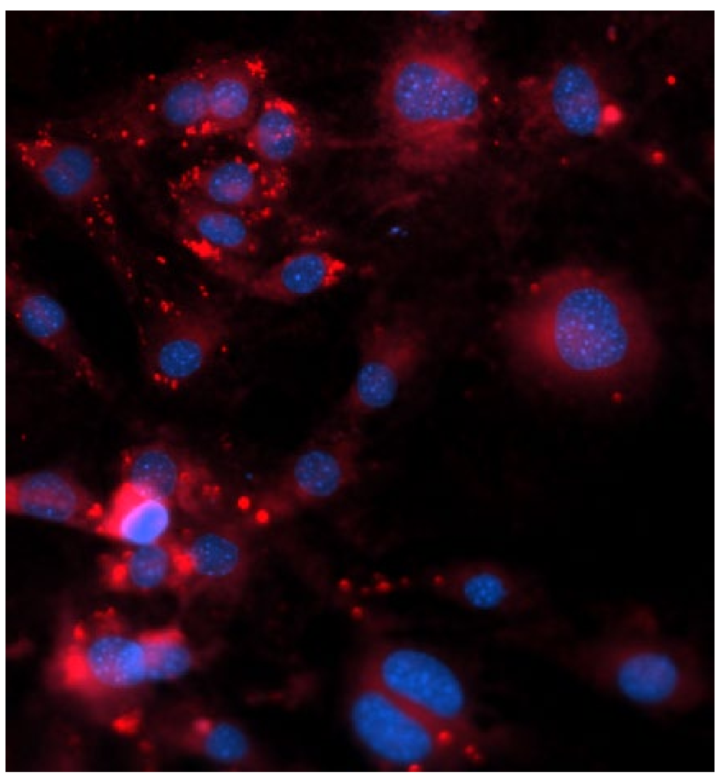
BMB 547

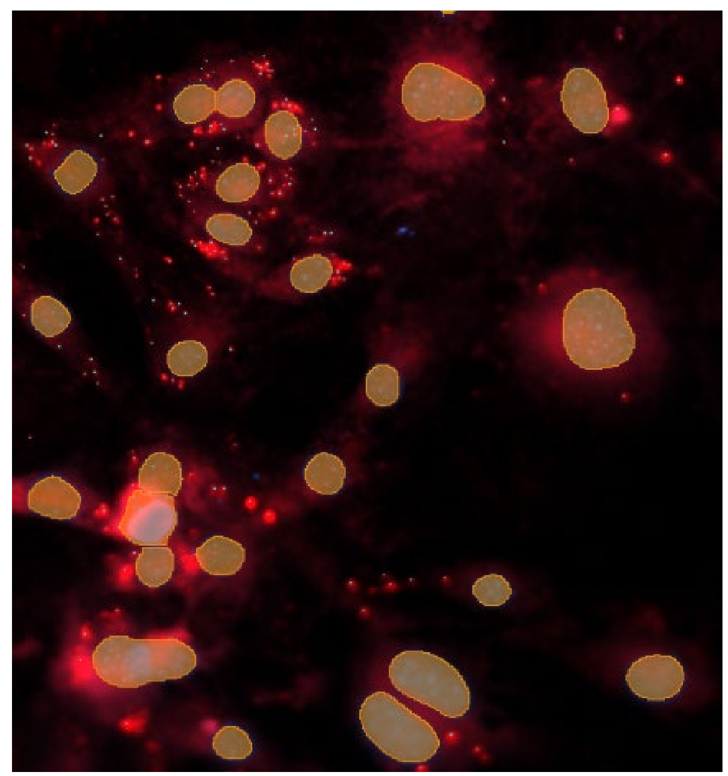
Jonathan Brewer

Why Image Analysis

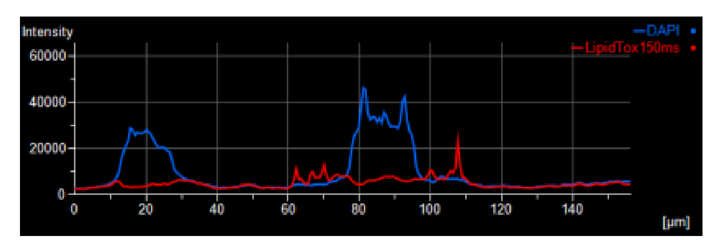
As scientists acquire images to get DATA

Image	Regionalize and Measure	Data

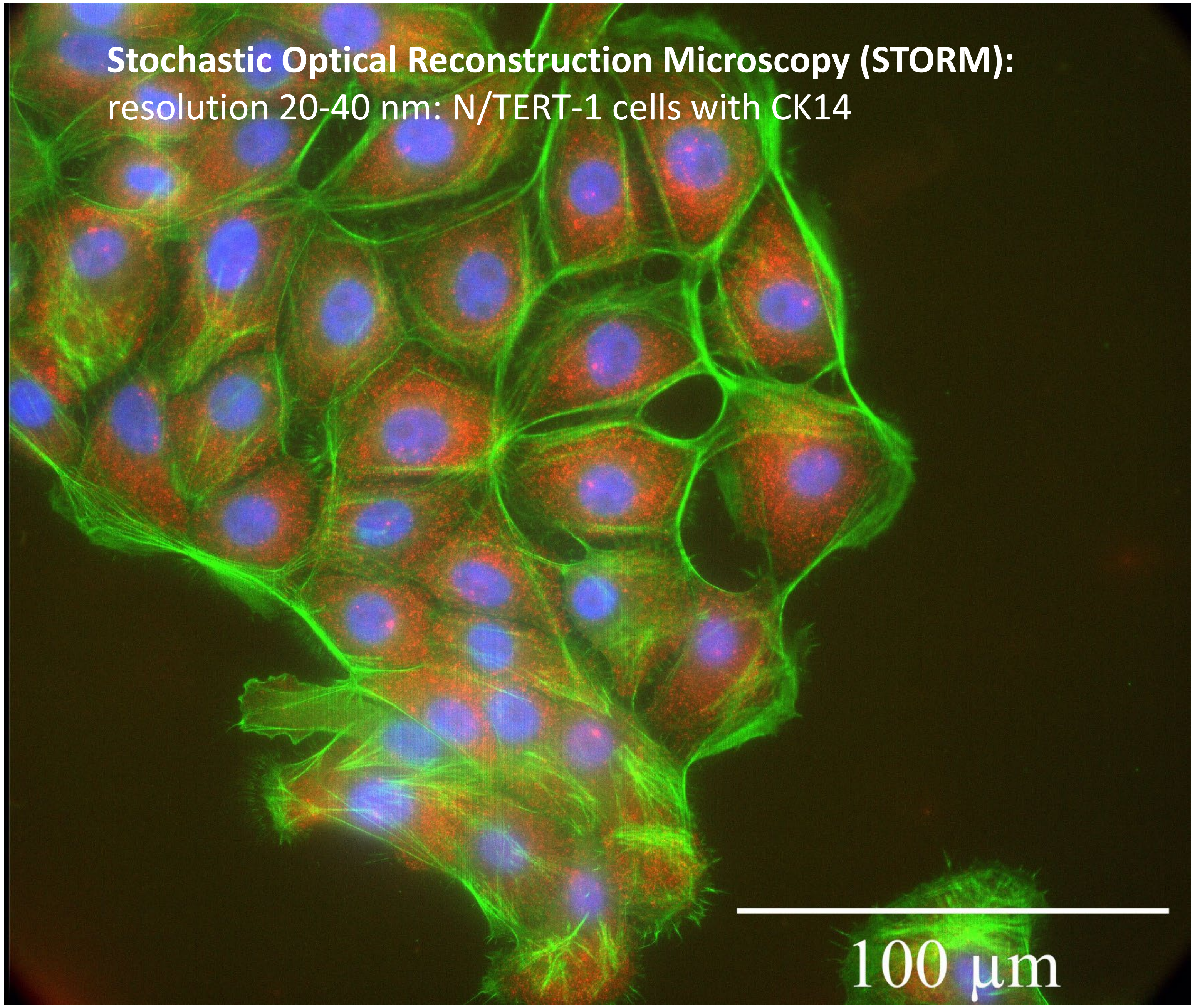




BinaryID	ObjID	FillArea
GA2_LipidsAroundNuc_Exercise4 (DAPI)	1	6.73
GA2_LipidsAroundNuc_Exercise4 (DAPI)	2	601.23
GA2_LipidsAroundNuc_Exercise4 (DAPI)	3	393.53
GA2_LipidsAroundNuc_Exercise4 (DAPI)	4	195.51
GA2_LipidsAroundNuc_Exercise4 (DAPI)	5	214.00
GA2_LipidsAroundNuc_Exercise4 (DAPI)	6	228.30
GA2_LipidsAroundNuc_Exercise4 (DAPI)	7	206.44
GA2_LipidsAroundNuc_Exercise4 (DAPI)	8	217.79
GA2_LipidsAroundNuc_Exercise4 (DAPI)	9	184.99
GA2_LipidsAroundNuc_Exercise4 (DAPI)	10	198.87
GA2_LipidsAroundNuc_Exercise4 (DAPI)	11	767.73
GA2_LipidsAroundNuc_Exercise4 (DAPI)	12	195.09
GA2_LipidsAroundNuc_Exercise4 (DAPI)	13	187.94
GA2_LipidsAroundNuc_Exercise4 (DAPI)	14	190.46
GA2_LipidsAroundNuc_Exercise4 (DAPI)	15	241.75
GA2_LipidsAroundNuc_Exercise4 (DAPI)	16	213.16
GA2_LipidsAroundNuc_Exercise4 (DAPI)	17	279.17
GA2_LipidsAroundNuc_Exercise4 (DAPI)	18	407.83
GA2_LipidsAroundNuc_Exercise4 (DAPI)	19	245.12
GA2_LipidsAroundNuc_Exercise4 (DAPI)	20	209.80
GA2_LipidsAroundNuc_Exercise4 (DAPI)	21	155.14
GA2_LipidsAroundNuc_Exercise4 (DAPI)	22	687.42
GA2_LipidsAroundNuc_Exercise4 (DAPI)	23	589.46

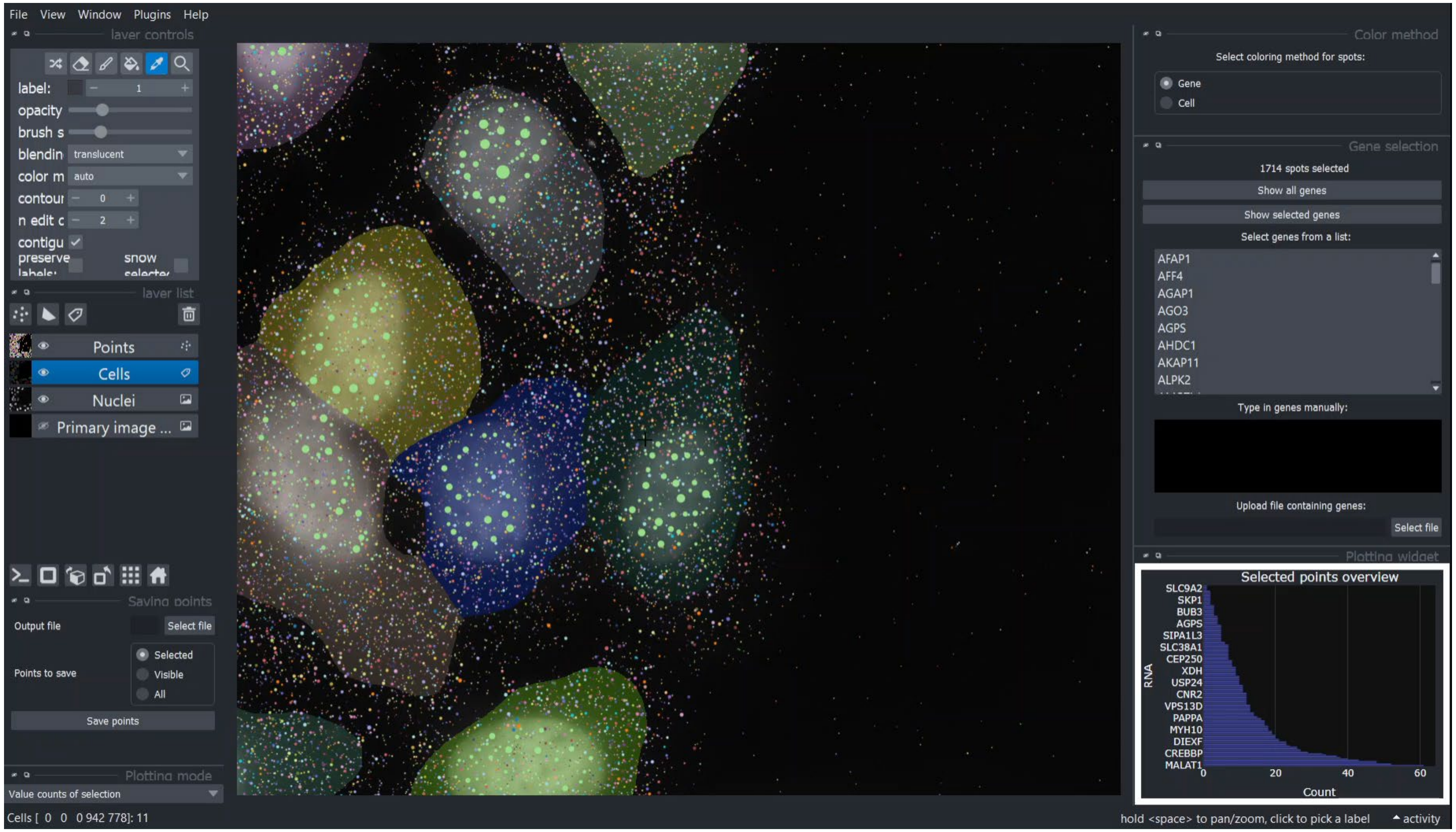


Spatial transcriptomics smFISH



Spatial transcriptomics

Single cell selection and change plotting mode



Topics:

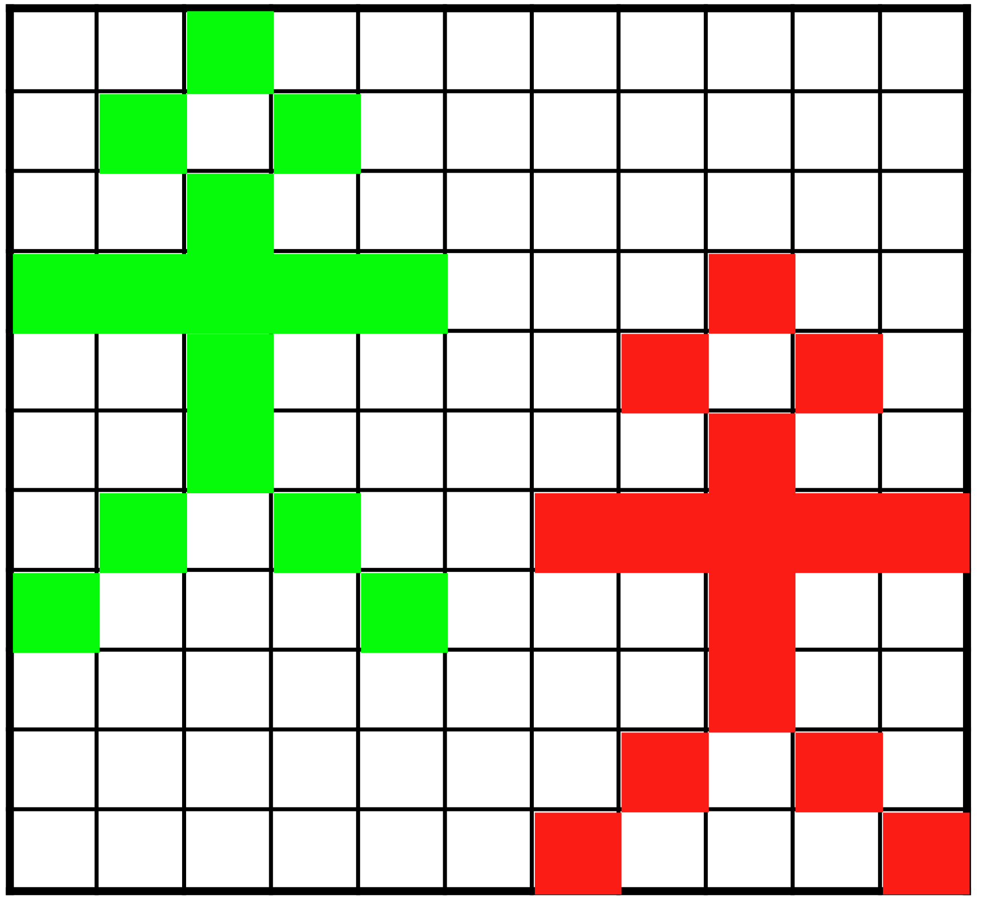
- Basic properties of digital images (sampling, formats, size, depth,)
- Image enhancement (how to make your image look better)
  - histograms, LUT, contrast enhancement
  - background subtraction
  - filter masks (smoothing and sharpening)
- Working with image stacks
- Colocalization
- Image segmentation by thresholding
- Deconvolution

Quantitative Imaging?
…what does that mean?

Art or Science?

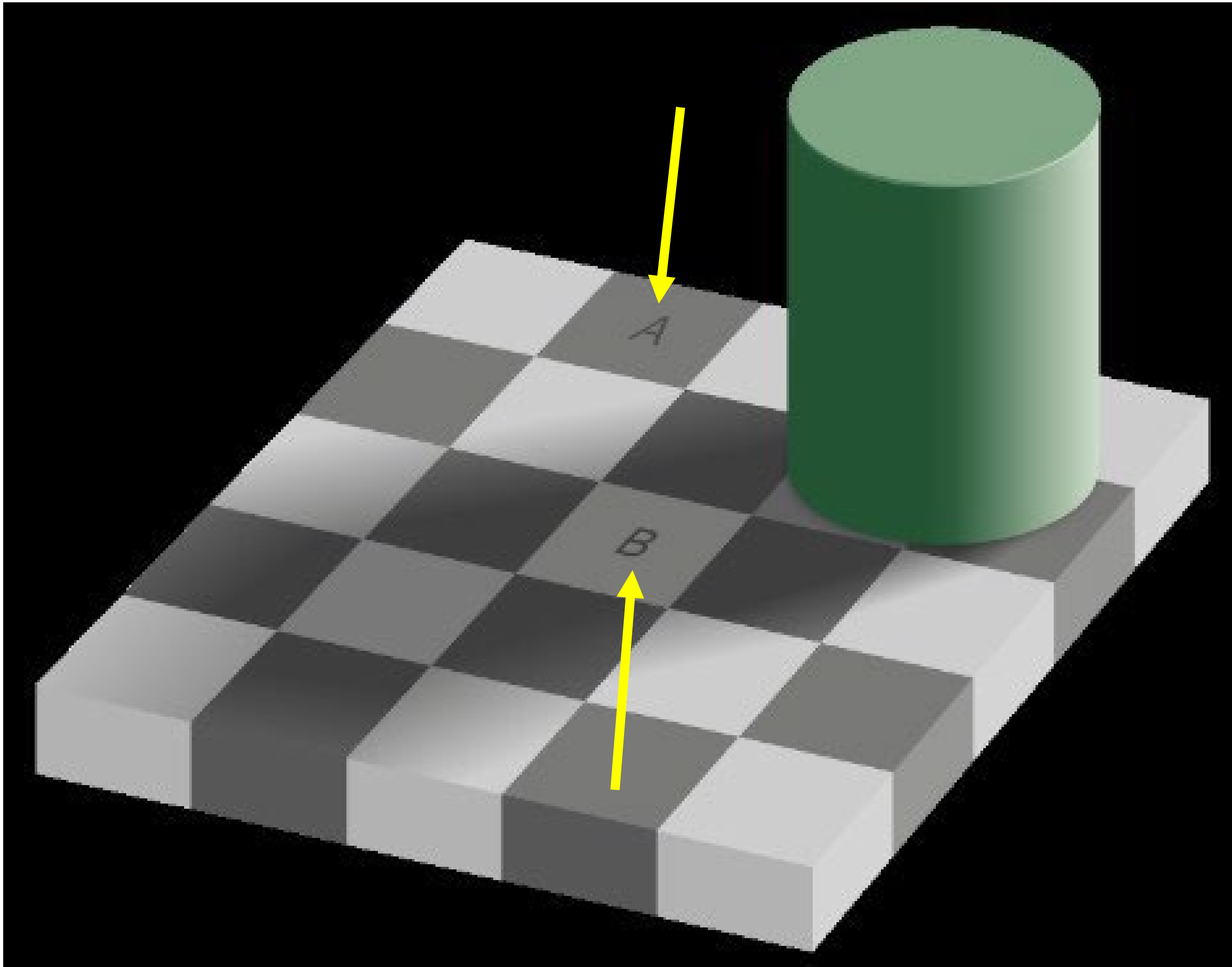
Science means to **measure** something!

- ✓Numerical Results
- ✓Statistics!
- ✓Computers become useful…

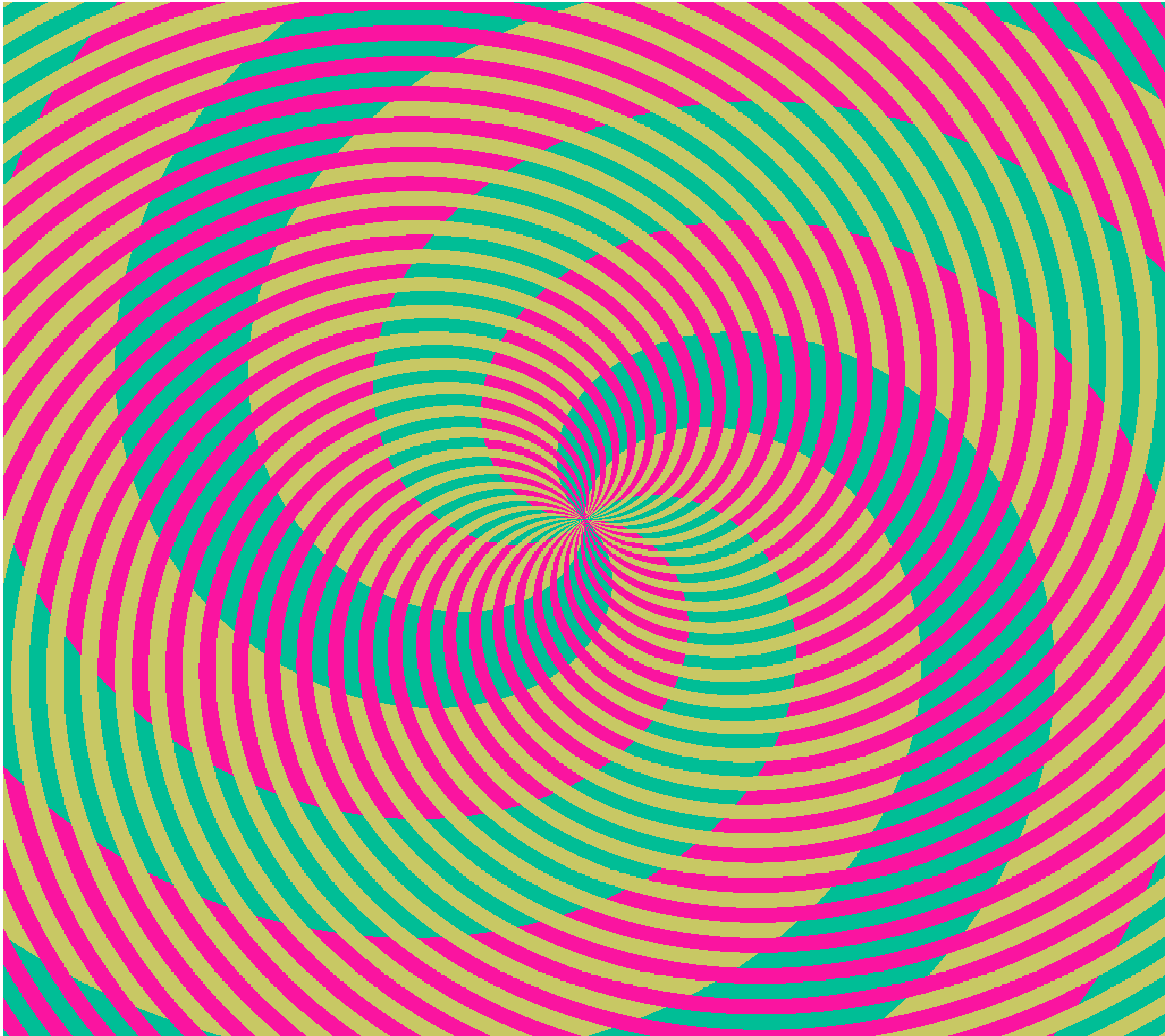


Which is brighter A or B?

Still sure you can interpret intensities in greyscale images with your eyes?

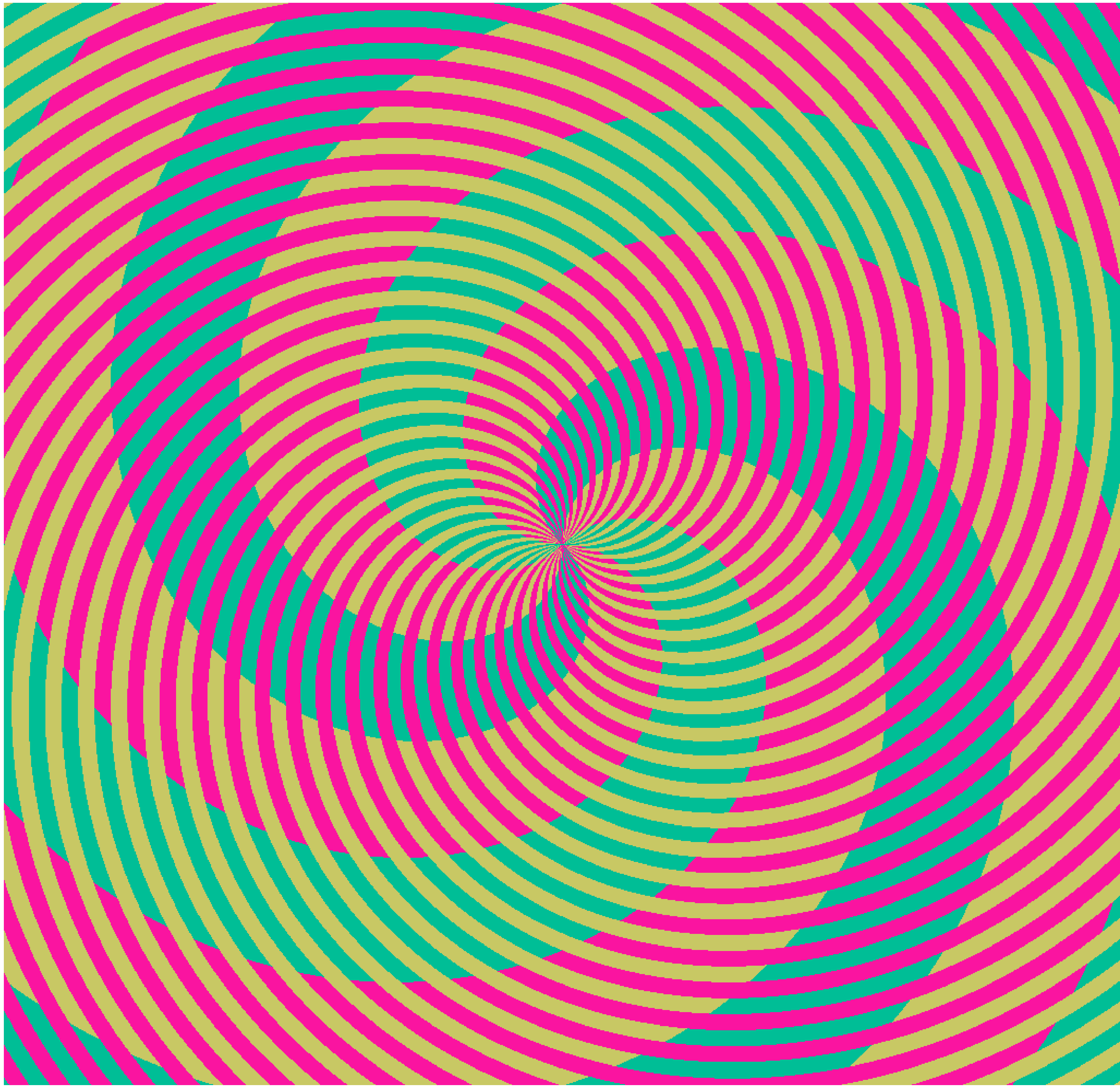


Which colours can you see???



“Colour Merge” images could ruin your life

You see spirals, of pink, orange, green and blue?



Actually,
the green and blue…
are the same color!

Moral of the story:
Don’t Trust Your Eyes!

What is Image Analysis / Quantification?

255	255	255	255	255	255	255	255	255	255
255	255	255	255	50	50	50	50	255	255
255	255	255	50	50	50	50	50	255	255
255	255	255	50	50	50	50	50	255	255
255	255	255	72	50	50	50	50	255	255
255	255	255	255	50	50	50	255	255	255
255	50	50	50	50	50	50	50	50	255
255	255	255	255	255	50	255	255	255	255
255	255	255	255	50	255	255	255	255	255
255	255	255	255	50	50	50	50	51	168
255	255	255	255	50	255	255	255	255	255
255	255	255	50	255	255	255	255	255	255
255	255	255	50	255	255	255	255	255	255
255	255	50	255	255	255	255	255	255	255

- Minimum: 50
- Maximum : 255
- Mean: 94.5
- Std.Dev.: 93.2
- Area: 10x14
- Pixels: 140
- Pix <255: 42

= Image Analysis/ Measurement

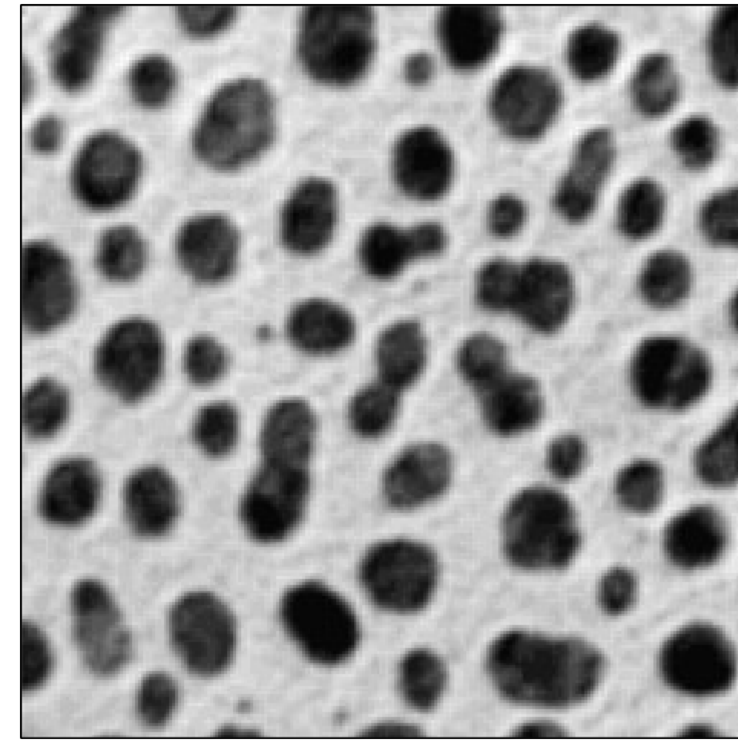
- Object: Stick man
- Body: 1
- Head: 1
- Legs: 2 (1 lifted)
- Arms: 2 (2 lifted)
- Walking left to right…

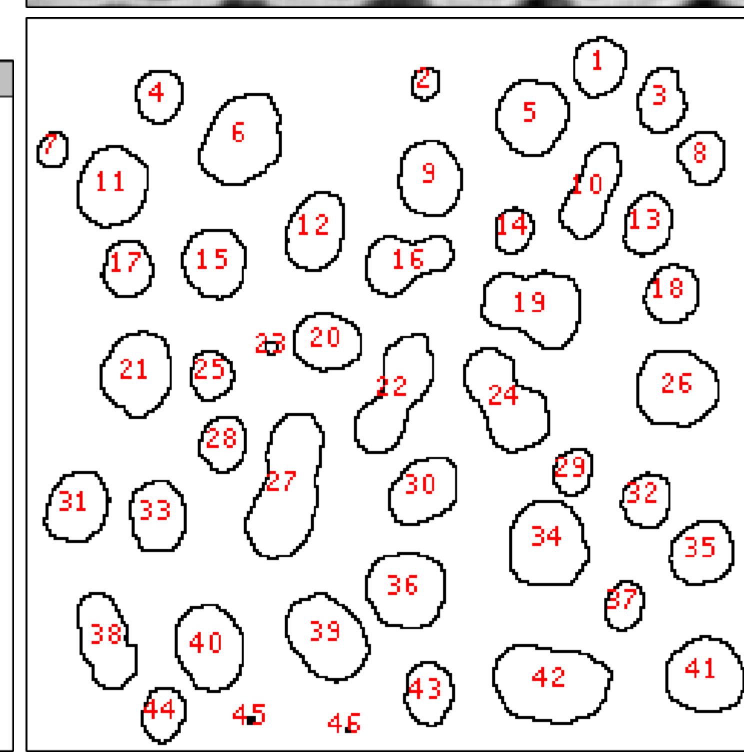
= Interpretation of Analysis Result

Image = Information

Images contain information!!!

- ✓ Quantify / Measure / Analyse
- ✓ Meta data (what, where, when, how)
- ✓ Noise / Background

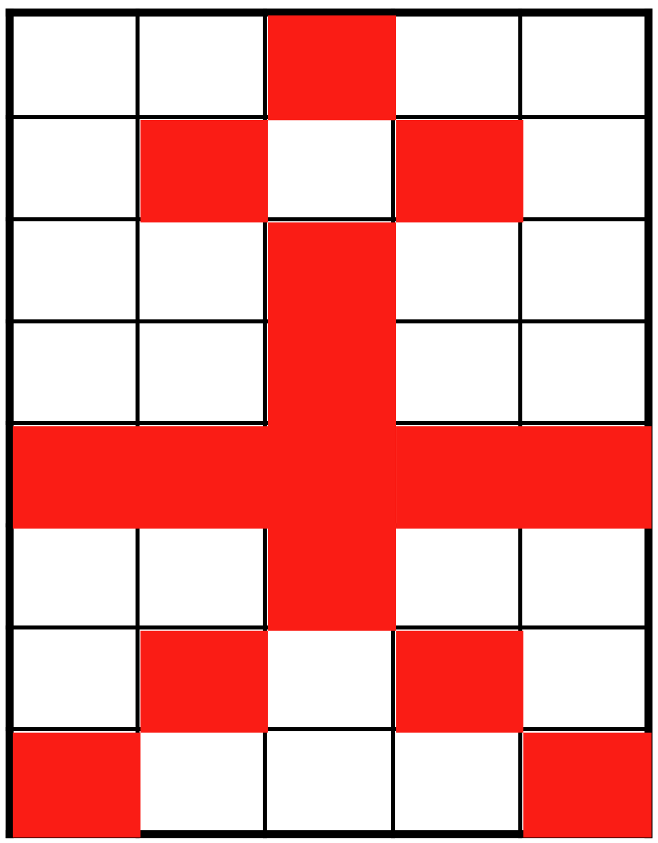




	Area	Mean	StdDev	Min	Max	IntDen	Median	XStart	YStart
1	285	255	0	255	255	72675	255	197	6
2	81	255	0	255	255	20655	255	136	17
3	278	255	0	255	255	70890	255	218	17
4	231	255	0	255	255	58905	255	42	18
5	501	255	0	255	255	127755	255	170	21
6	660	255	0	255	255	168300	255	75	26
7	99	255	0	255	255	25245	255	7	39
8	228	255	0	255	255	58140	255	231	39
9	448	255	0	255	255	114240	255	137	42
10	401	255	0	255	255	102255	255	198	43
11	520	255	0	255	255	132600	255	27	44
12	425	255	0	255	255	108375	255	99	60
13	271	255	0	255	255	69105	255	215	60
14	159	255	0	255	255	40545	255	168	65
15	412	255	0	255	255	105060	255	60	73
16	426	255	0	255	255	108630	255	123	75
17	260	255	0	255	255	66300	255	31	77
18	289	255	0	255	255	73695	255	222	85
19	676	255	0	255	255	172380	255	178	87

Slice	Count	Total Area	Average Size	Area Fraction
blobs.gif	46	17686.000000	384.478261	27.2

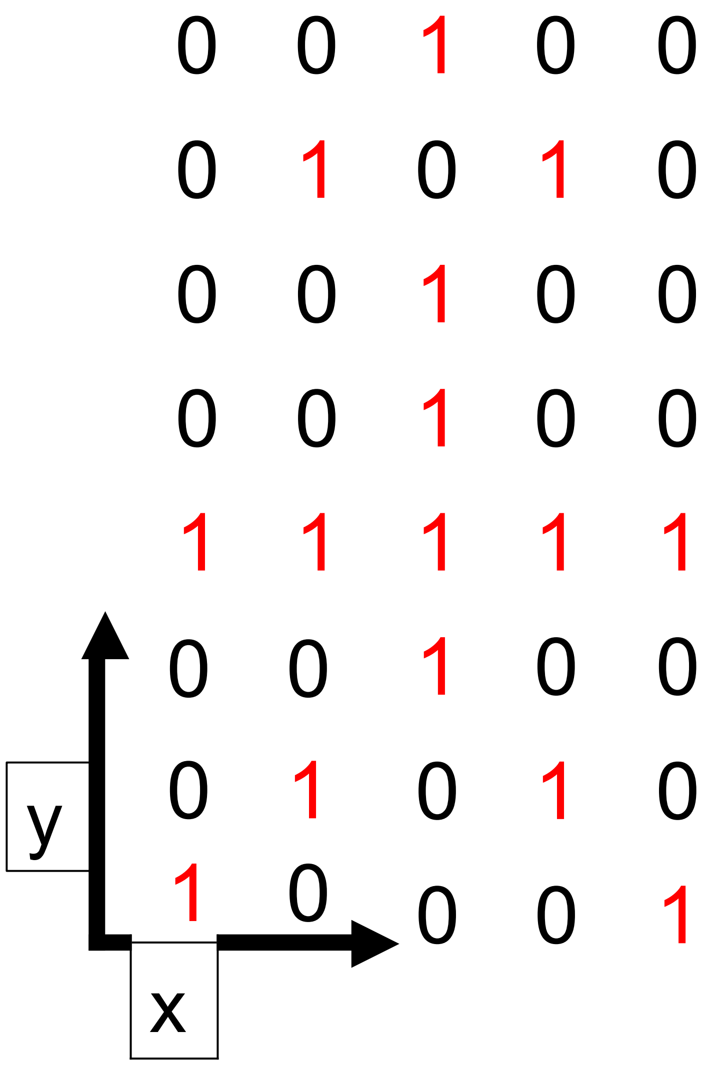
Photographer or Spectroscopist?



This

Is simply a way to **"Visualise"**

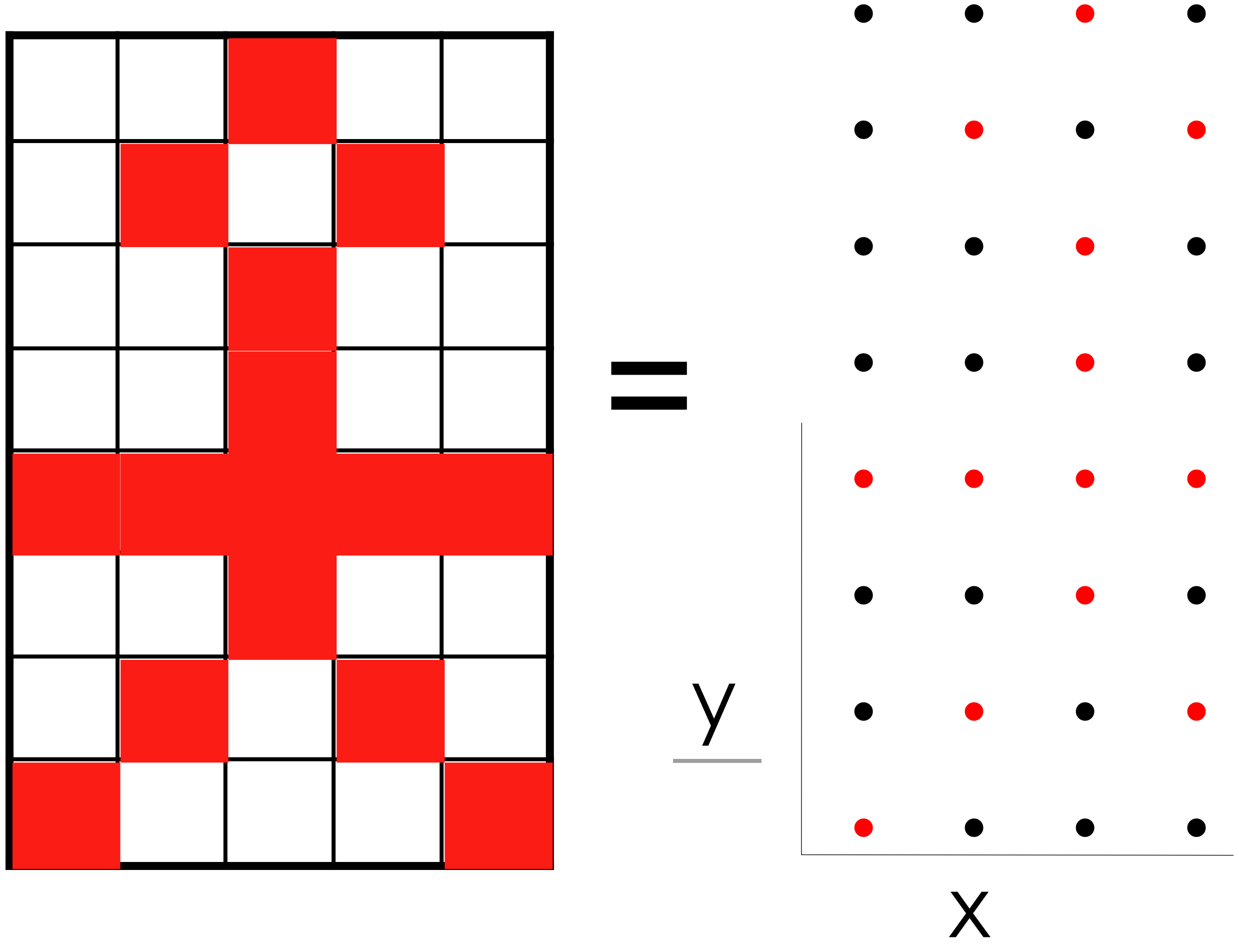
This



What is a (Digital) Image anyway..?

- ✓ it's a digital "**representation**" of reality!
- ✓ it's an **artifact** that contains less info than the object!
- ✓ it's just **numbers**! NOT analogue art!

The Image of a point is NOT a point!!!
(Point Spread Function – PSF)

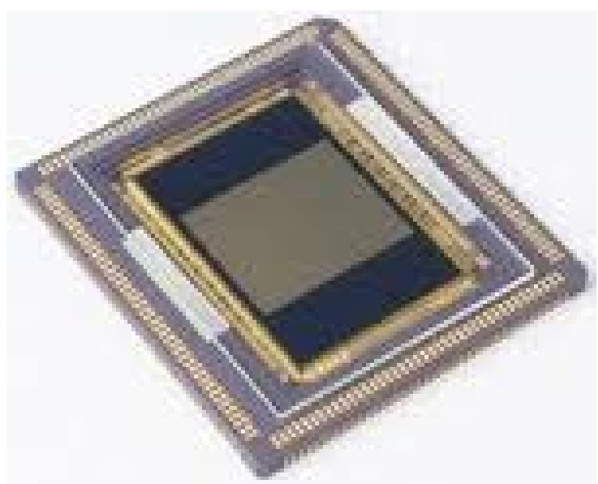


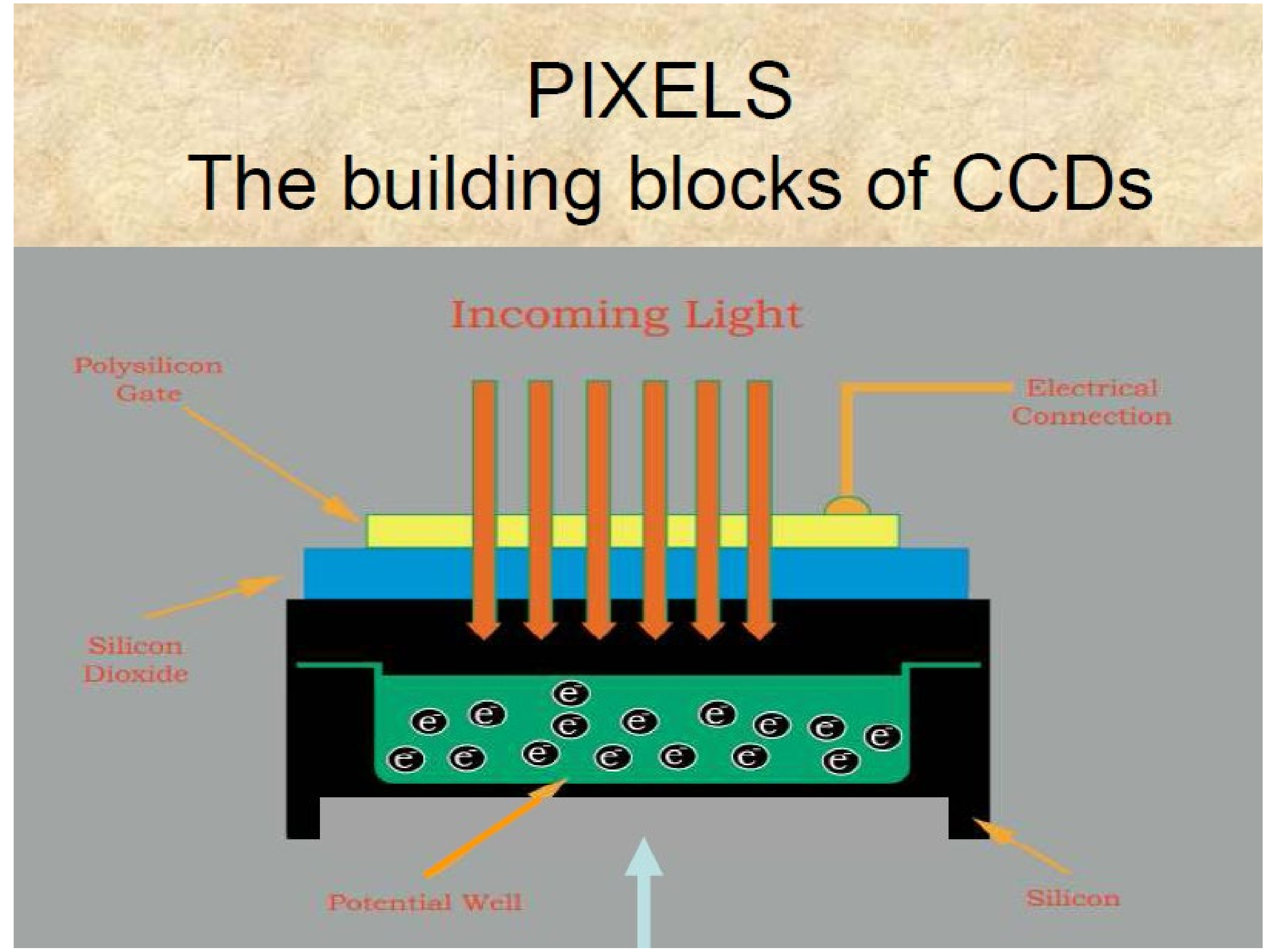
A digital image of ???

Image Analysis
(Brain or Computer)

A stick man?
How do I know?
How can computer know
- algorithm?

The digital image: pixel size

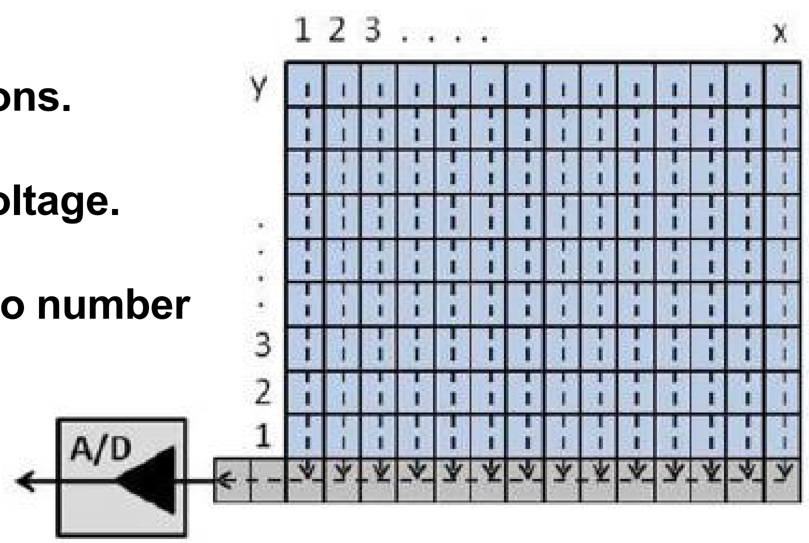




Photons induce transport of electrons.

Transport of electrons generates voltage.

Generated voltage is proportional to number of incoming photons.



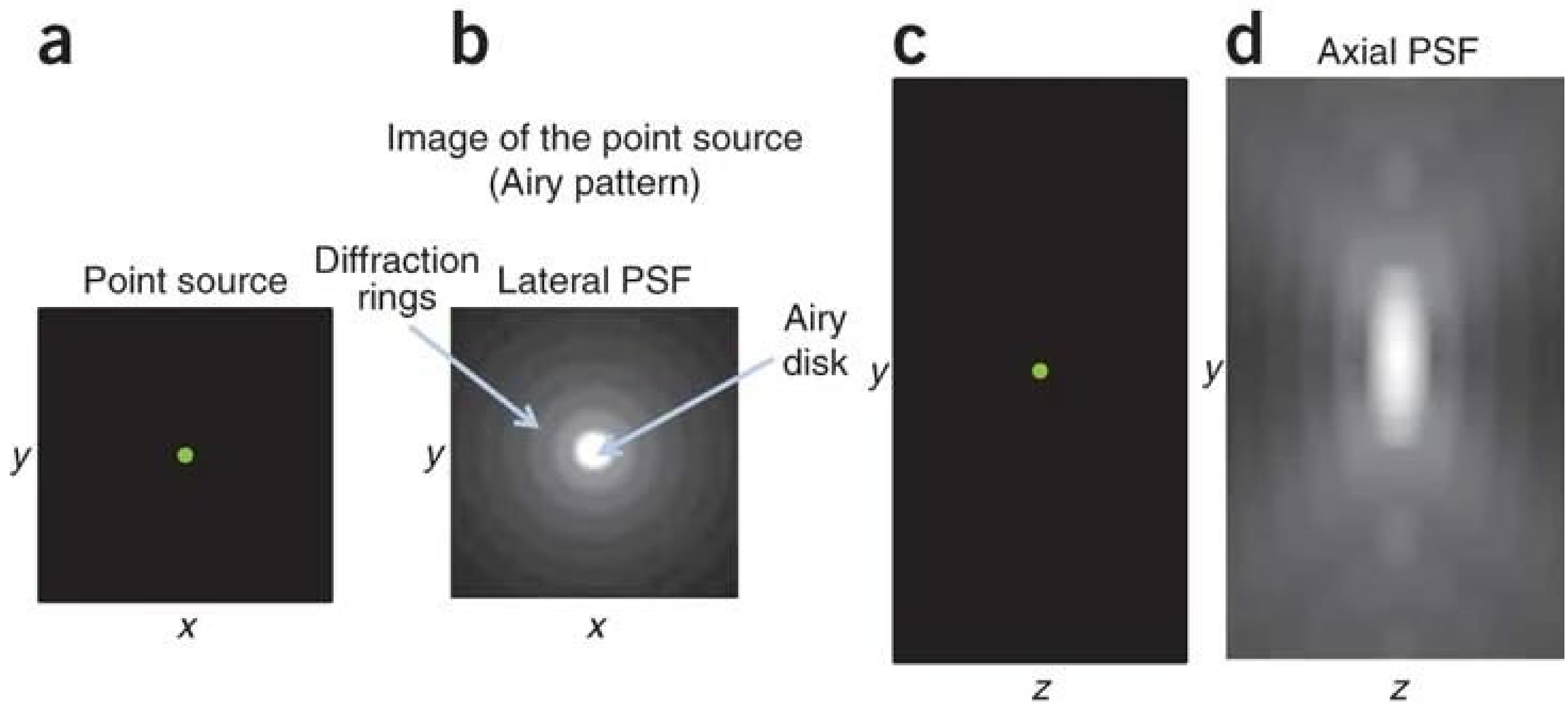
$S = \{a_{11}, a_{12}, ..a_{1x}, a_{21}, ..a_{2x}, ..a_{yx}\}$

Pixel Quantization

249	244	240	230	209	233	227	251	255
248	245	210	93	81	120	97	193	254
250	170	133	94	137	120	104	145	253
241	116	118	107	134	138	96	92	163
277	142	121	113	124	115	107	71	179
234	106	84	125	97	108	125	106	204
241	202	102	132	75	73	141	246	252
253	252	244	239	178	199	242	250	245
255	249	244	250	226	231	240	251	253

The point spread function (PSF)

The **point spread function** (**PSF**) describes the response of a microscope to a point source or point object.



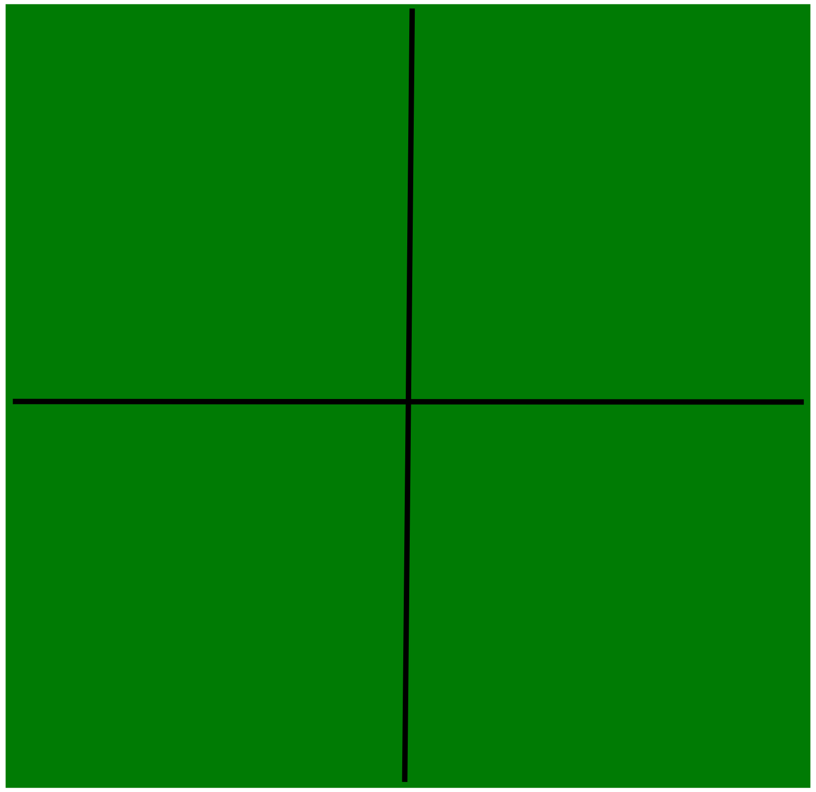
Pixel Size / Image Resolution

"Correct" image size? 64x64, 512x512, 2048x2048, ...

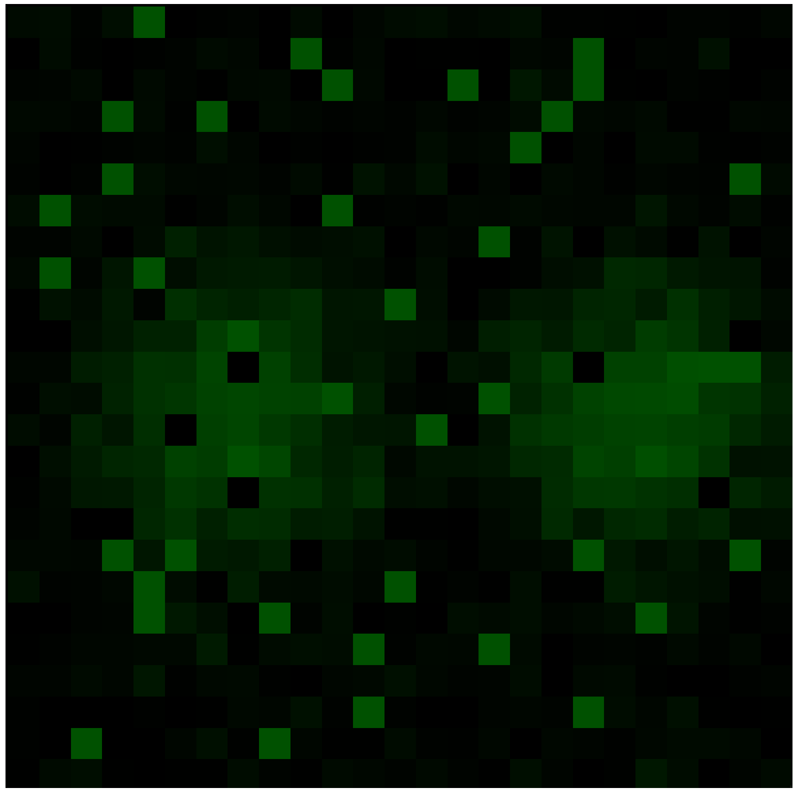
Nyquist – Shannon sampling theory: Proper spatial sampling

2.3 – 3 times smaller than optical resolution (x, y, AND z)

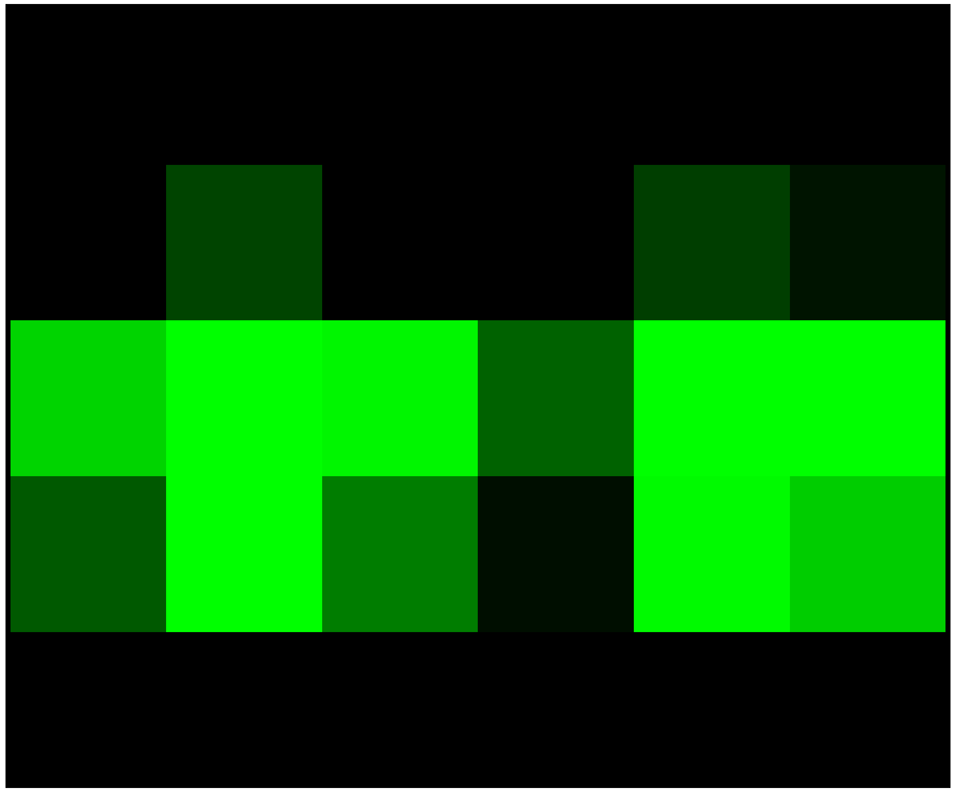
Adjust zoom, binning, and image size (no of pixels)



under sampled



over sampled



correct sampling

1 Airy unit

Nyquist sampling criterion

General form

Digital sampling frequency > analogue frequency x 2

Spatial representation

Image pixel size x 2.3 = smallest resolvable distance

Microscopy

Image pixel size x 2.3 = optical resolution (d)

Pixel size / Spatial Calibration

Pixel size is determined by the microscope system!

✓ CCD photodiode "pixel" size - Magnification X

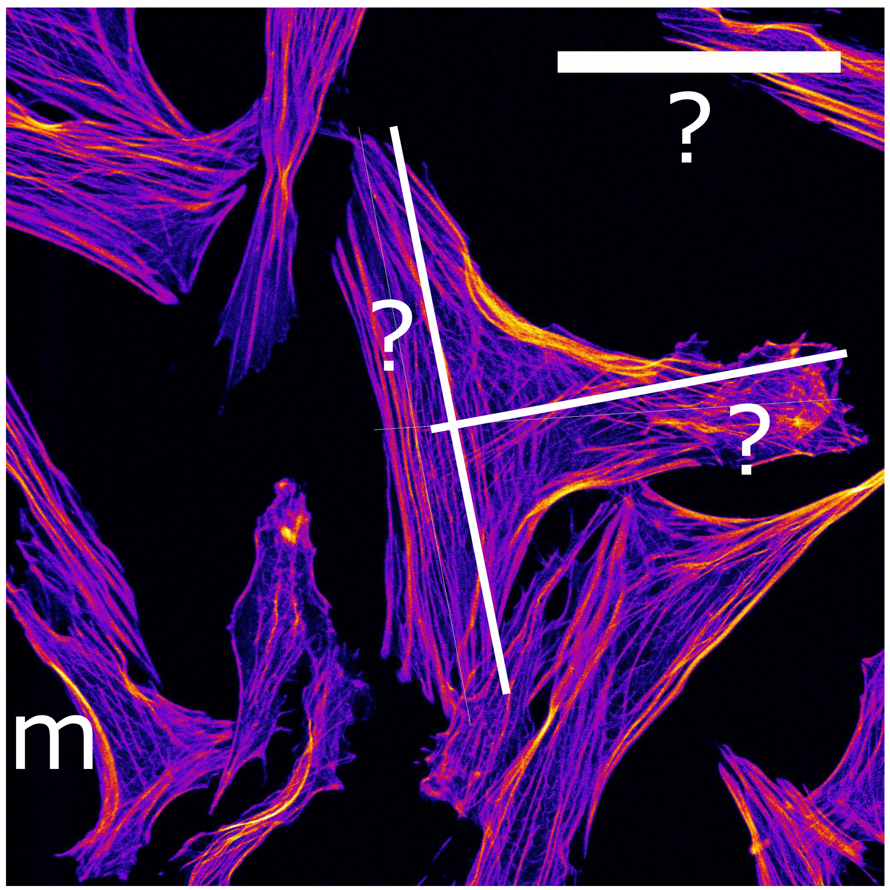
✓ Point scanner settings – zoom and image size

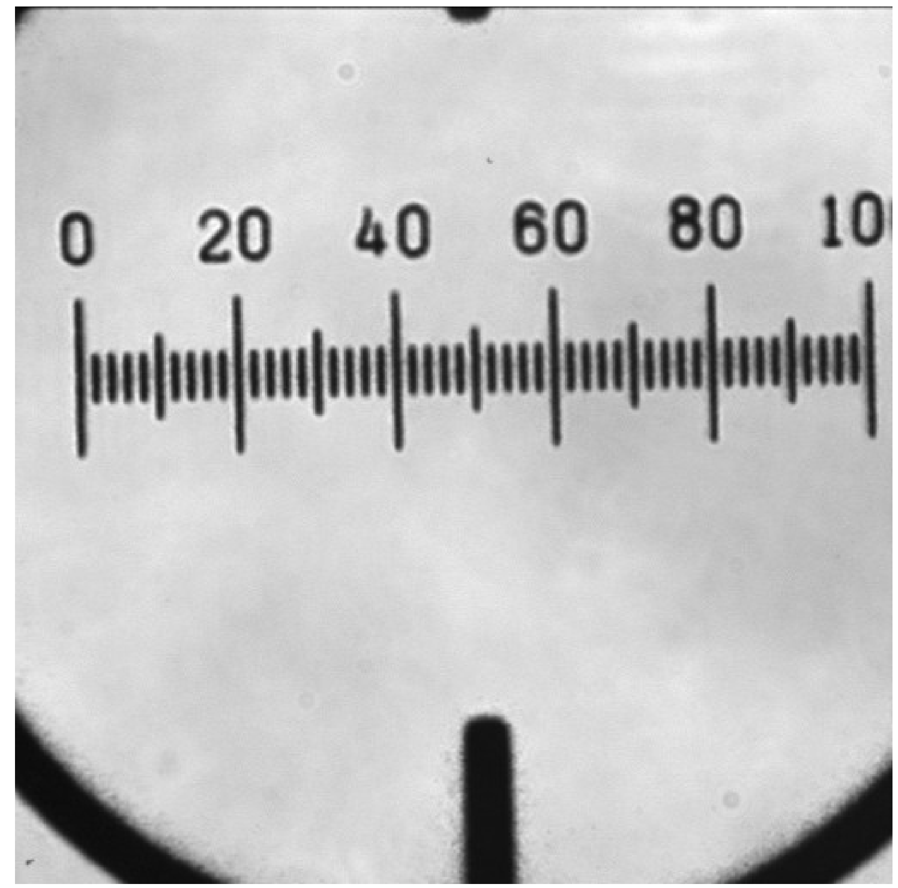
✓ Field of View Size - No. of Samples or "pixels"

It might be changed / lost during processing

It is stored in the "**Meta Data**"

Pixel Size / Spatial Calibration

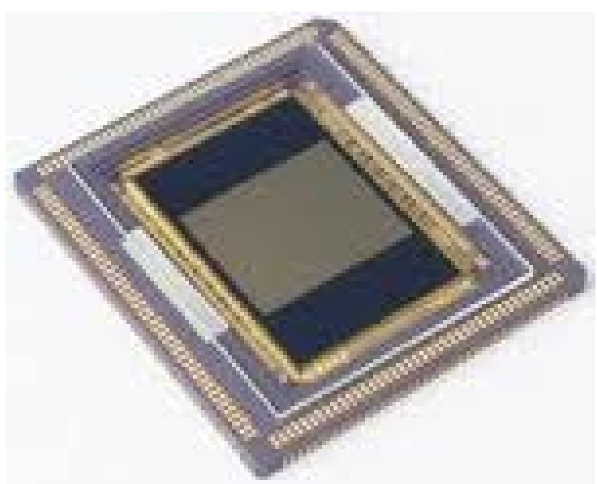


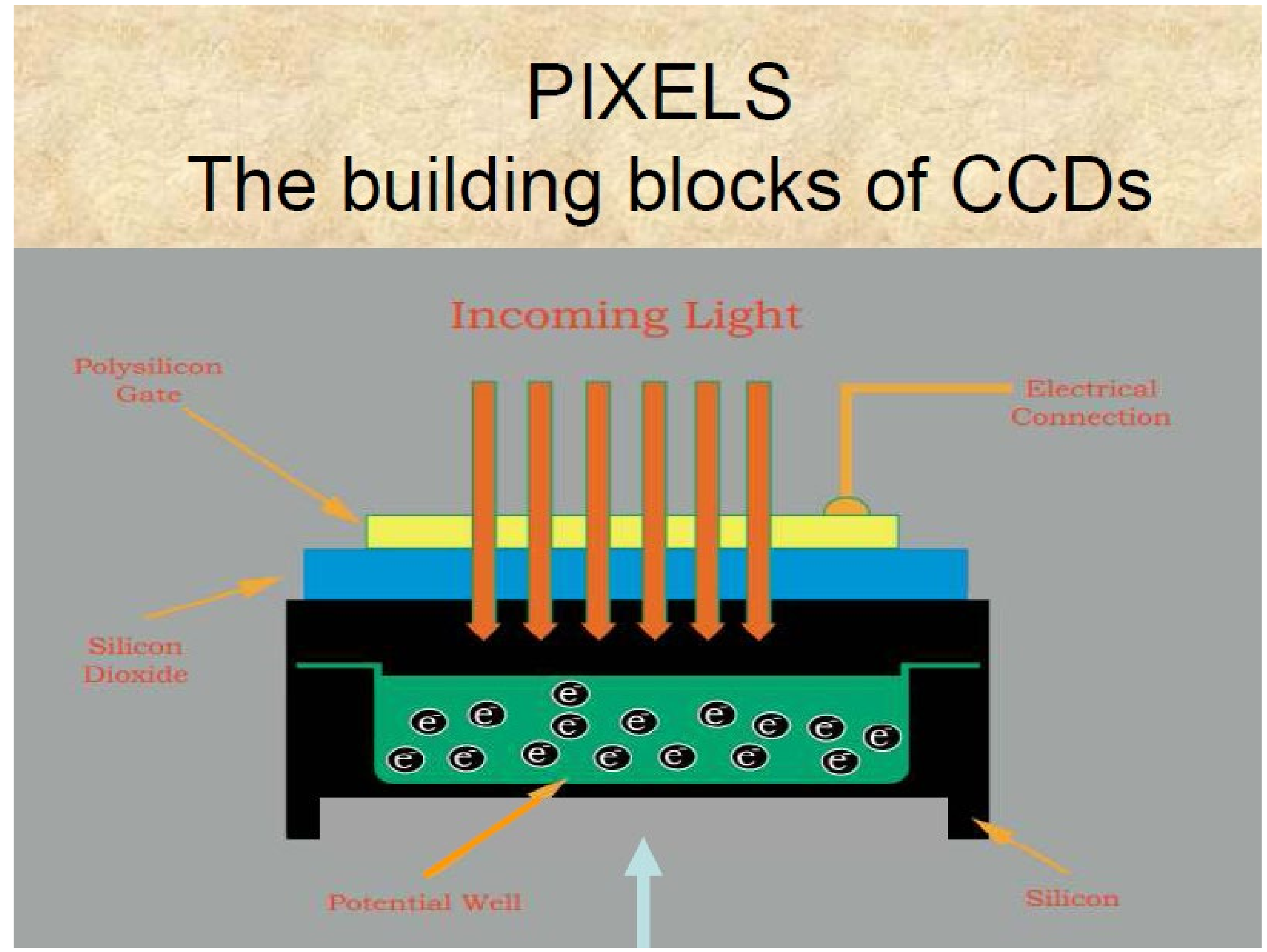


?

?

The digital image: Bit Depth





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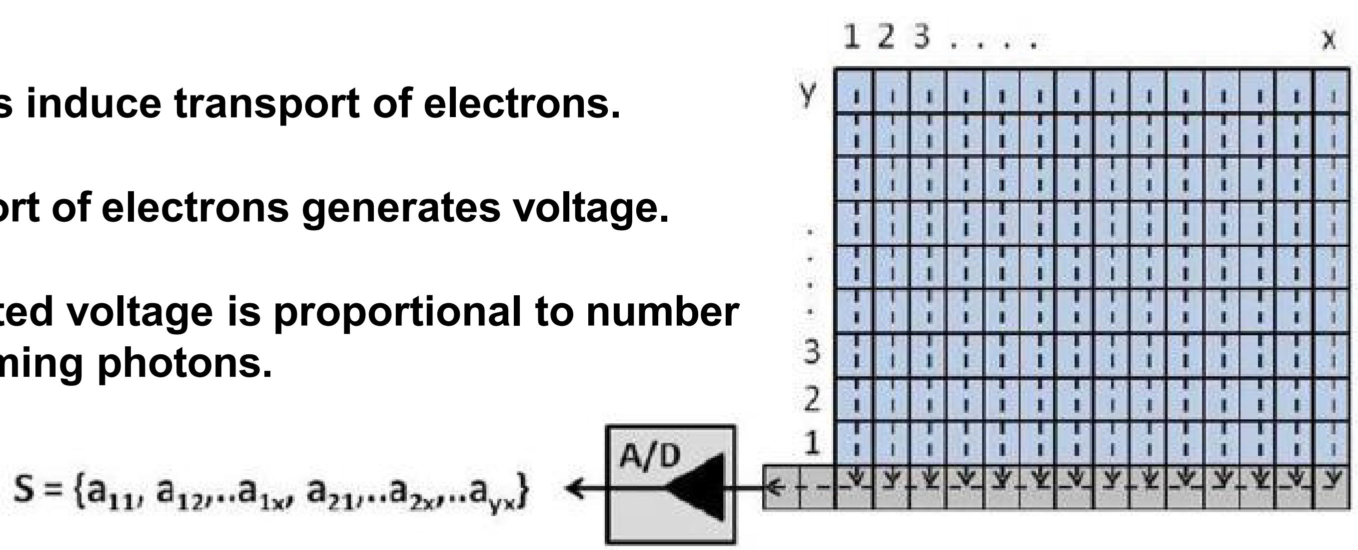

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248	245	210	93	81	120	97	193	254
250	170	133	94	137	120	104	145	253
241	116	118	107	134	138	96	92	163
277	142	121	113	124	115	107	71	179
234	106	84	125	97	108	125	106	204
241	202	102	132	75	73	141	246	252
253	252	244	239	178	199	242	250	245
255	249	244	250	226	231	240	251	253

"Intensity" Digitisation

Bit Depth

Measured intensity by detector

digitization

Corresponding level in image

"Bucket" holds
0-9 electrons

5 electrons counted

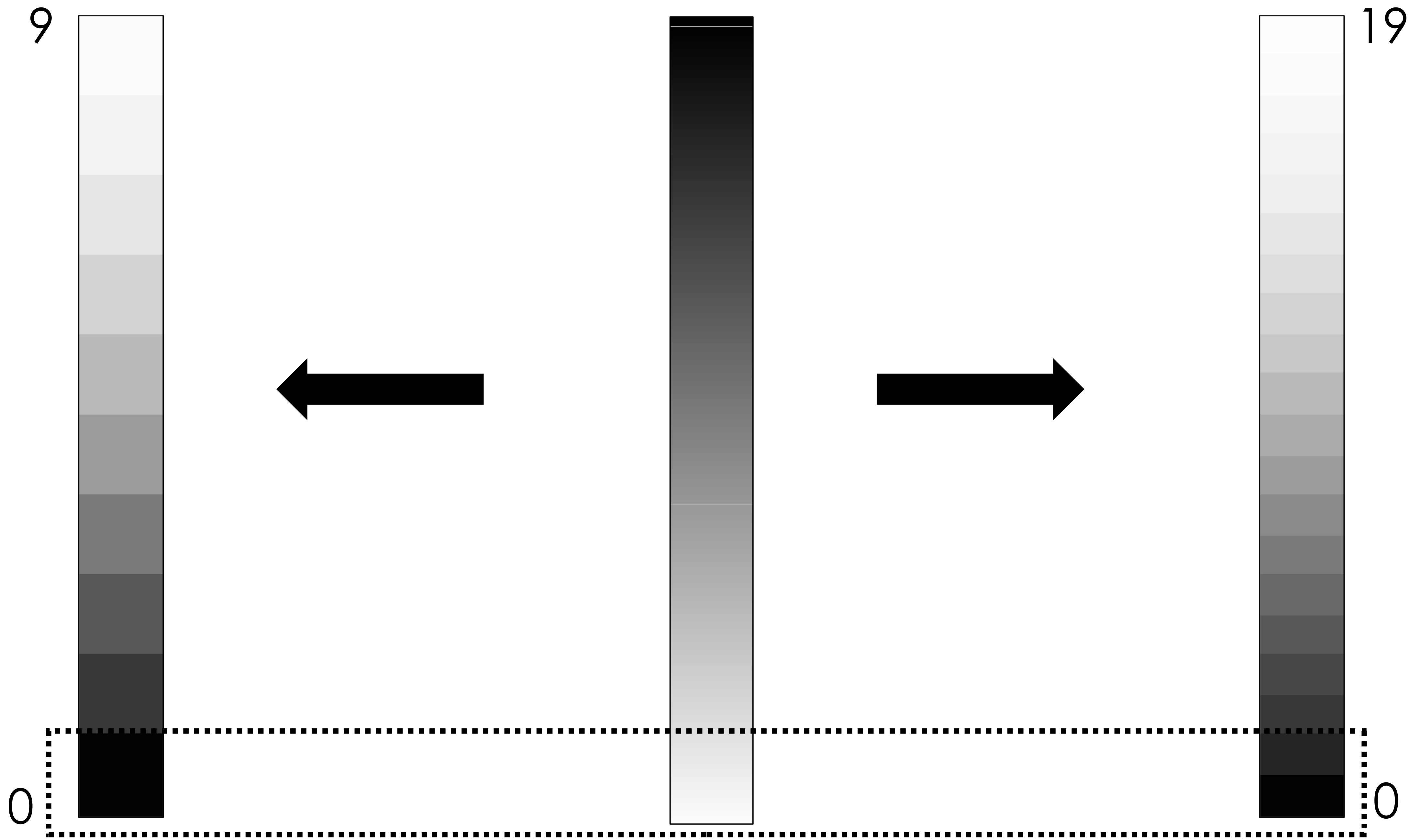
Bit depth: 10 (0 to 9)levels

Level 5 selected
for
RAW data "image"

“Intensity” Digitisation

Bit Depth

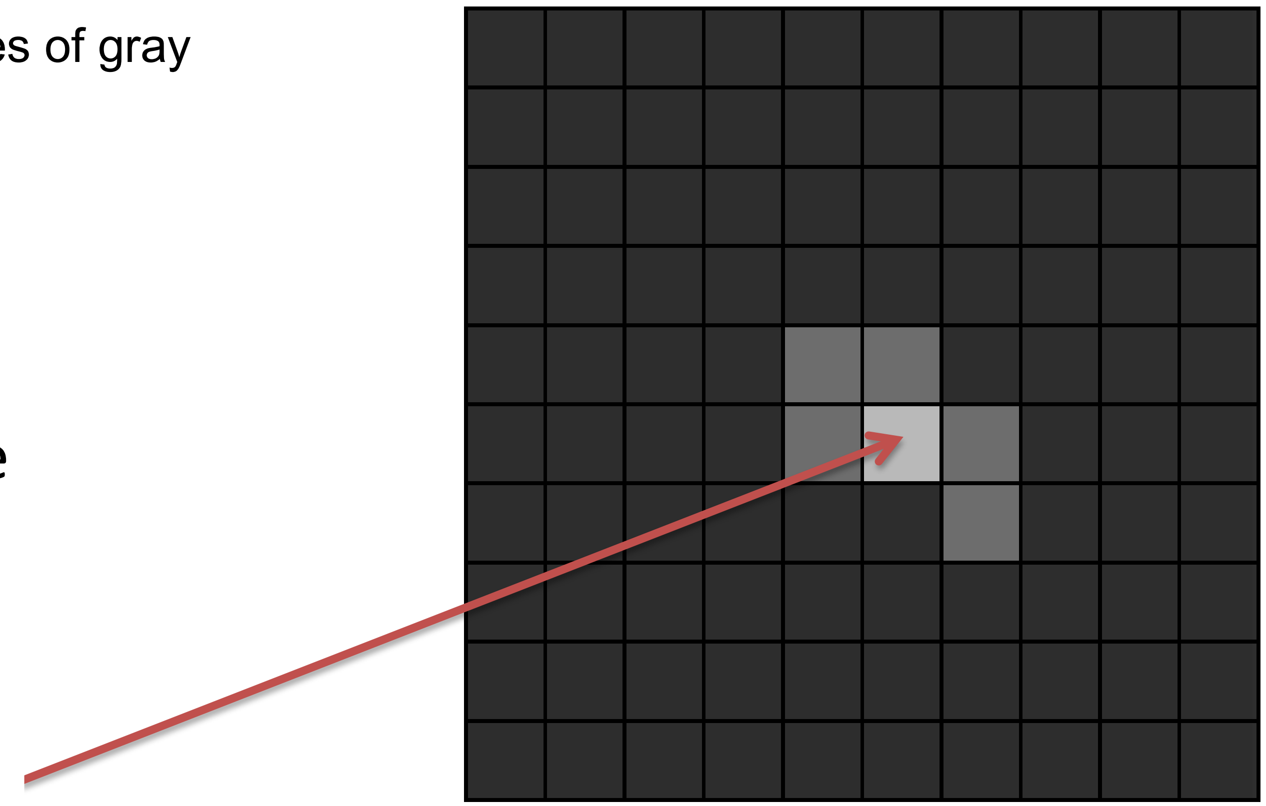
“digital“ intensity resolution: 10	“real” analogue intensities	“digital“ intensity resolution: 20



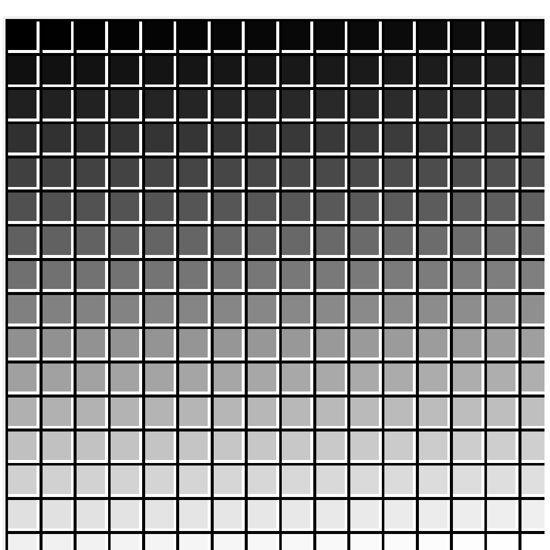
Brief explanation of digital imaging

$8 \text{ bit} = 2^8 = 256$ → Black, white, 254 shades of gray

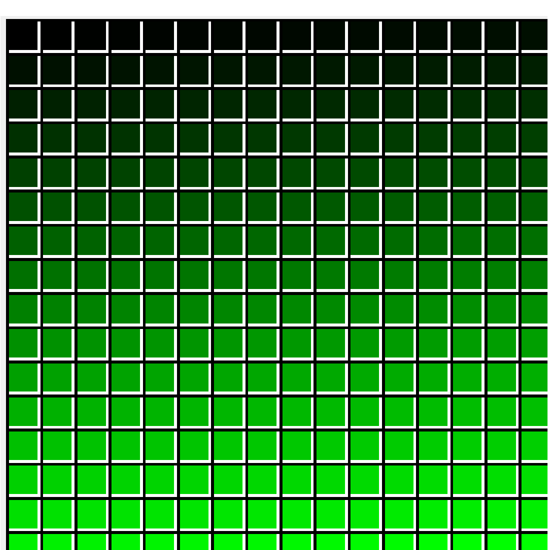
Binary	Hex	Decimal	Color
00000000	00	0	Black
11111111	FF	255	White



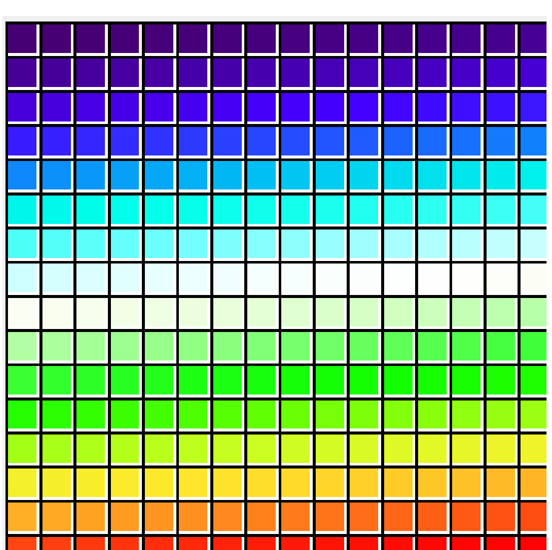
Color-table dependent!



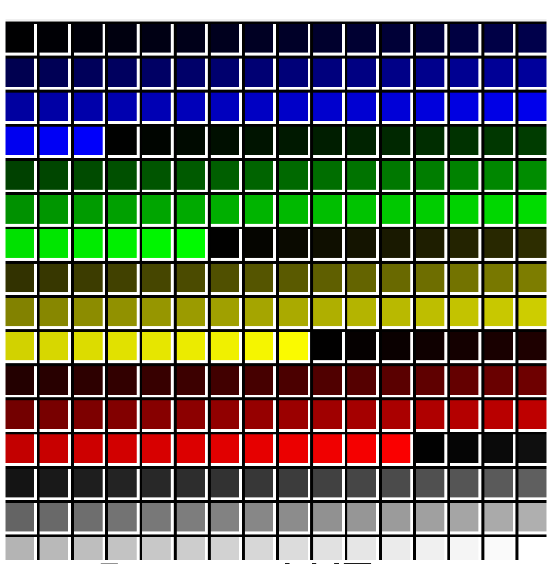
Gray LUT



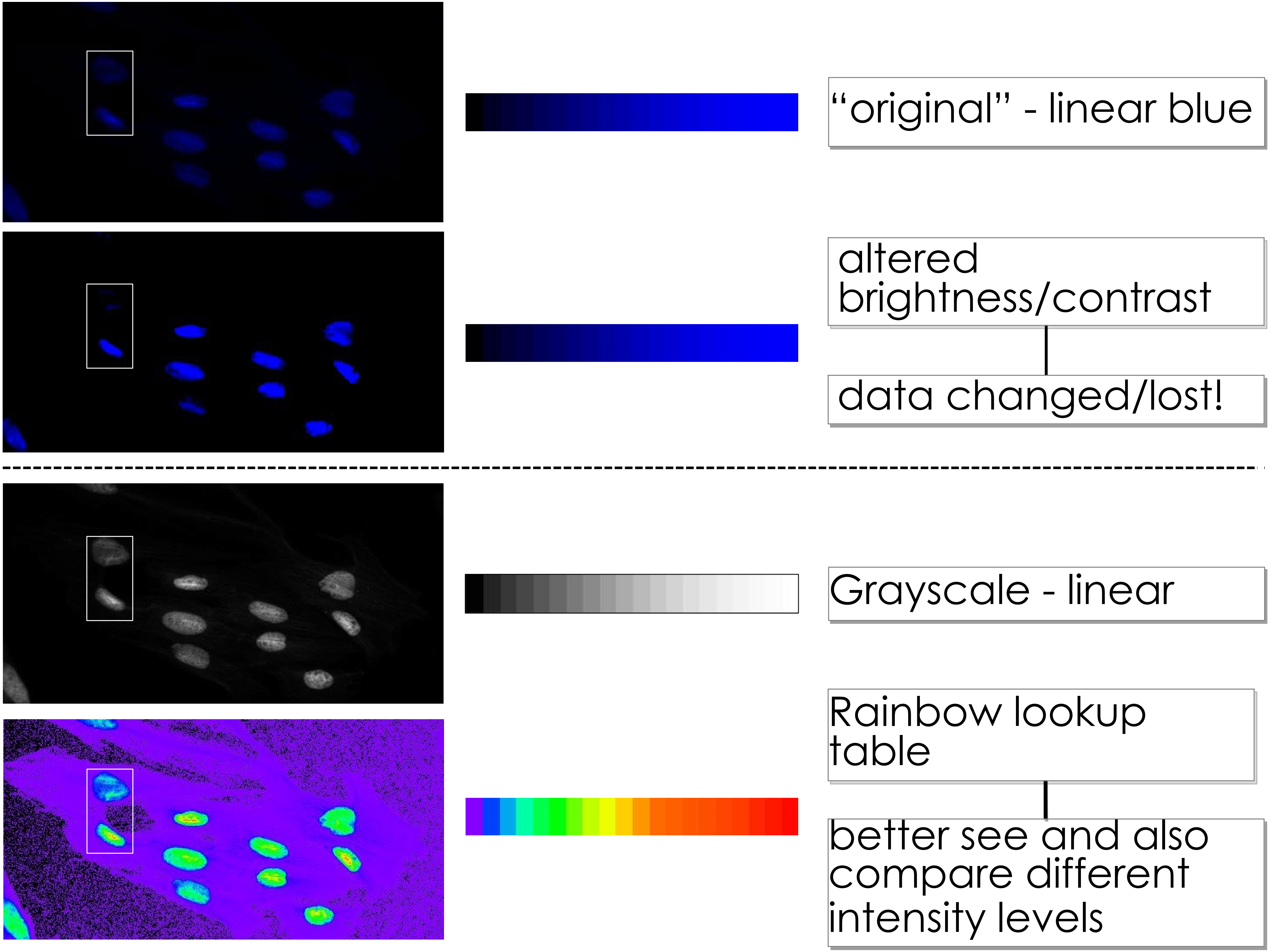
Green LUT



Thermal LUT



5 ramps LUT



“original” - linear blue
altered brightness/contrast
data changed/lost!
Grayscale - linear
Rainbow lookup table
better see and also compare different intensity levels

Image depth

Bit Depth and Gray Levels in Digital Images

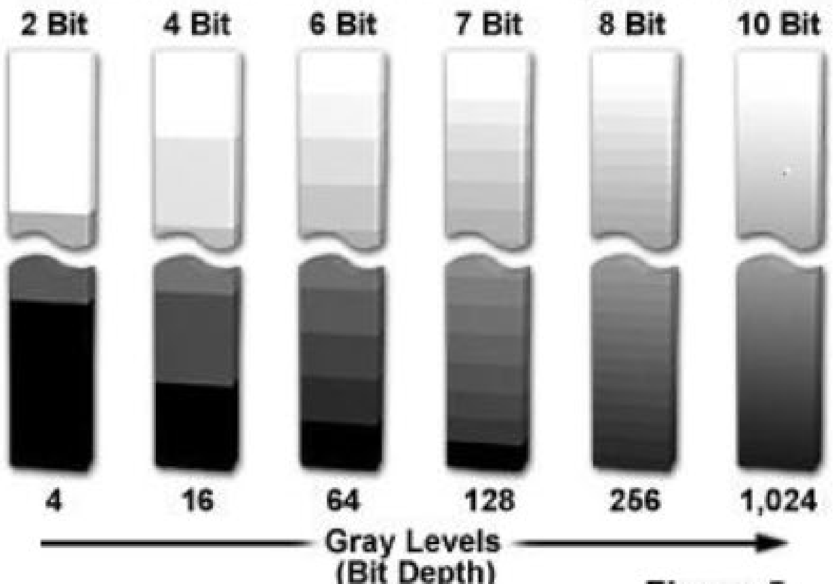


Figure 5

Grayscale Resolution and Digital Image Appearance

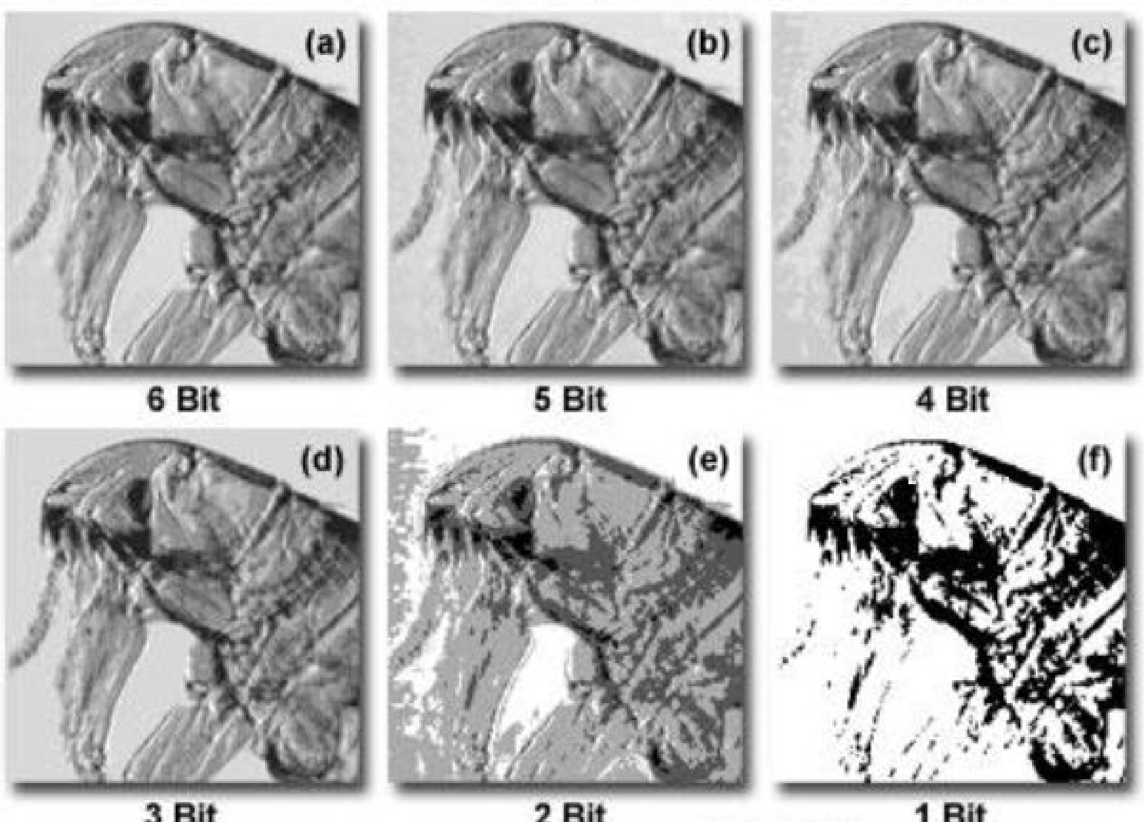


Figure 6

Bit Depth	Grayscale Levels	Dynamic Range (Decibels)
1	2	6 dB
2	4	12 dB
3	8	18 dB
4	16	24 dB
5	32	30 dB
6	64	36 dB
7	128	42 dB
8	256	48 dB
9	512	54 dB
10	1,024	60 dB
11	2,048	66 dB
12	4,096	72 dB
13	8,192	78 dB
14	16,384	84 dB
16	65,536	96 dB
18	262,144	108 dB
20	1,048,576	120 dB

The histogram

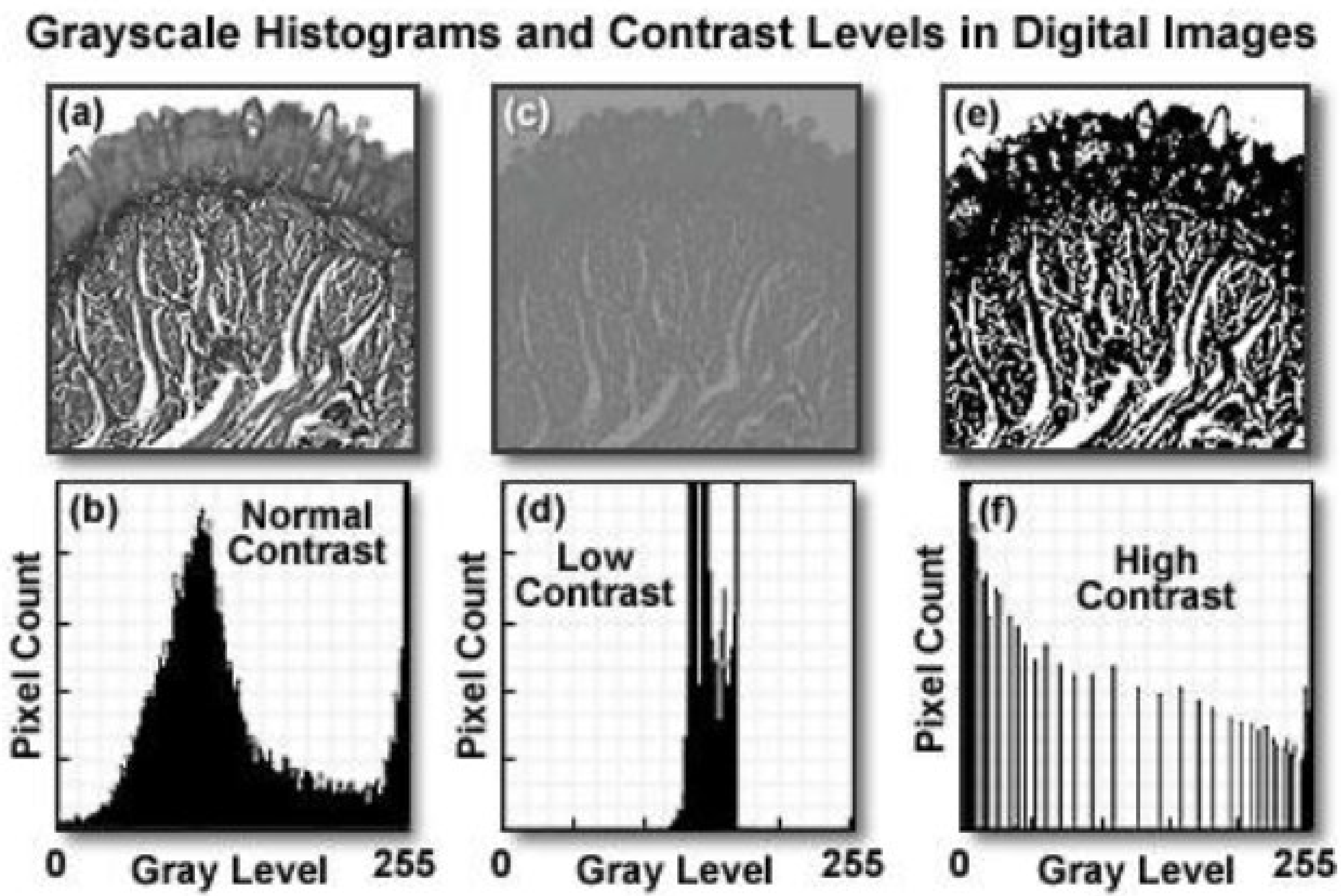


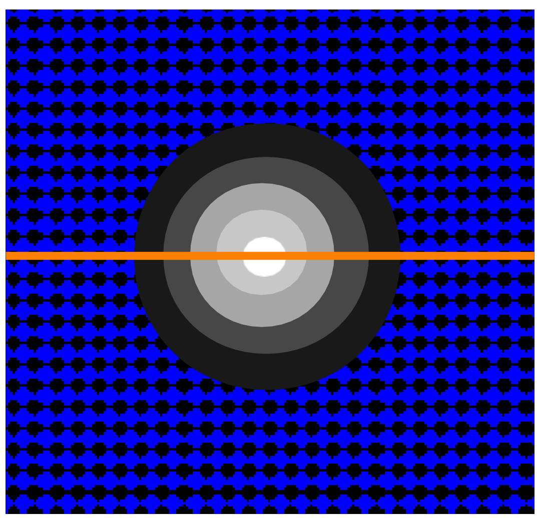
Figure 7

Intensity / Exposure / Saturation

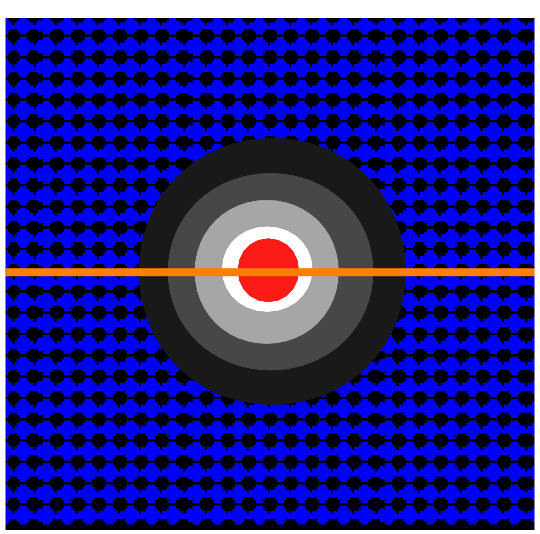
Do NOT over expose / saturate your image!!!

Why not? → Lost Information!

Use "Look Up Tables (LUT) / palettes to check the saturation



in range



clipped
overexposed
saturated

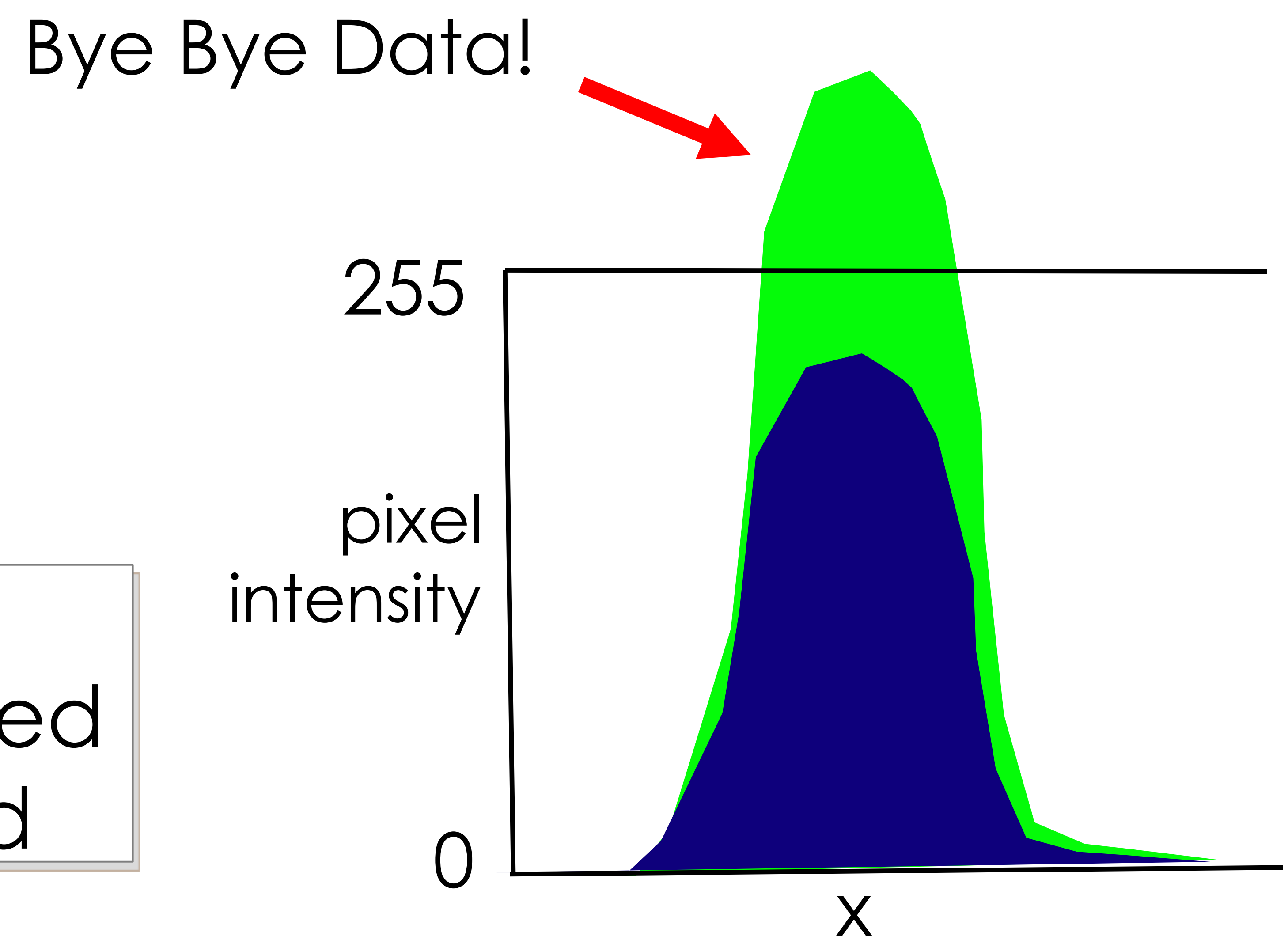
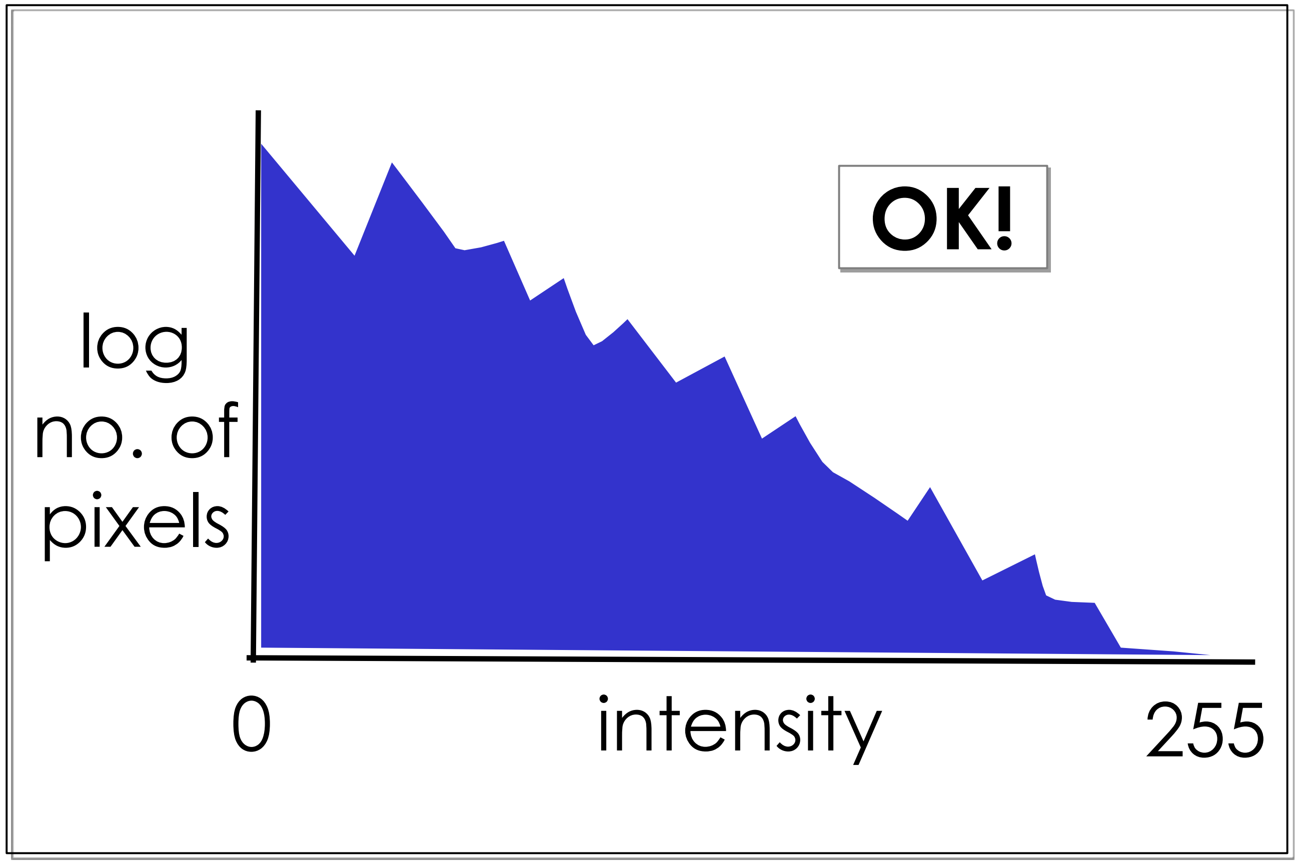
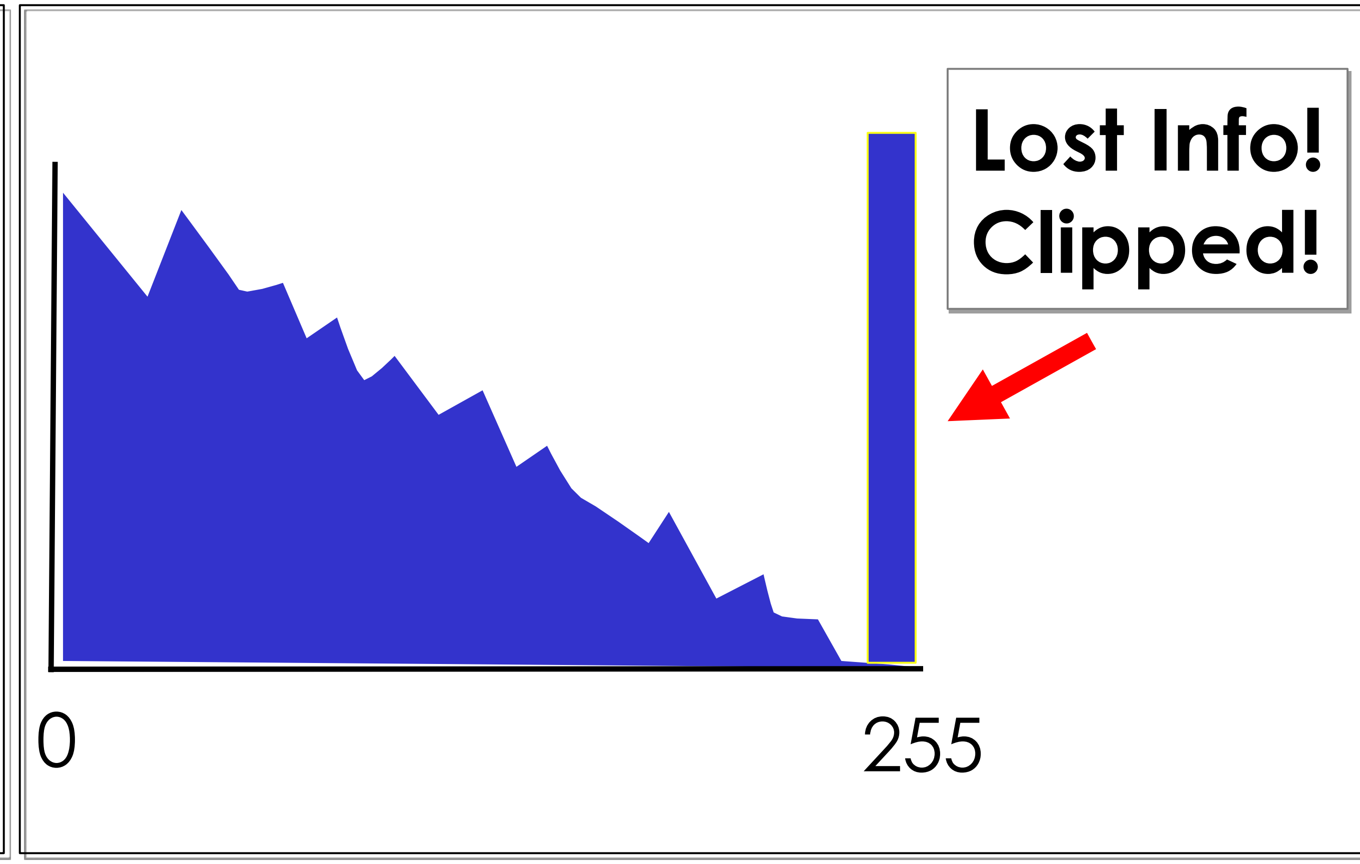


Image Intensity Histograms - Use them!

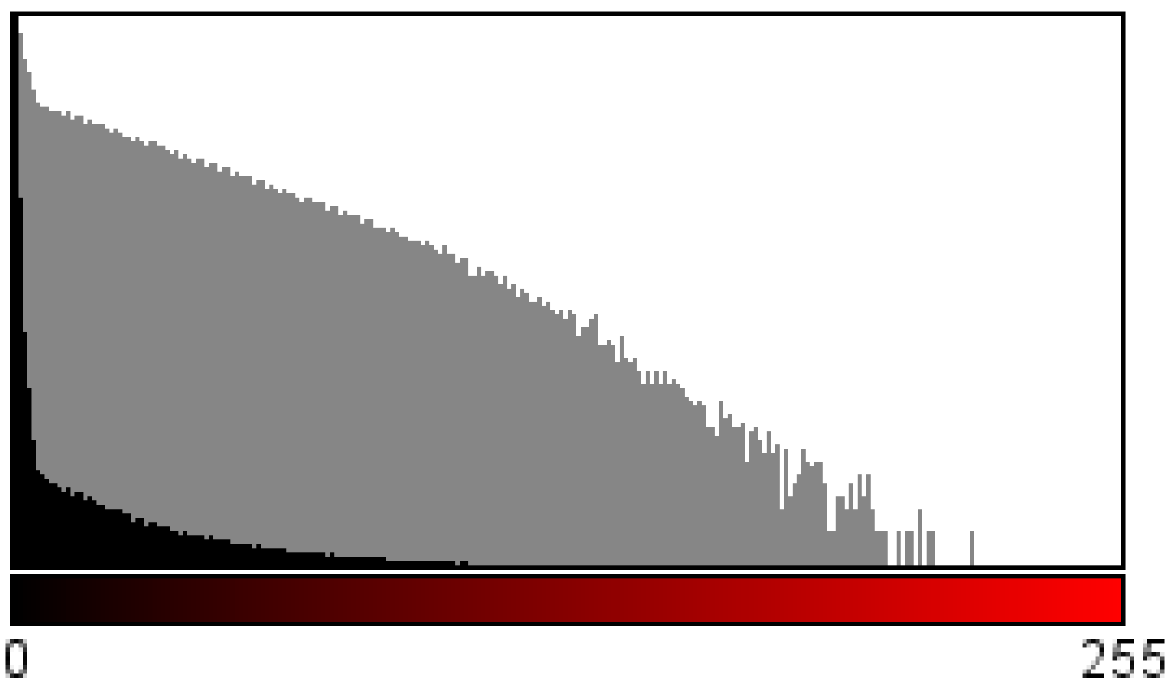




In Histograms:
easily see problems
for image
quantification!

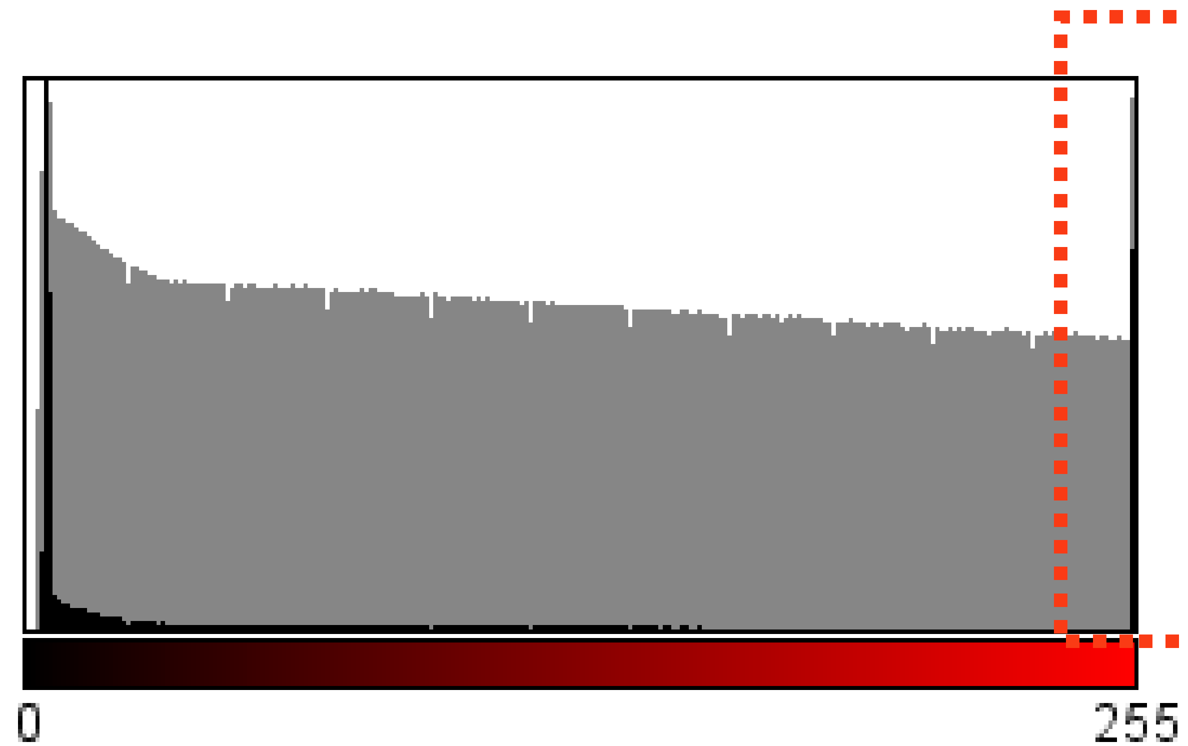
Intensity Histogram

Fluorescence Microscopy



Count: 524288
Mean: 18.561
StdDev: 26.465
Min: 0
Max: 235
Mode: 0 (174427)

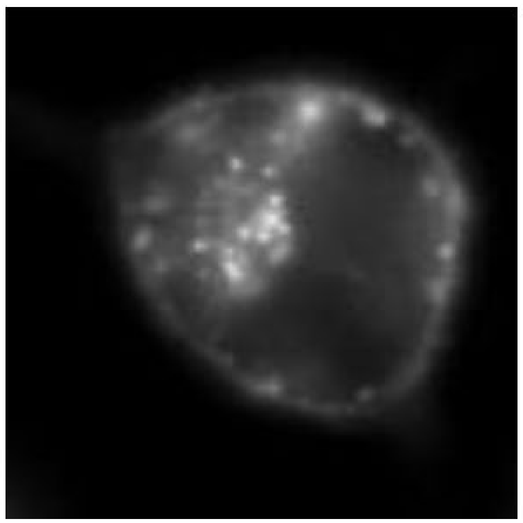
OK

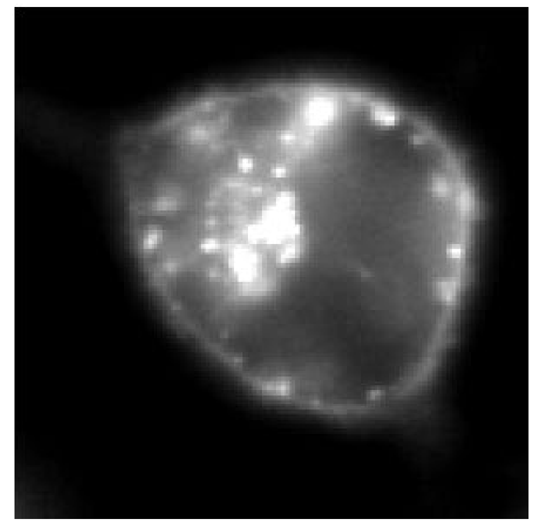


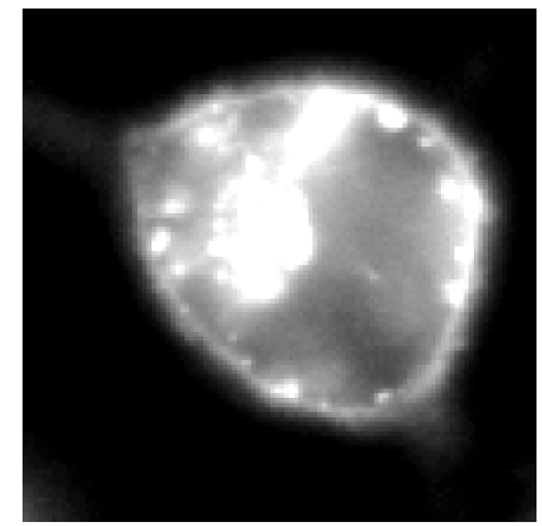
Count: 524288
Mean: 82.504
StdDev: 93.452
Min: 2
Max: 255
Mode: 4 (101652)

not OK - why?

Intensity / Exposure / Saturation

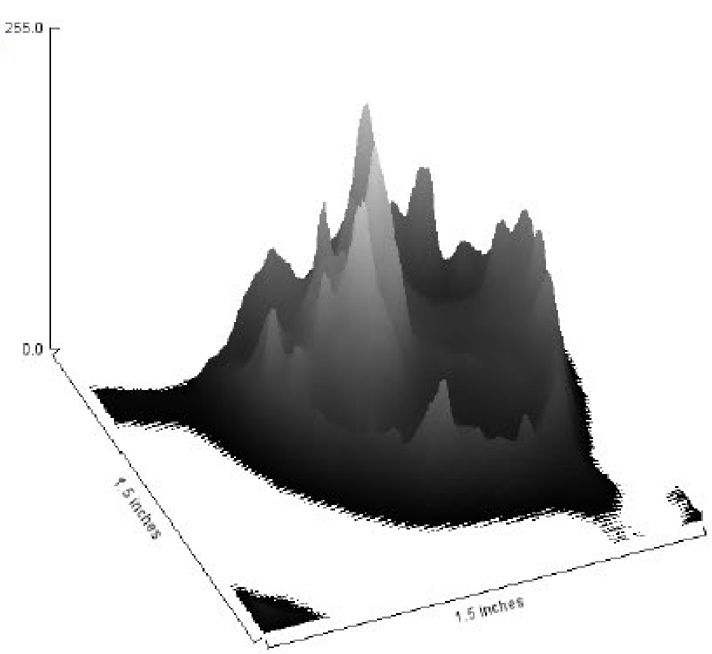


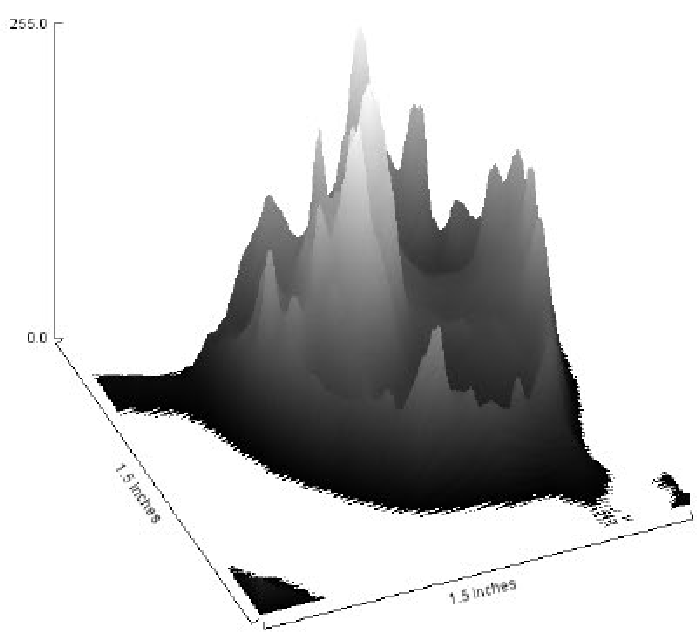


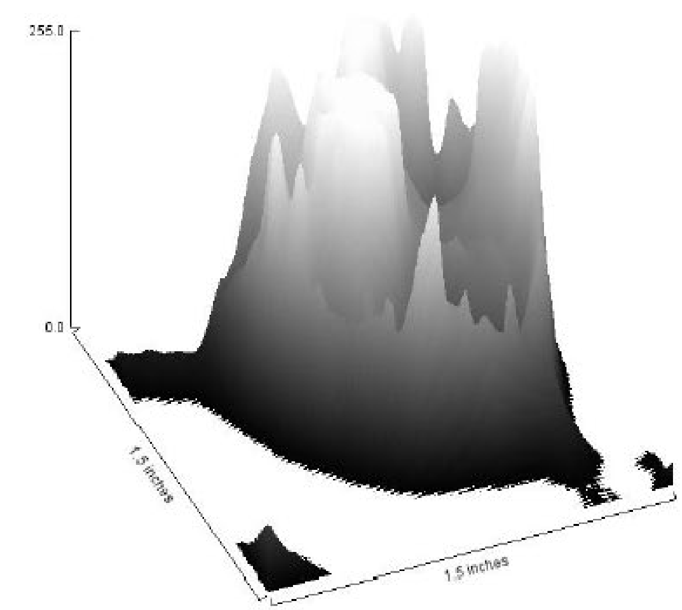


256

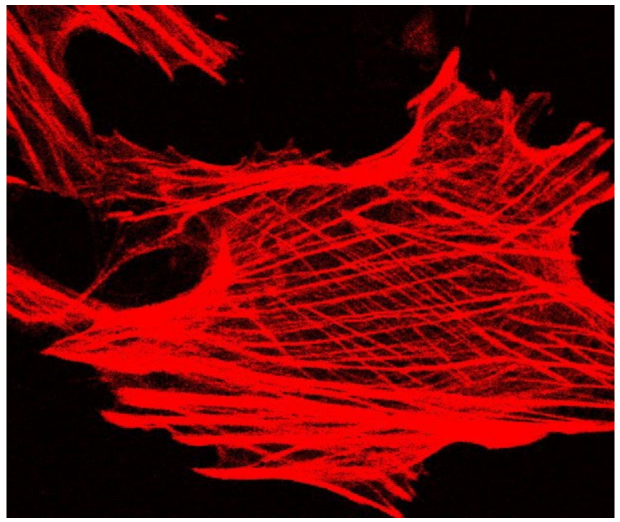
0

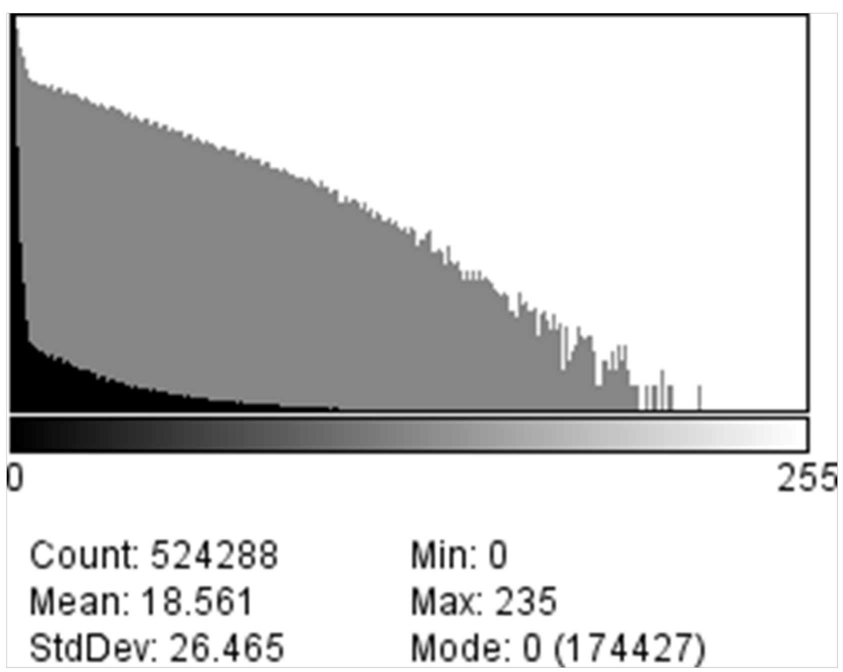




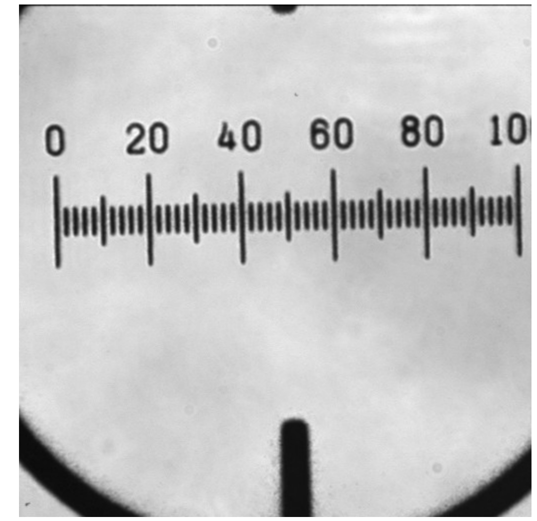


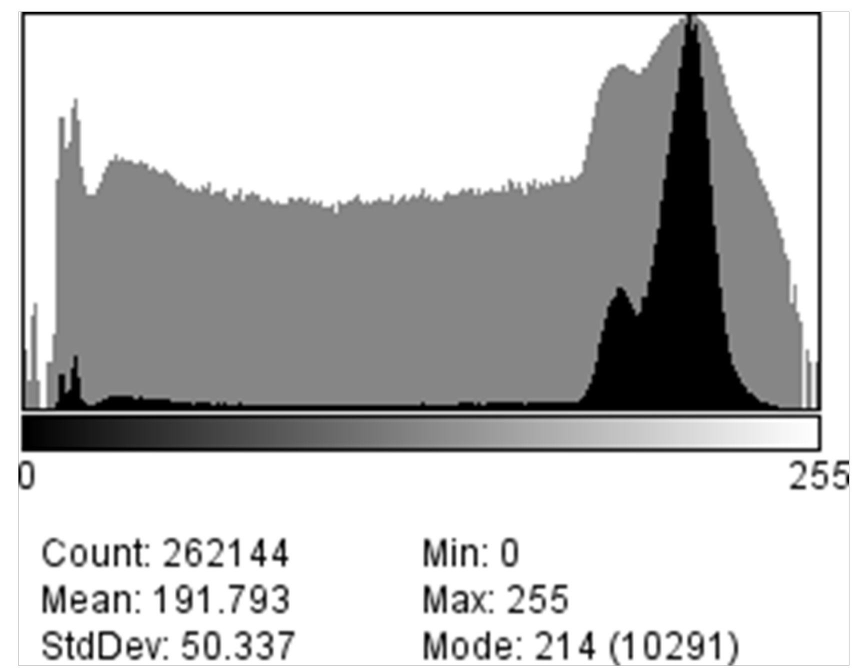
Intensity Histogram





fluorescence





brightfield

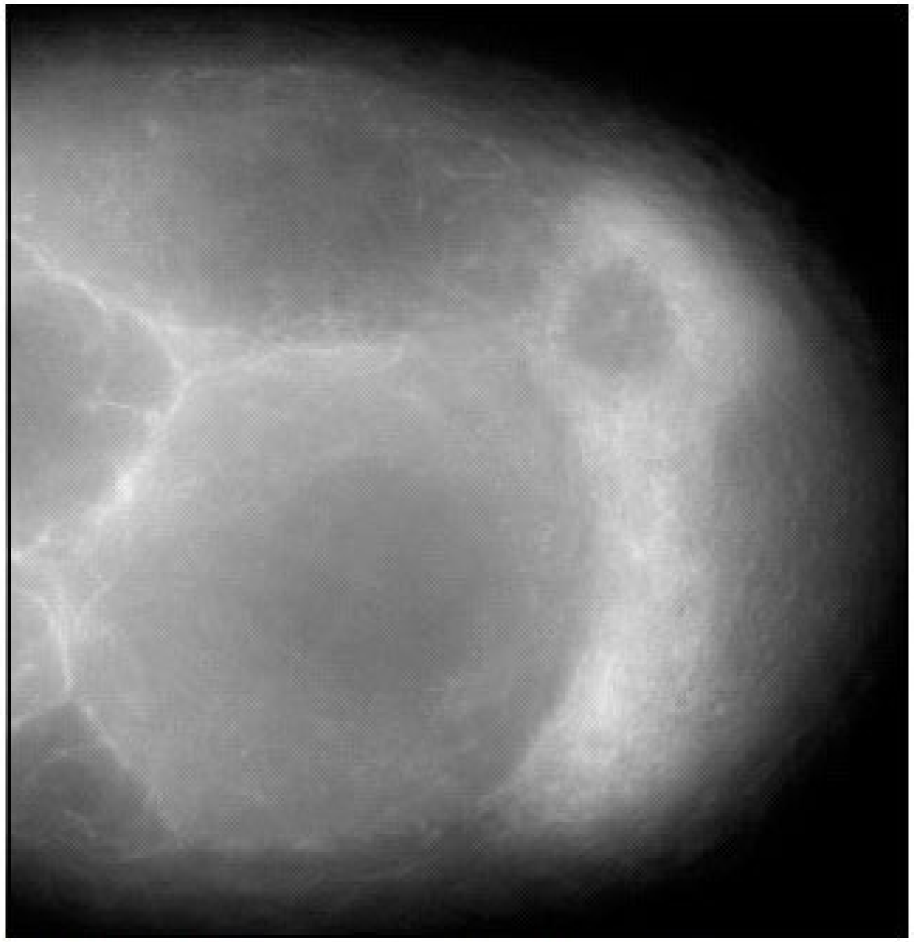
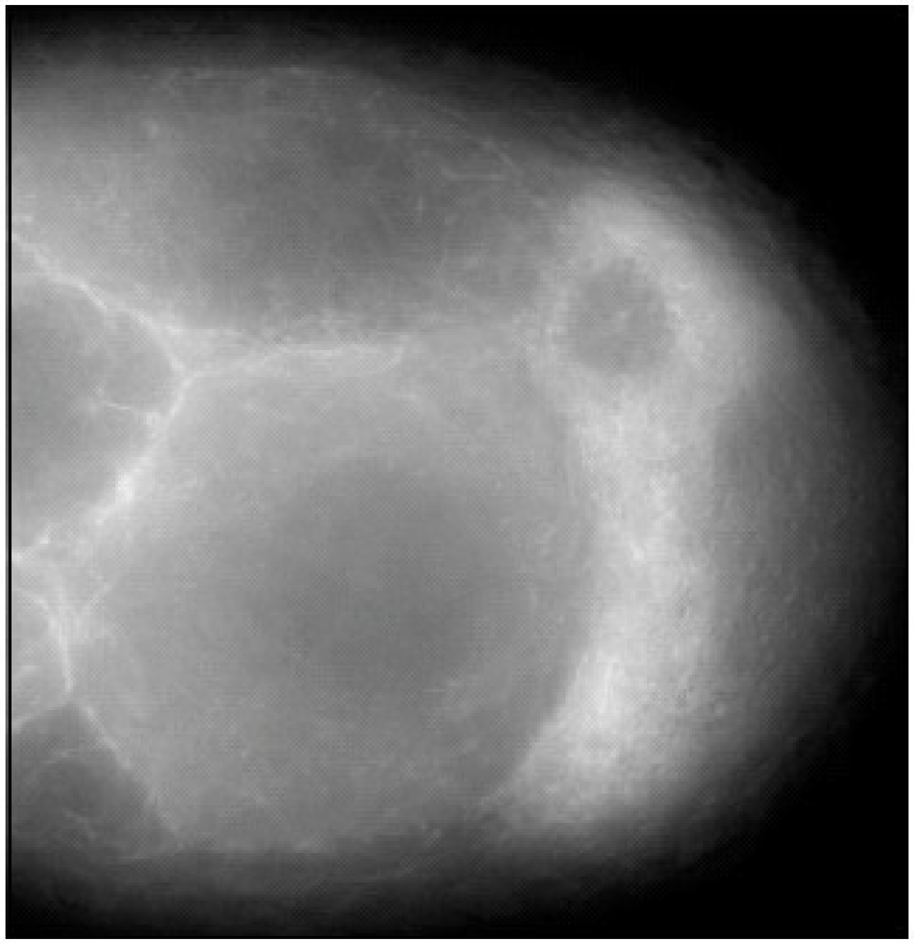


Image plus meta data



Wide-field fluorescence 490 ex 520 em
X60 1.2 WI
xy 212 nm; 60 (z step 200 nm)
Bin 2x2
250 ms exposure
Contrast stretched to fill 8 bit display
Tau-GFP Oocyte

- Type of imaging – Wide-field fluorescence
- Excitation and Emission wavelengths 490/520
- Optics used – x60 NA 1.2 water immersion
- Image pixel dimensions – 212x212 (x200) nm
- Depth or Dynamic range – 8 bit; 256 greys
- Any processing performed
  - 12 bit to 8 bit conversion
  - contrast adjustment
- Display parameters - range 0-255, grey scale
  - gamma = 1
- The Biology - Drosophila stage 8 egg chamber
  - Tau GFP, labelling microtubules

http://www.openmicroscopy.org/

ImageJ using FIJI

What is Fiji and ImageJ

Open source https://imagej.net/downloads
Advantages and disadvantages

- Free
- Fairly easy to use
- Programmable
- Thousands of all ready developed ‘plugins’
- Google probably know what you are looking for.
- Poorly documented
- Work/click intensive
- The different plugins can be very inconsistent

Currently, if one works with microscopy, ImageJ is unavoidable!

File - Open Samples - Embryos

Spatial Scaling:

Can you measure the length and area of objects?

→ See Fiji Tutorial - SpatialCalibration (search Wiki)

- ✓ ***Analyze - Set Scale, Analyze-Tools-Scale Bar***
- ✓ ***Line and ROI selection - ctrl M (cmd M)***
- ✓ ***Rectangle, Oval, Polygon, Freehand, Angle, Point, Wand.***
- ✓ ***Analyze - Set Measurements (Results – Edit - summarize)***

File - Open Samples - Neuron

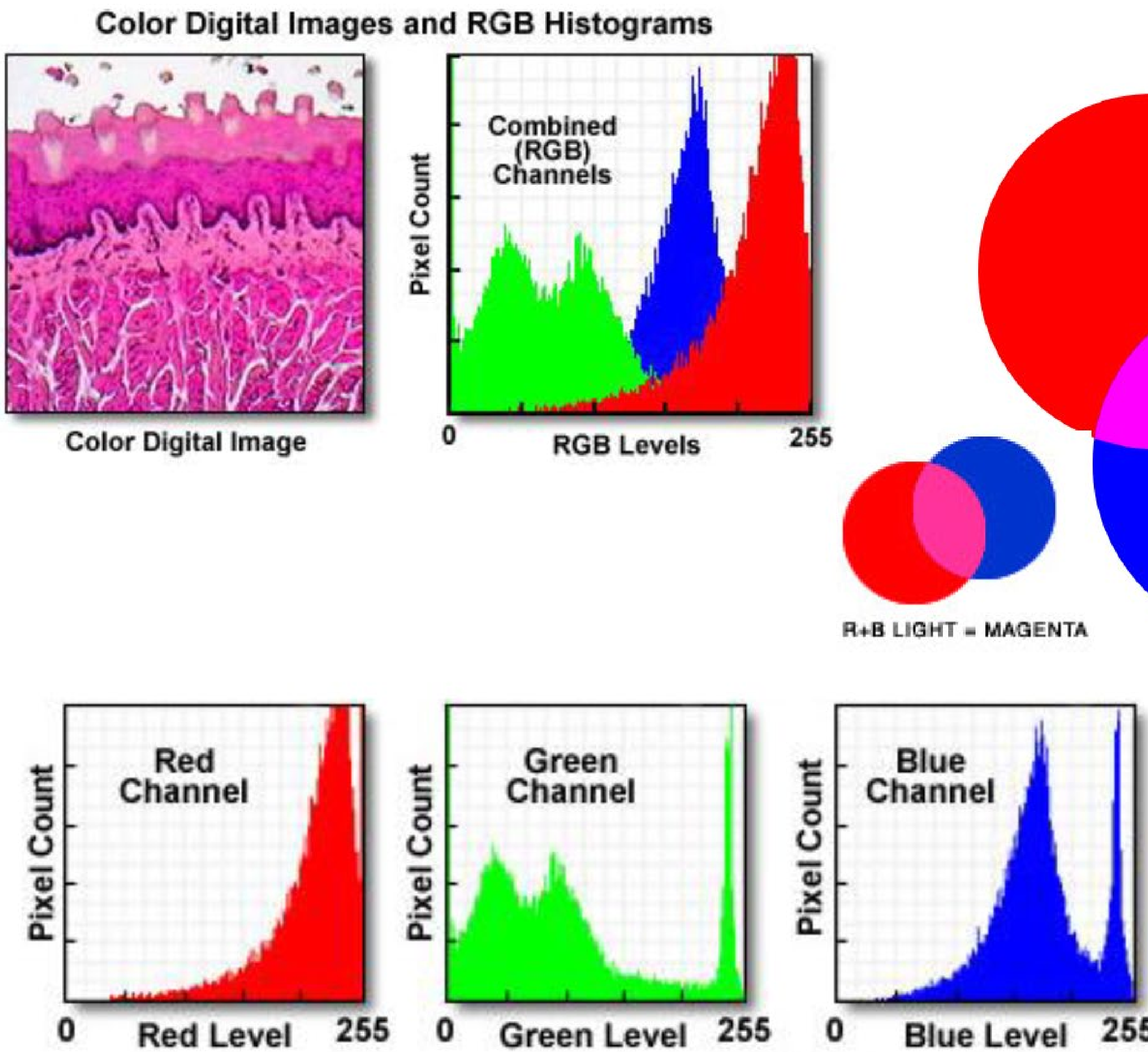
Intensity clipping/ saturation and offset:

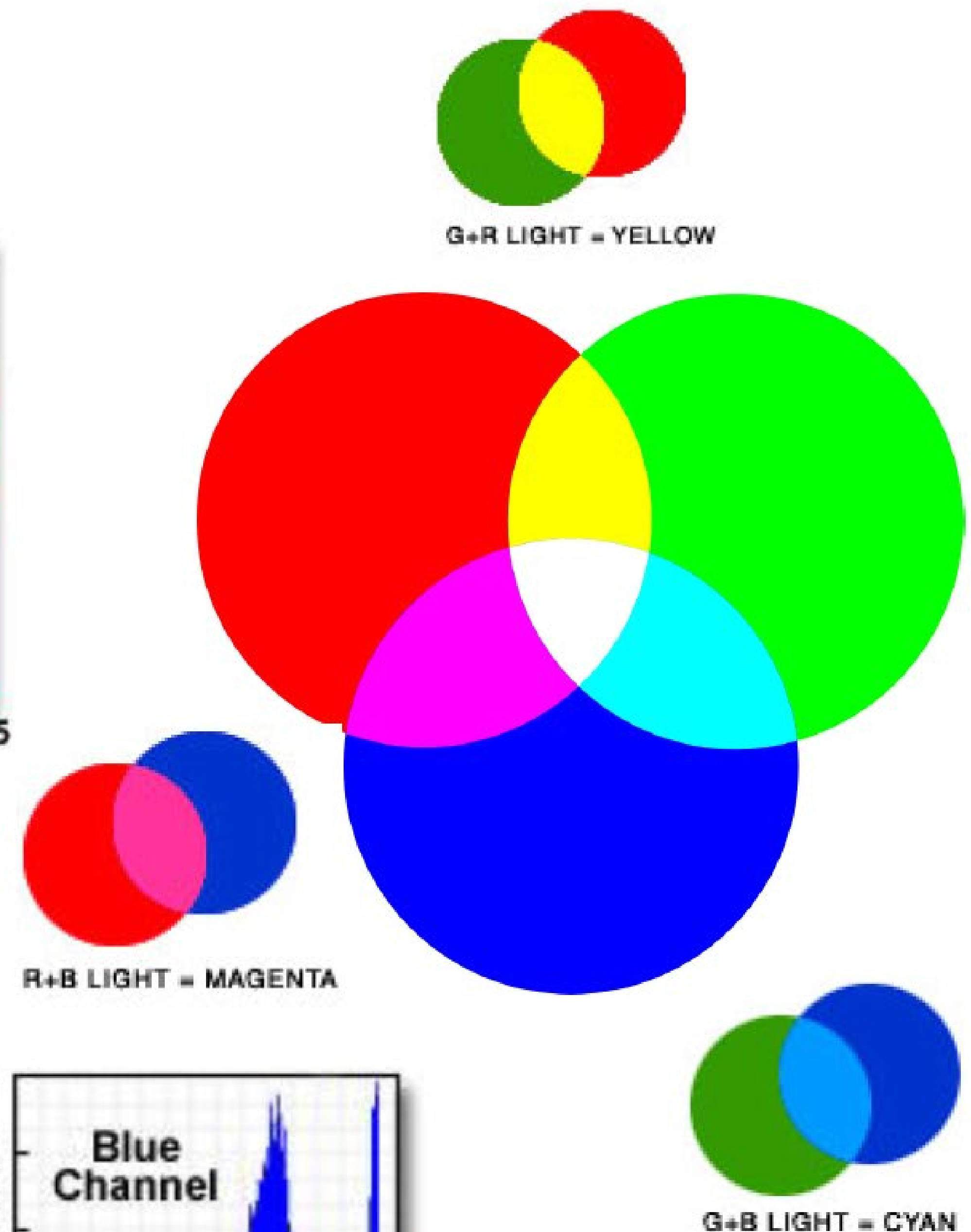
✓ <u>**Bit Depth**</u> **–** ***change from 16 to 8. What happens to the numbers?***

✓ <u>**Brightness/Contrast**</u>**:** ***Image-Adjust-Brightness/Contrast. Realize: you can loose data using "Apply"!***

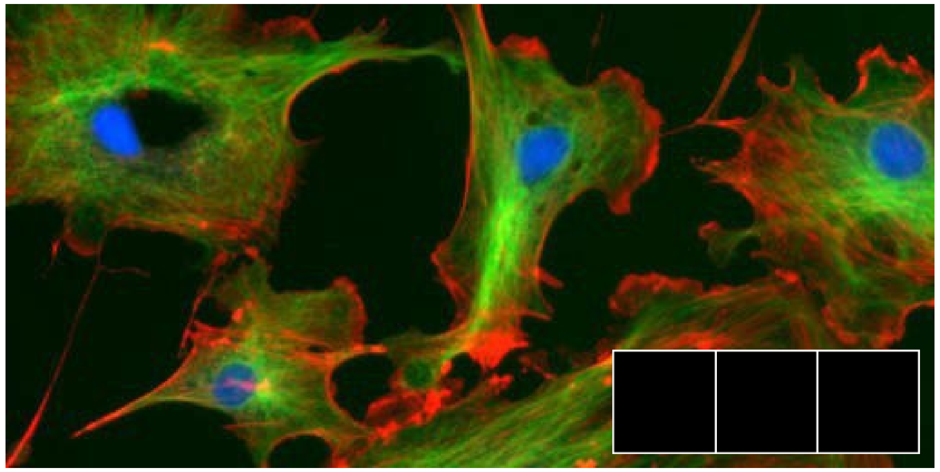
✓ <u>**Intensity Histograms**</u>**:** ***log scale for fluorescence***

Color images

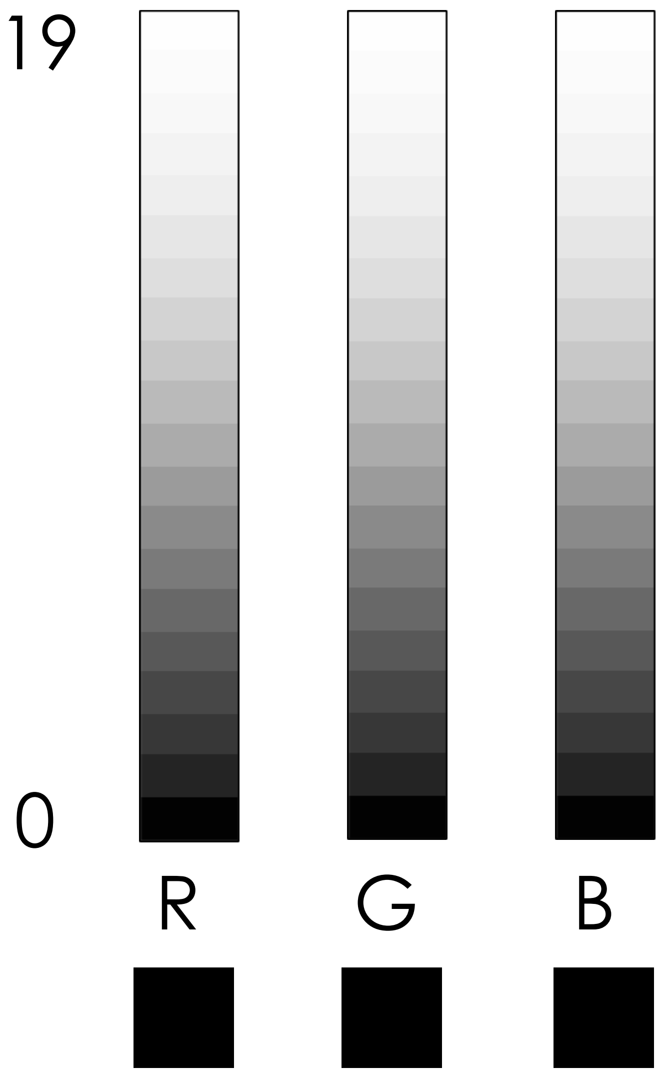




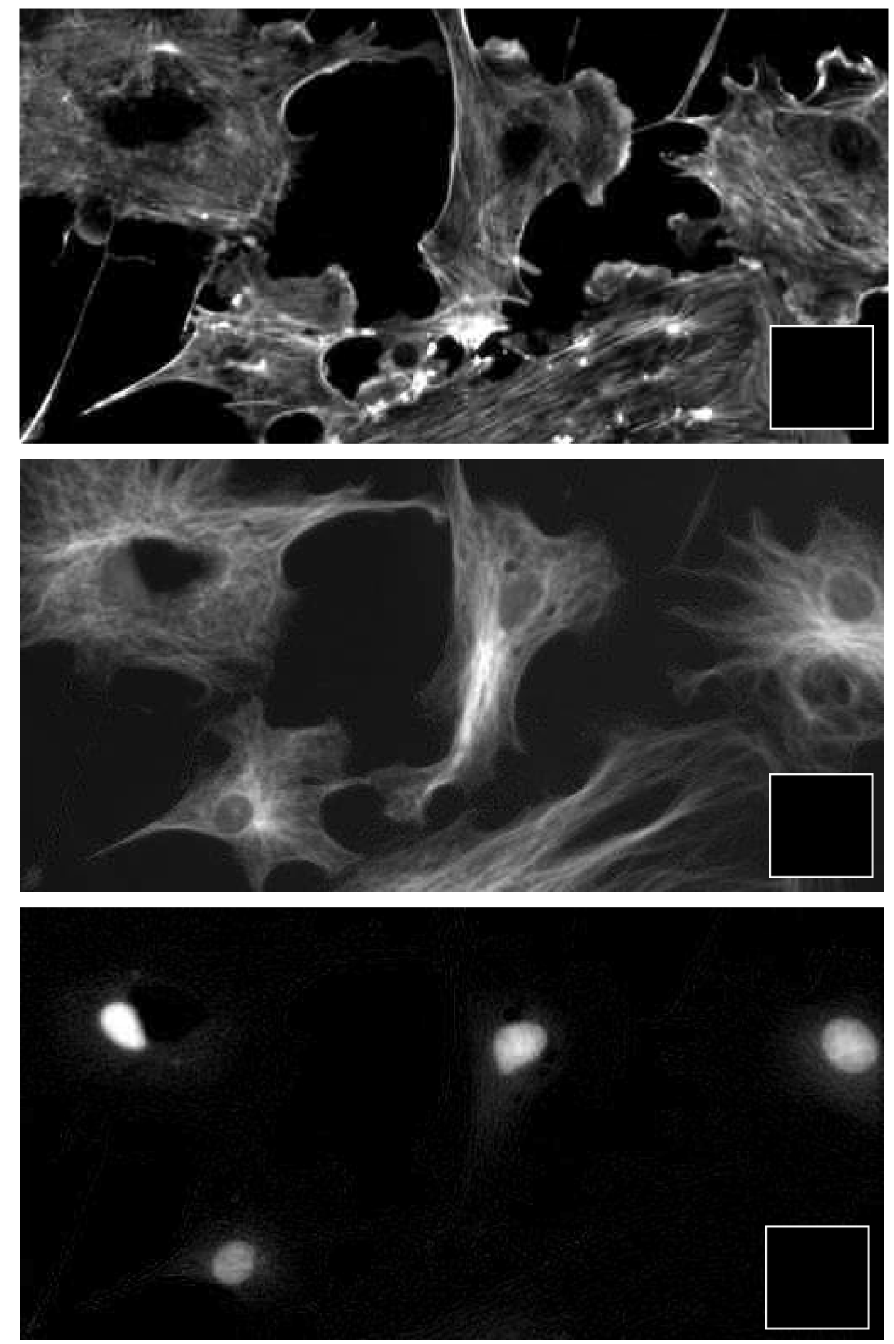
RGB Color Space



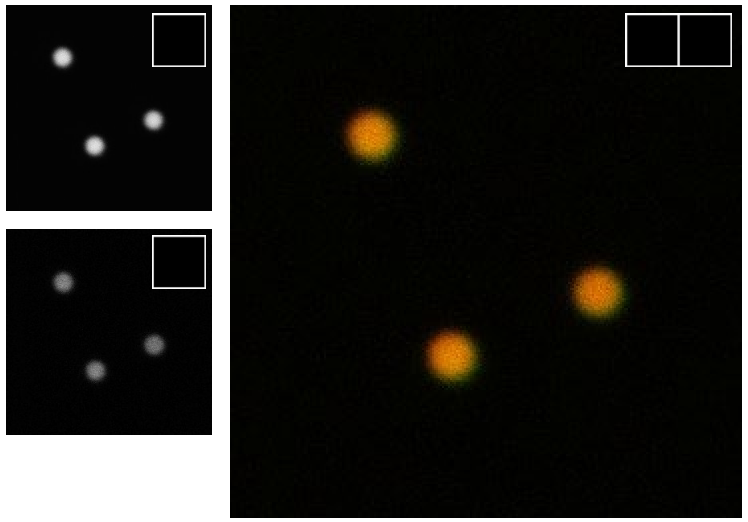
Why RGB? ... because we have red, green and blue sensitive photo receptors in our eyes!



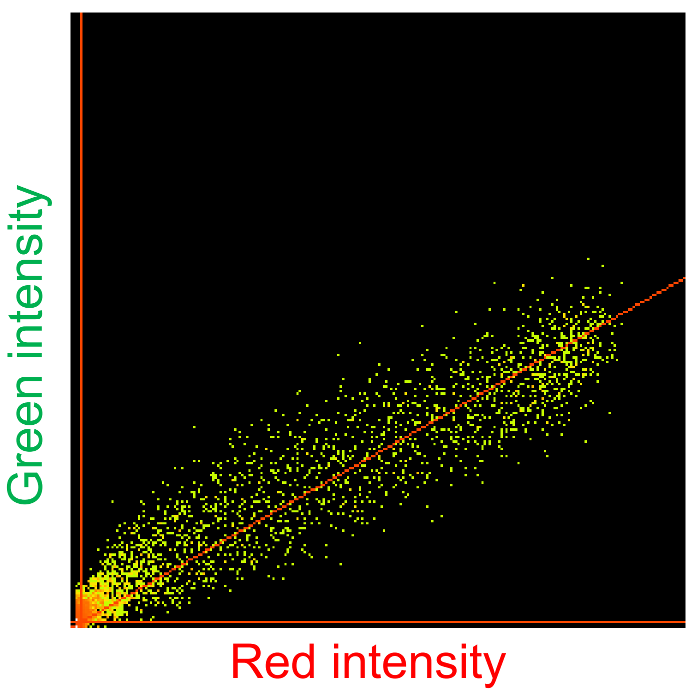
Each "colour" is really just single greyscale numbers!

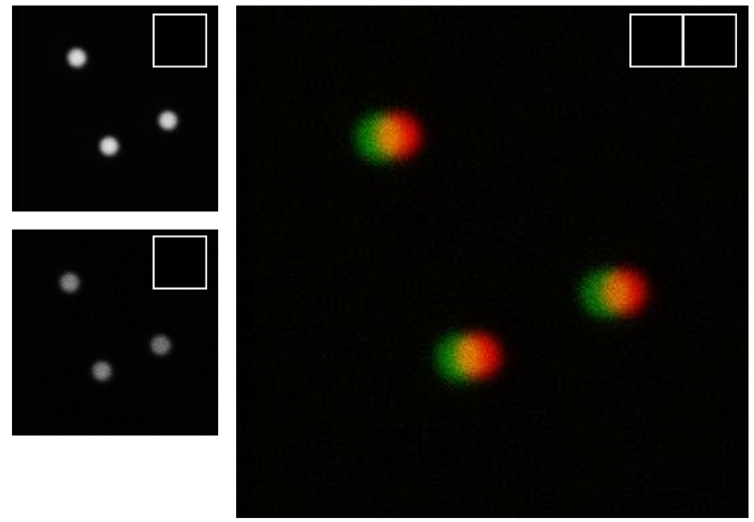


Scatterplot / 2D Histogram/ Colocalization Analysis

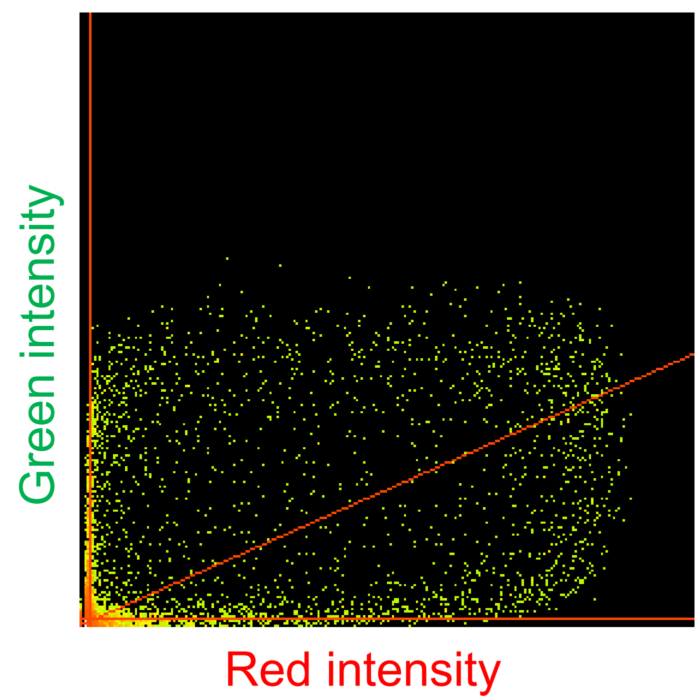


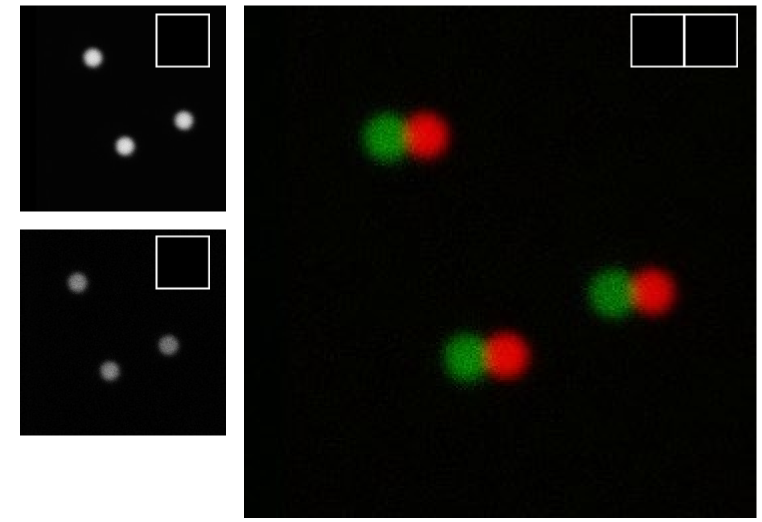
original R+G



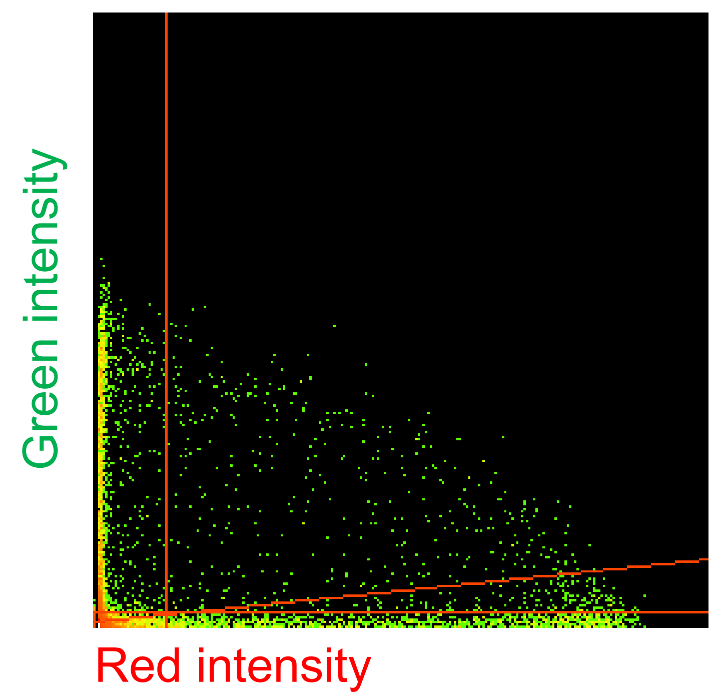


R shifted +10 pix





R shifted +20 pix



Find a way to visualise what you actually want to see:
Here, we don't care WHERE the beads are; We care if they are in the same place or not!

Pearson's Image Correlation Coefficient (Manders et al., 1993)

$$r = \frac{\sum_i (R_i - R_{av}) \cdot (G_i - G_{av})}{\sqrt{\sum_i (R_i - R_{av})^2 \cdot (G_i - G_{av})^2}}$$

Correlation between images, *r* ranges from -1 to +1

+1 means full correlation (images are the same)

0 means no correlation (random)

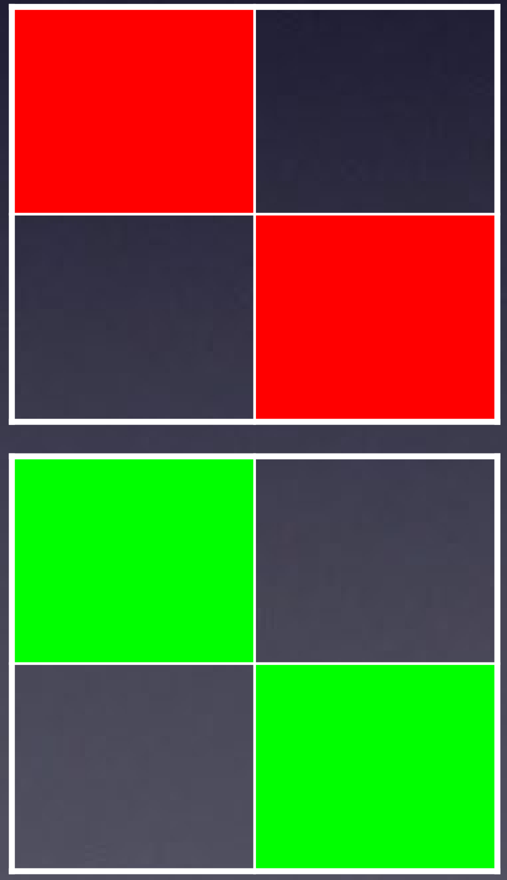
-1 means full anti correlation (no red where there is green)

Pearson's Image Correlation Coefficient

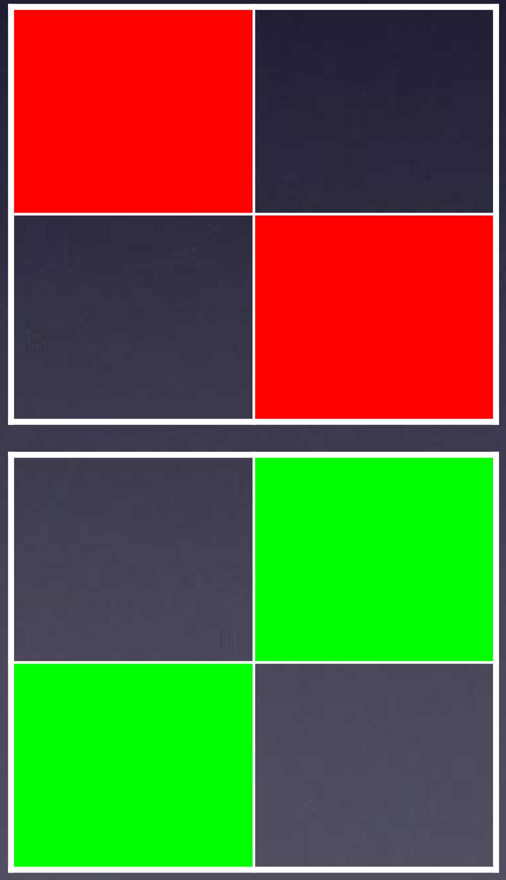
In English…per pixel and summed for the whole image:

$$r = \frac{\text{sum of (red intensity - average red) x (green intensity - average green)}}{\text{sqrt of squares of above}}$$

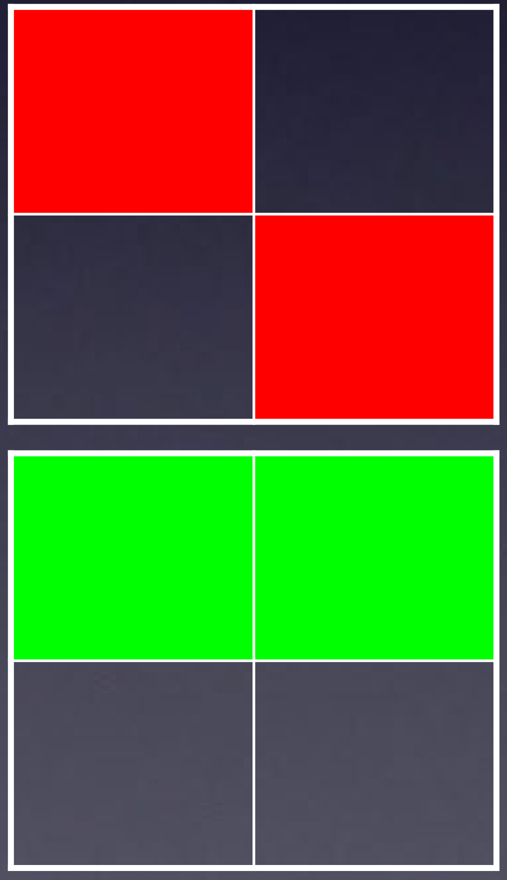
$r = +1$



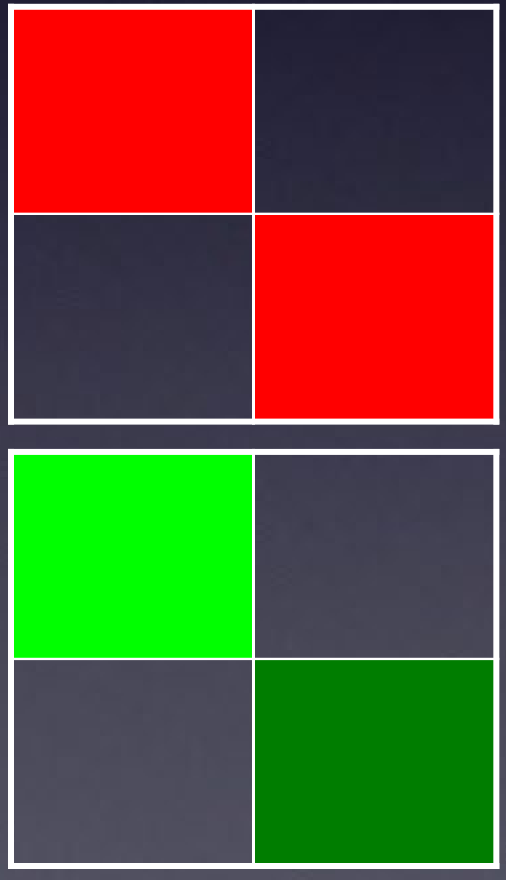
$r = -1$



$r = 0$



$r > 0$



Pearson's Image Correlation Coefficient is…

- Insensitive to diff. intensity of the 2 images.
- Insensitive to intensity offset.
- If red is 1/2 as bright as green…
  - Still can get $r = 1$
  - … so Pearsons r is is robust for biological imaging…

File - Open Samples - Neuron

RGB colour space:

✓ <u>Colour channels</u> – *Image-Colour-Channels Tool, Split channels etc.*

✓ <u>LookUp Tables/Paletts</u>: *Image - Lookup tables, or LUT toolbar icon*

✓ <u>Histogramm</u>: Analyze-Histogram

✓<u>Intensity Scale</u>: *Analyze – Tools - Calibration bar*

Working with multi dimensional images

More than one frame, time point or channel
Ortho view, montage and other ways of looking at >2D image

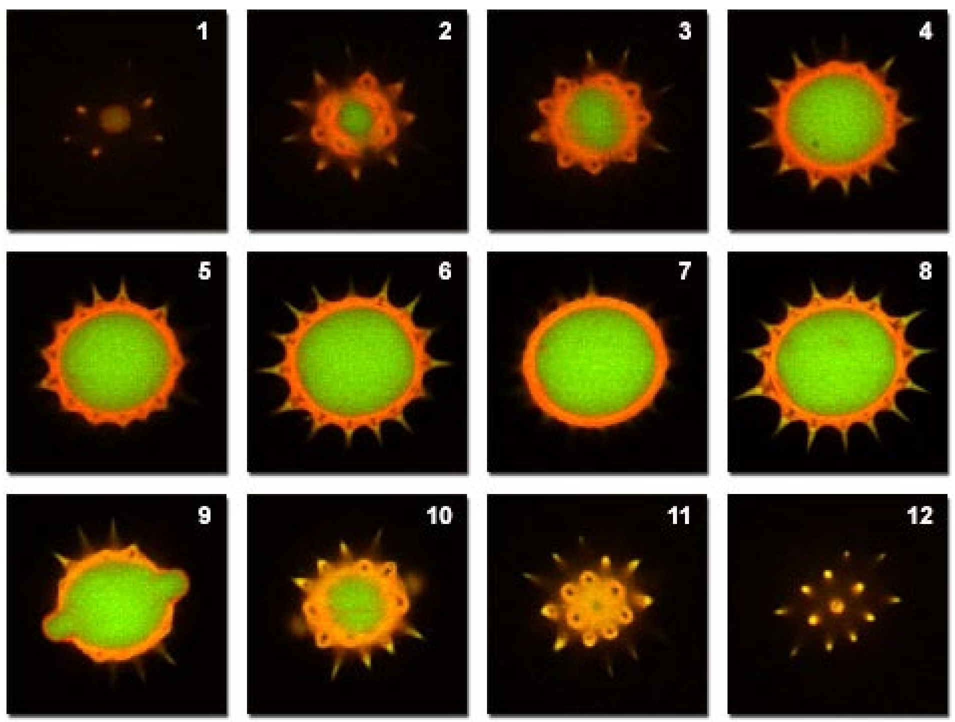
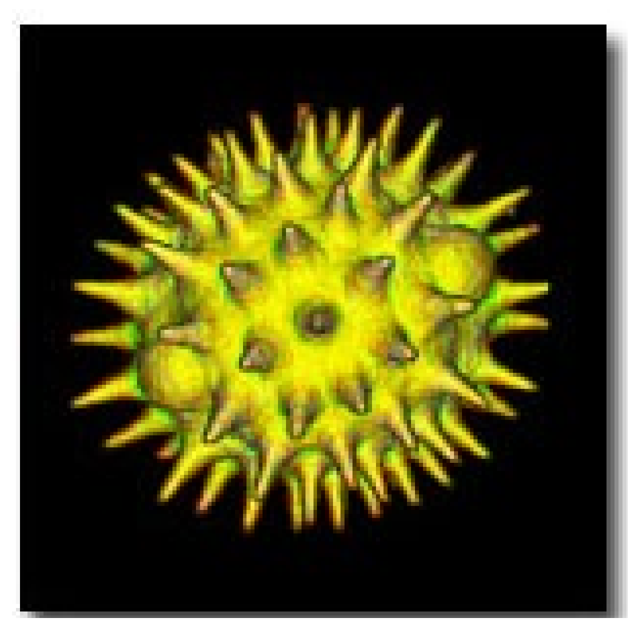


Figure 6

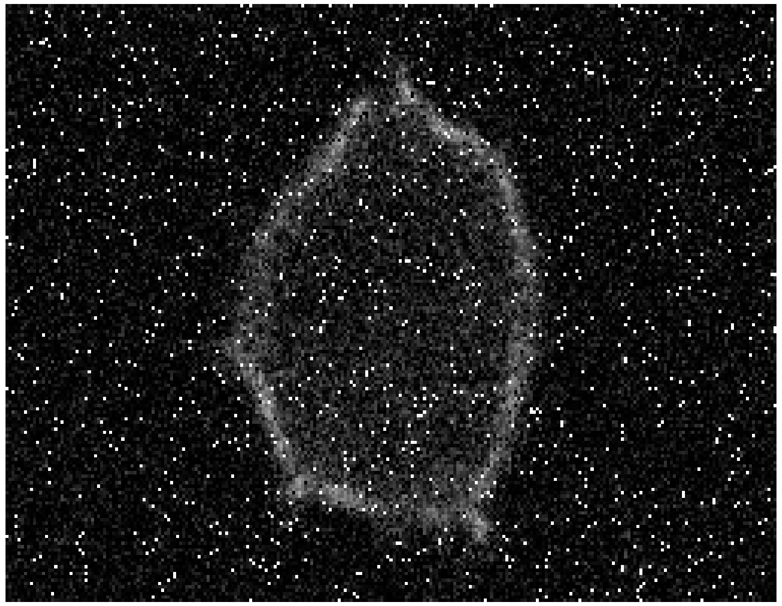
3D volume

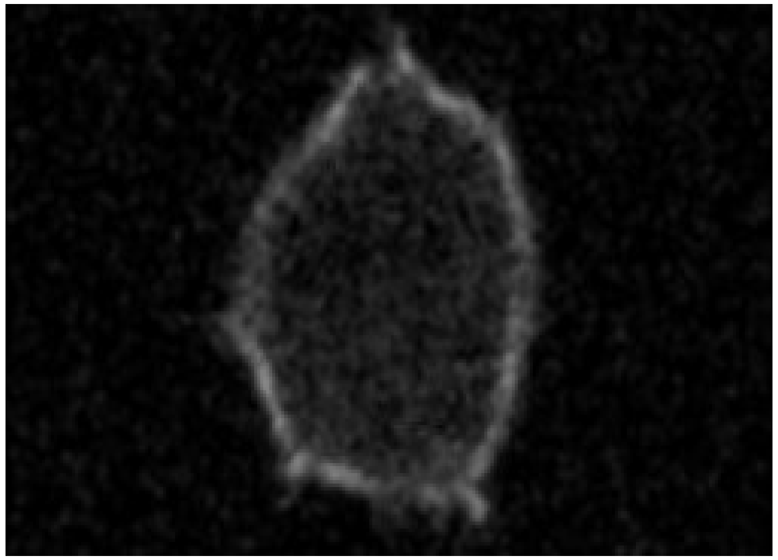


File - Open Samples - confocal-series.tif

Make a montage: Stk→ Make montage

Image processing masks and filters





Introduction

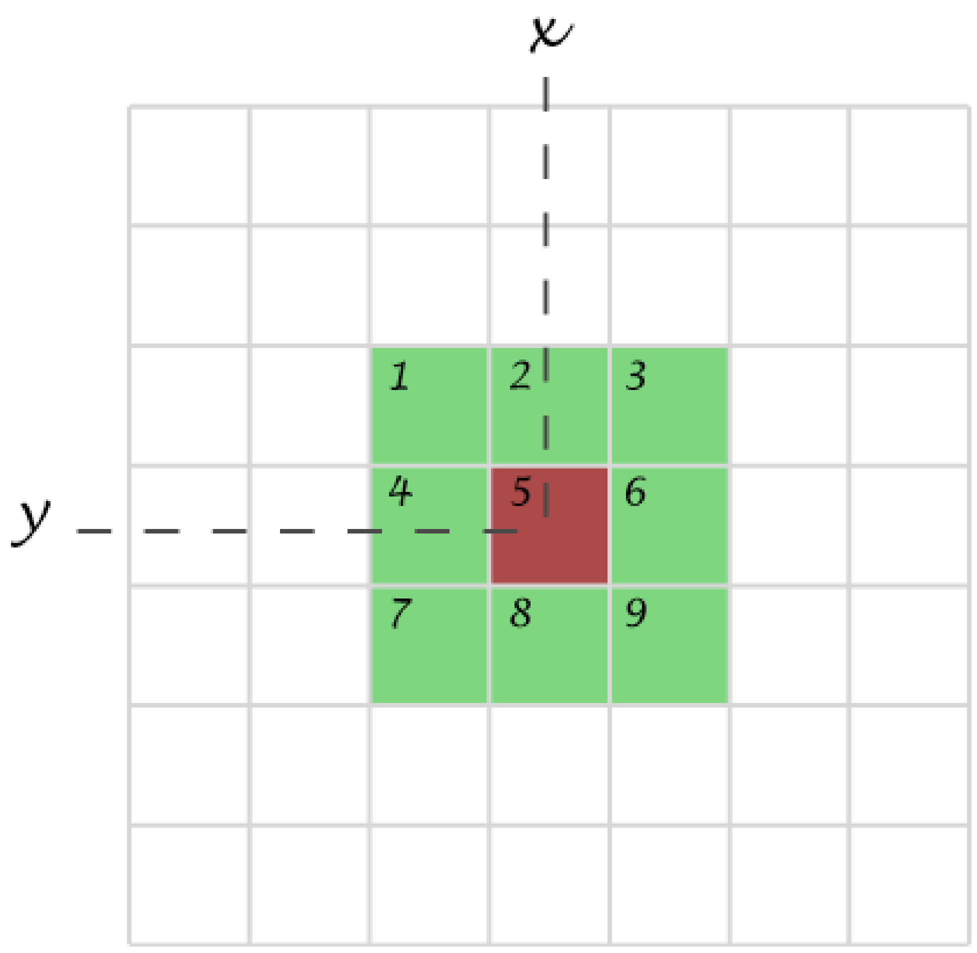
- Neighbourhood
- Operation on neighbourhood

Spatial filters

- Mean and Median filter
- Edge detection

A. Introduction

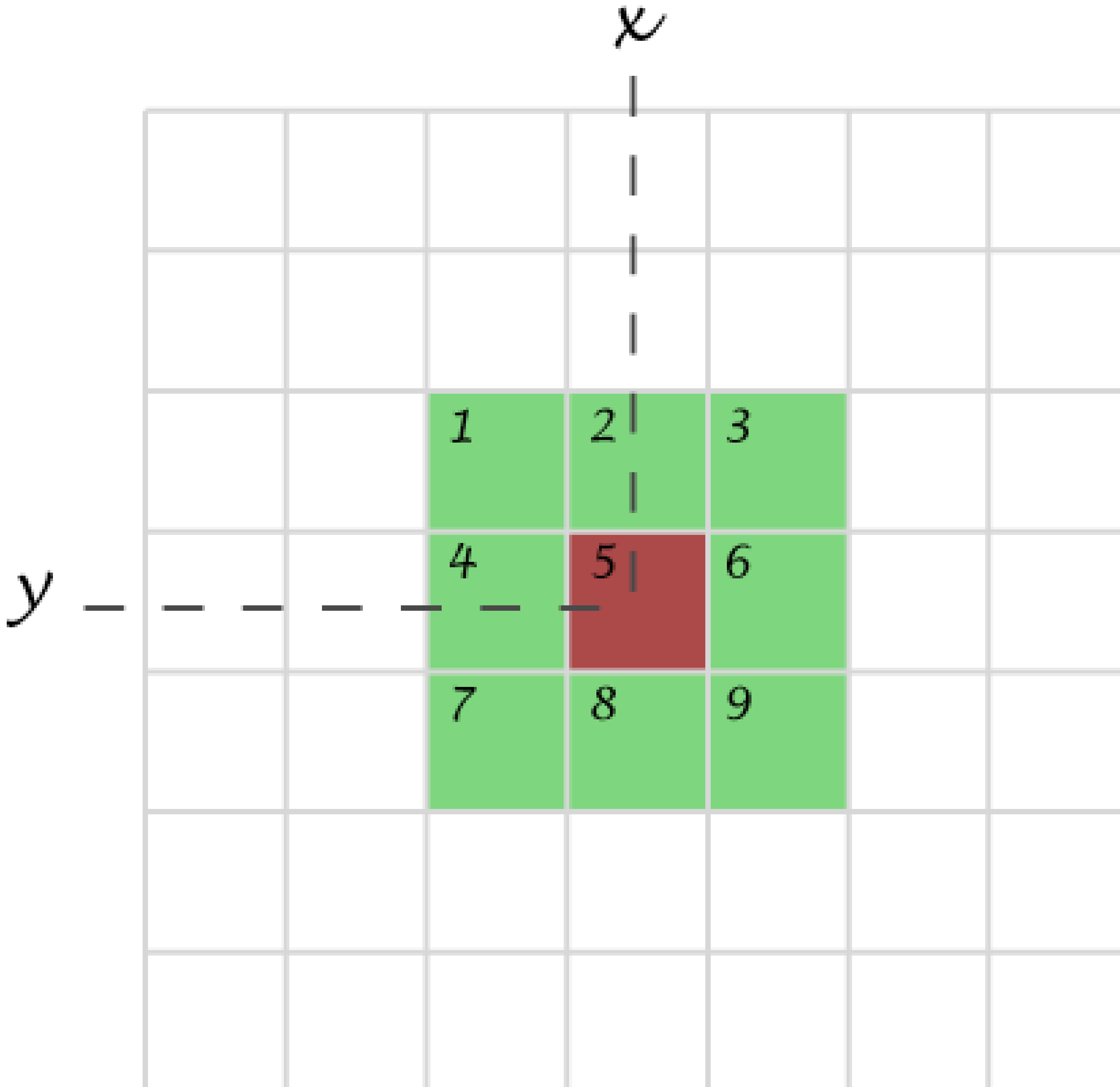
"Transformation or set of transformations where a new image is obtained by neighbourhood operations."



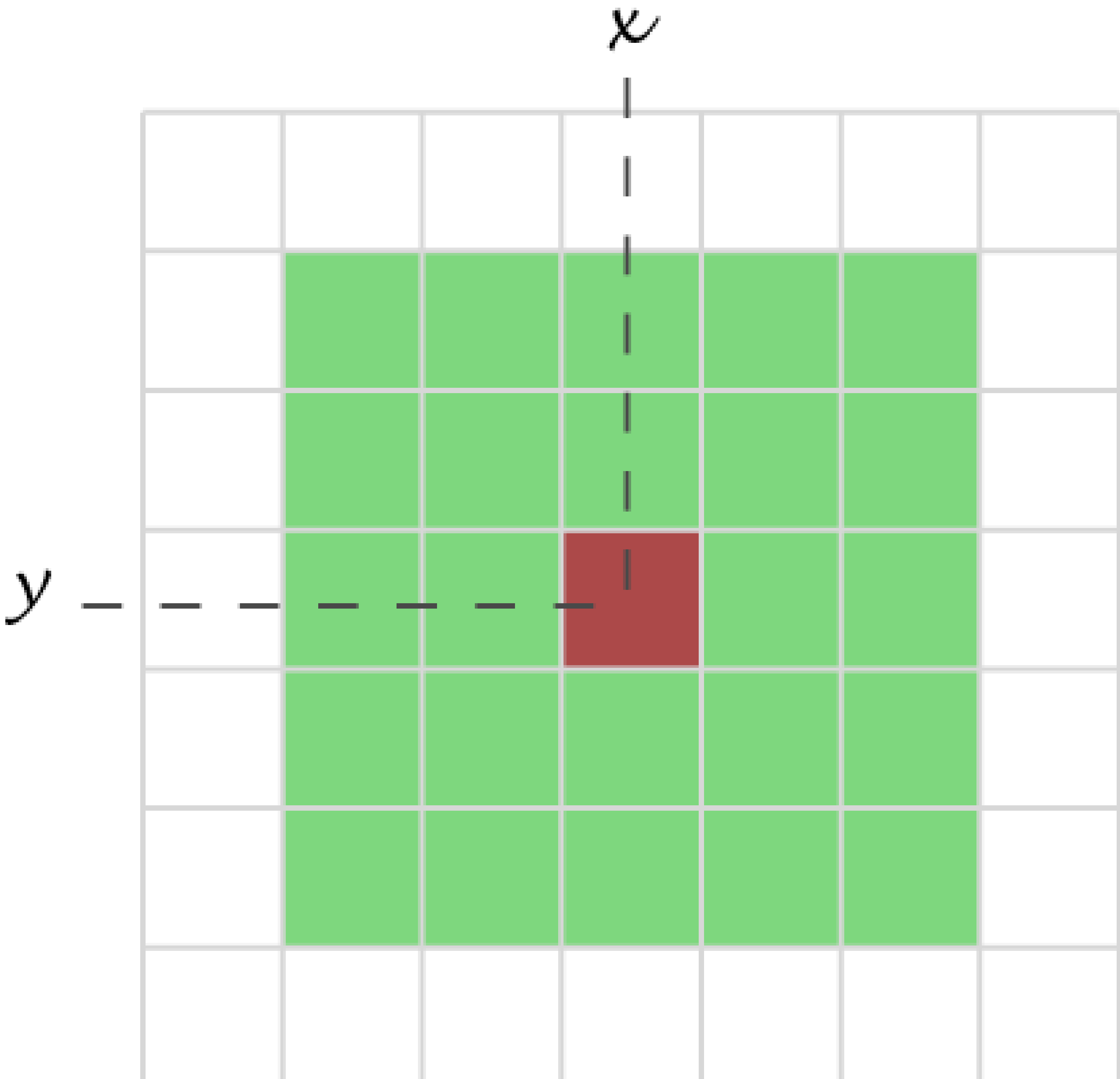
The Intensity of a pixel in the new image depends on the intensity values of "neighbour pixels"

Neighbourhood (or kernel): pixels that matter

3 x 3

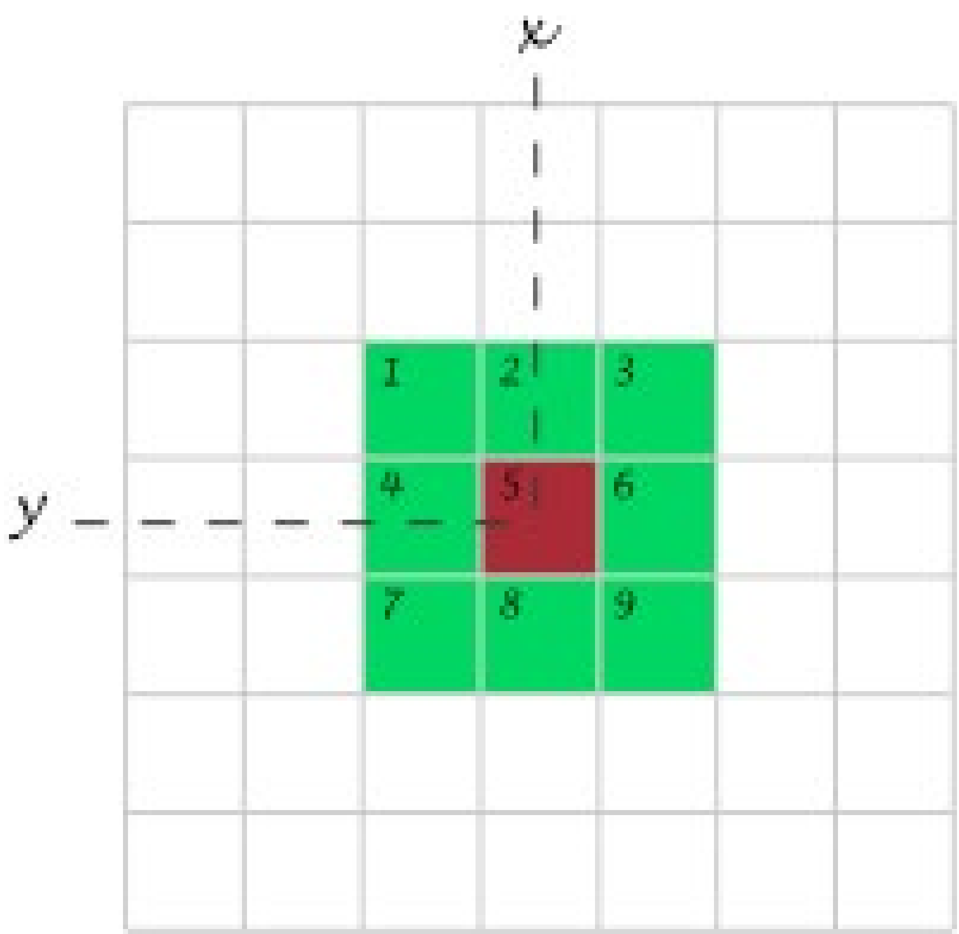


5 x 5



B: Filtering - the mean filter

Simplest filter: The value of a pixel is replaced by the intensity mean of the neighbourhood pixels.



3x3 example:

$$a_5^* = \frac{1}{9}(a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7 + a_8 + a_9)$$

The mean filter

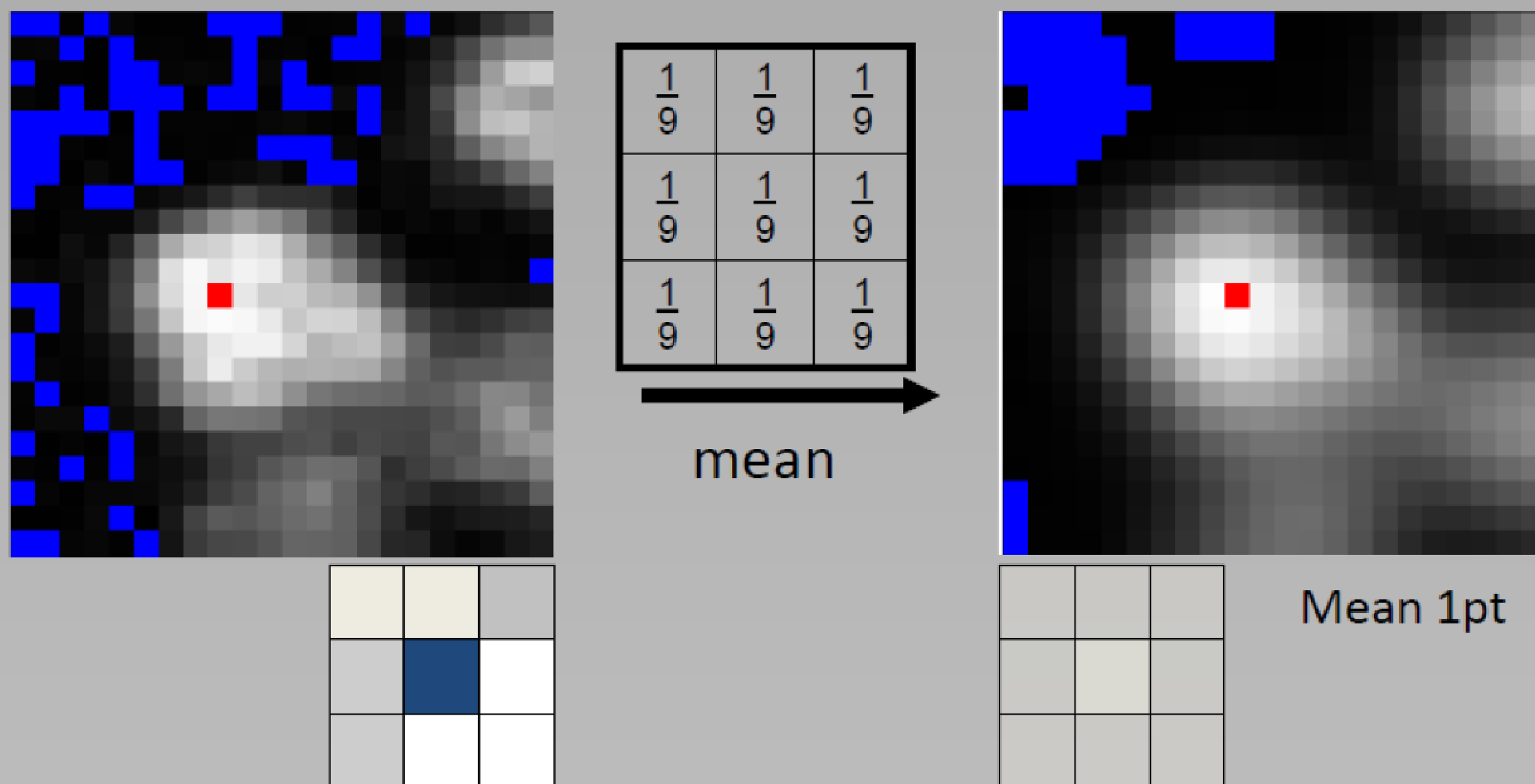
The mean filter is a linear filter!

$\alpha_{1,1}$	$\alpha_{1,2}$	$\alpha_{1,3}$
$\alpha_{2,1}$	$\alpha_{2,2}$	$\alpha_{2,3}$
$\alpha_{3,1}$	$\alpha_{3,2}$	$\alpha_{3,3}$

"The new pixel value depends on a linear combination of neighbourhood pixel values"

(The order of several linear filters in sequence does not matter)

Noise Reduction: Mean



The mean filter

Main property: low-pass filter (smooths small objects)
• kernel size influence
• number of successive applications

+ simplest filter – fast

+ it's a linear filter

+ averages noise, does not eliminate it

+ works against Gaussian and Poisson noise

- blurs images - small details are lost (low pass filter)

- smoothes edges dramatically

- fails for salt & pepper noise

The median filter

The value of a pixel is replaced by the median of the pixel intensity in neighbour pixels

Take neighbourhood (e.g. 3x3)

5	112	86
235	88	211
137	233	108

Sort it

5
86
88
108
112
137
211
233
235

Take median

112

The median filter

noise elimination

Original:

5	9	6	6	9	5	9
9	5	9	7	8	7	9
8	9	8	6	7	9	9
9	9	7	200	9	6	9
6	5	8	6	9	6	7
9	7	9	9	8	6	7
7	9	5	6	7	6	6

outlier

Median filtered:

0	5	6	6	6	7	0
5	8	7	7	7	9	7
8	9	8	8	7	9	7
6	8	8	8	7	9	6
6	8	8	9	8	7	6
6	7	7	8	6	7	6
0	7	6	6	6	6	0

The outlier value has been completely removed from the dataset

The median filter

+ Typically good for “Salt & pepper” noise removal

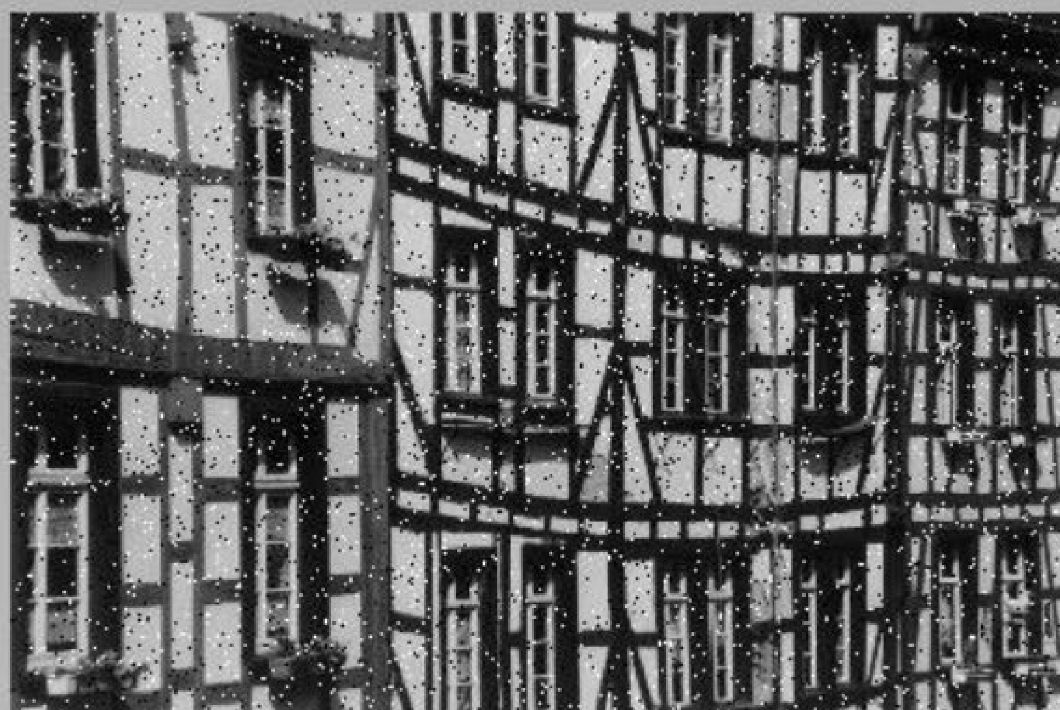
+ Eliminates noise

+ Edge-preserving

-Slower than mean (not such a problem anymore... computers are fast)

- NOT linear

Noise Reduction: Median, Mean



Median, 1pt

Mean, 1pt

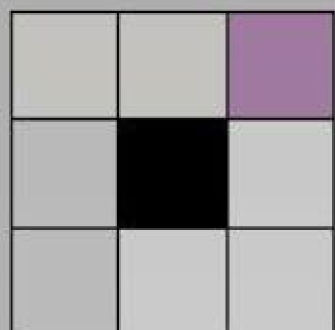
Sharpen / Blur

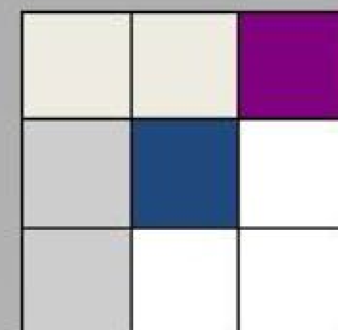
$$\begin{matrix} -1 & -1 & -1 \\ -1 & 9 & -1 \\ -1 & -1 & -1 \end{matrix}$$

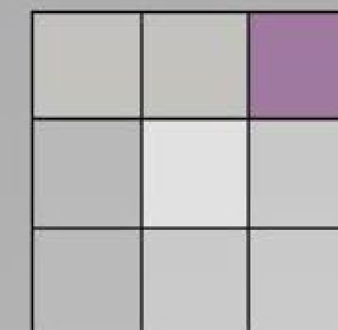
sharpen

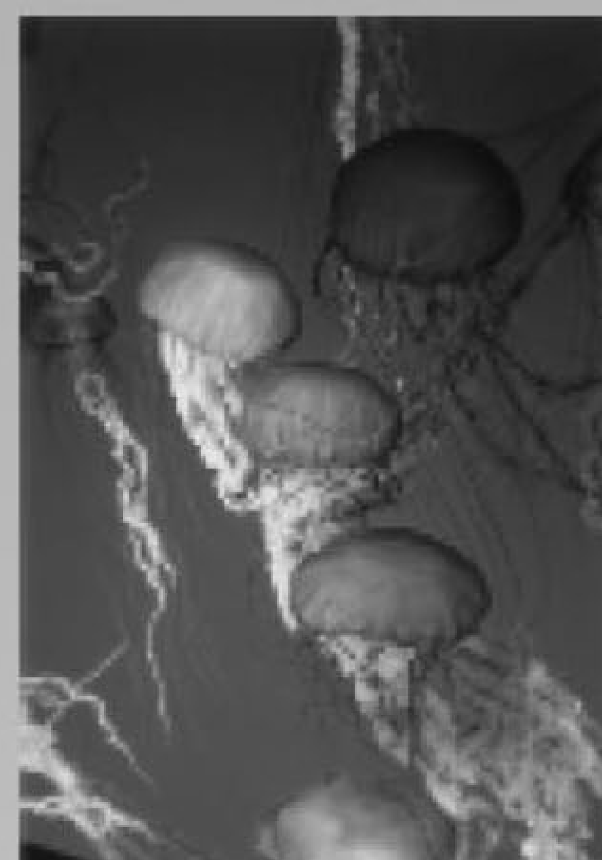
$$\begin{matrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{matrix}$$

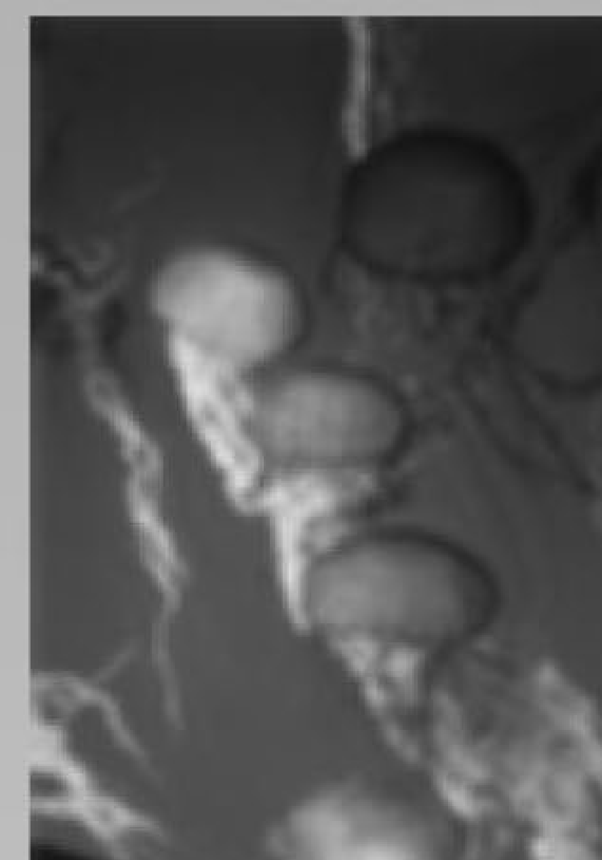
blurring











Simple Image Filtering

(1) File - Open Samples – Neuron

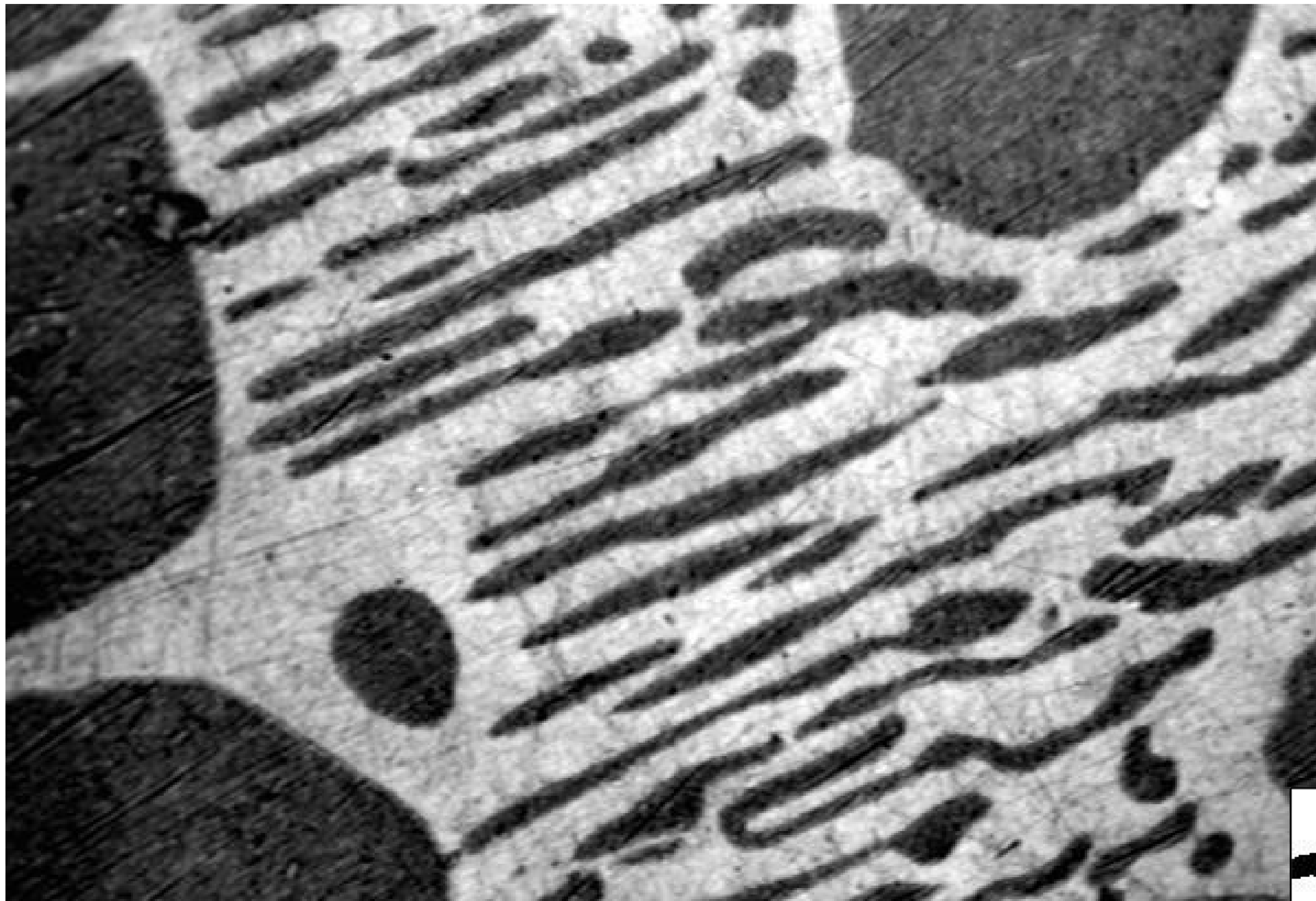
(1) Convolve a simple binary image

✓ Process – Filters – Convolve (play with different kernels)

(2) Noisy sample image

✓ Mean and Median Filter (change pixel number, kernel size)

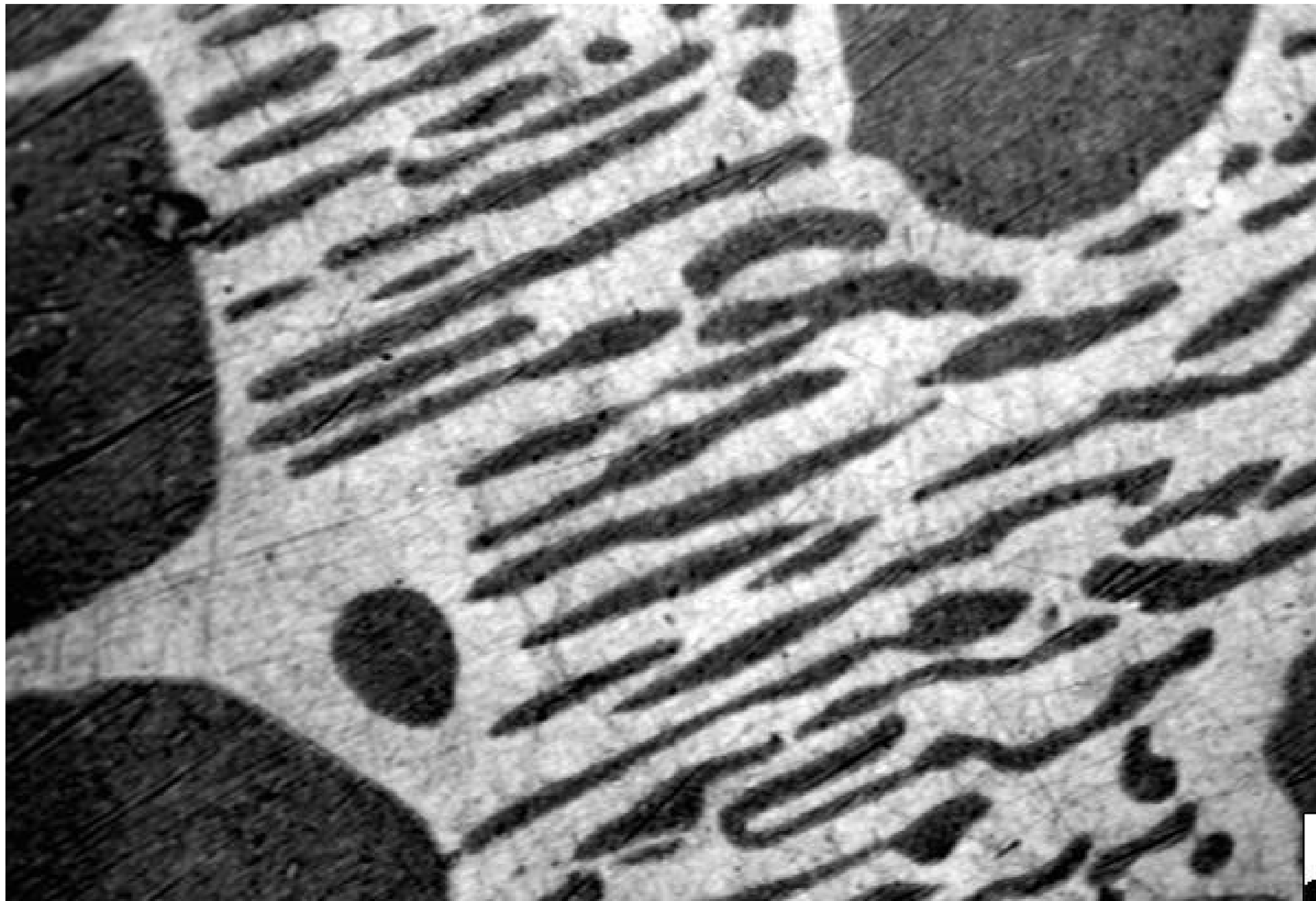
What is “Image Segmentation”?



“Greyscale” image

Foreground background

What is "Image Segmentation"?



"Scalar Intensity" image

"Binary" image

What is "Image Segmentation"?

1	65	13	55	2
2	3	34	2	1
4	0	31	1	2
1	33	3	54	3
56	3	2	1	34

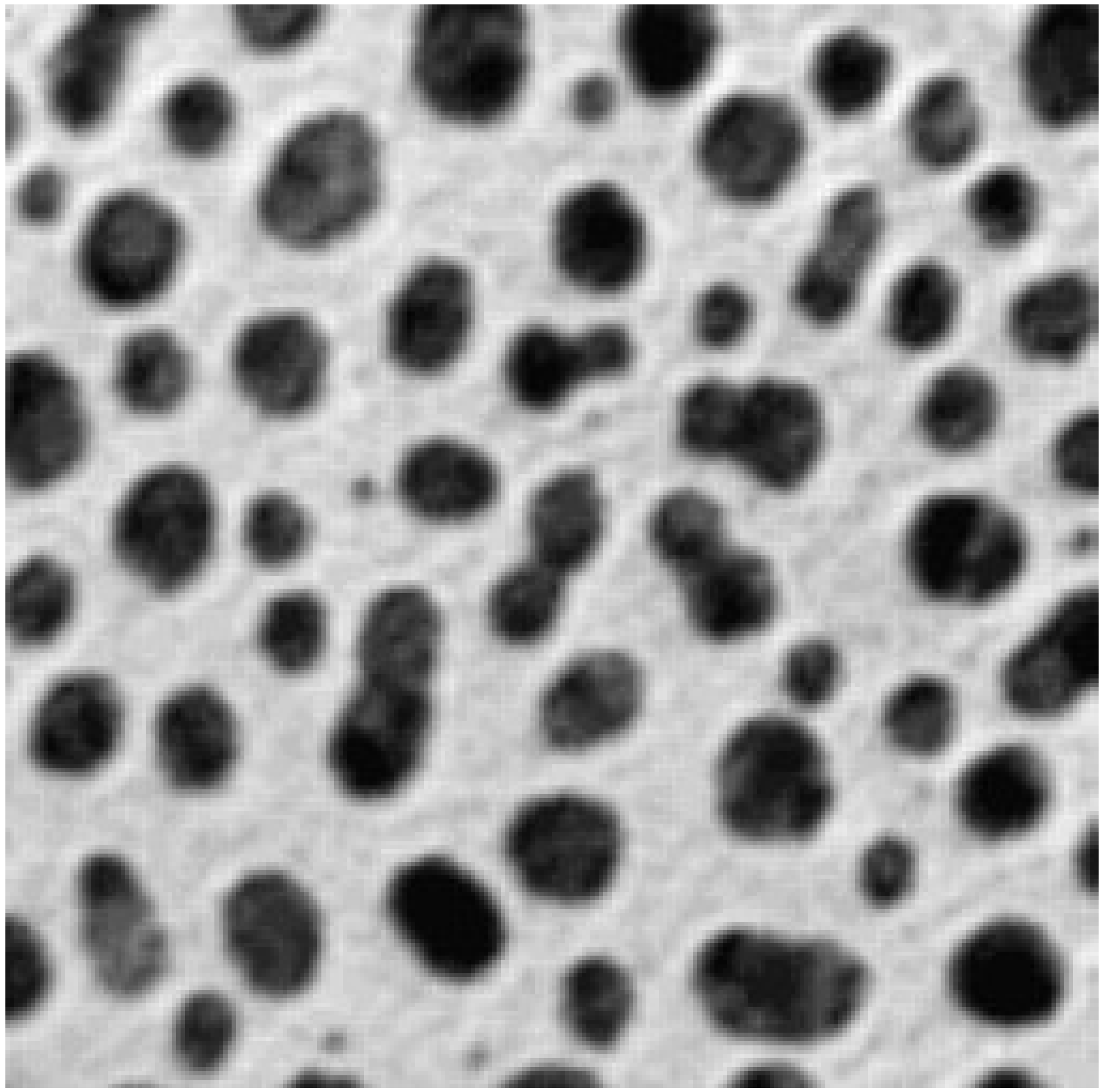
"Scalar Intensity" image

→

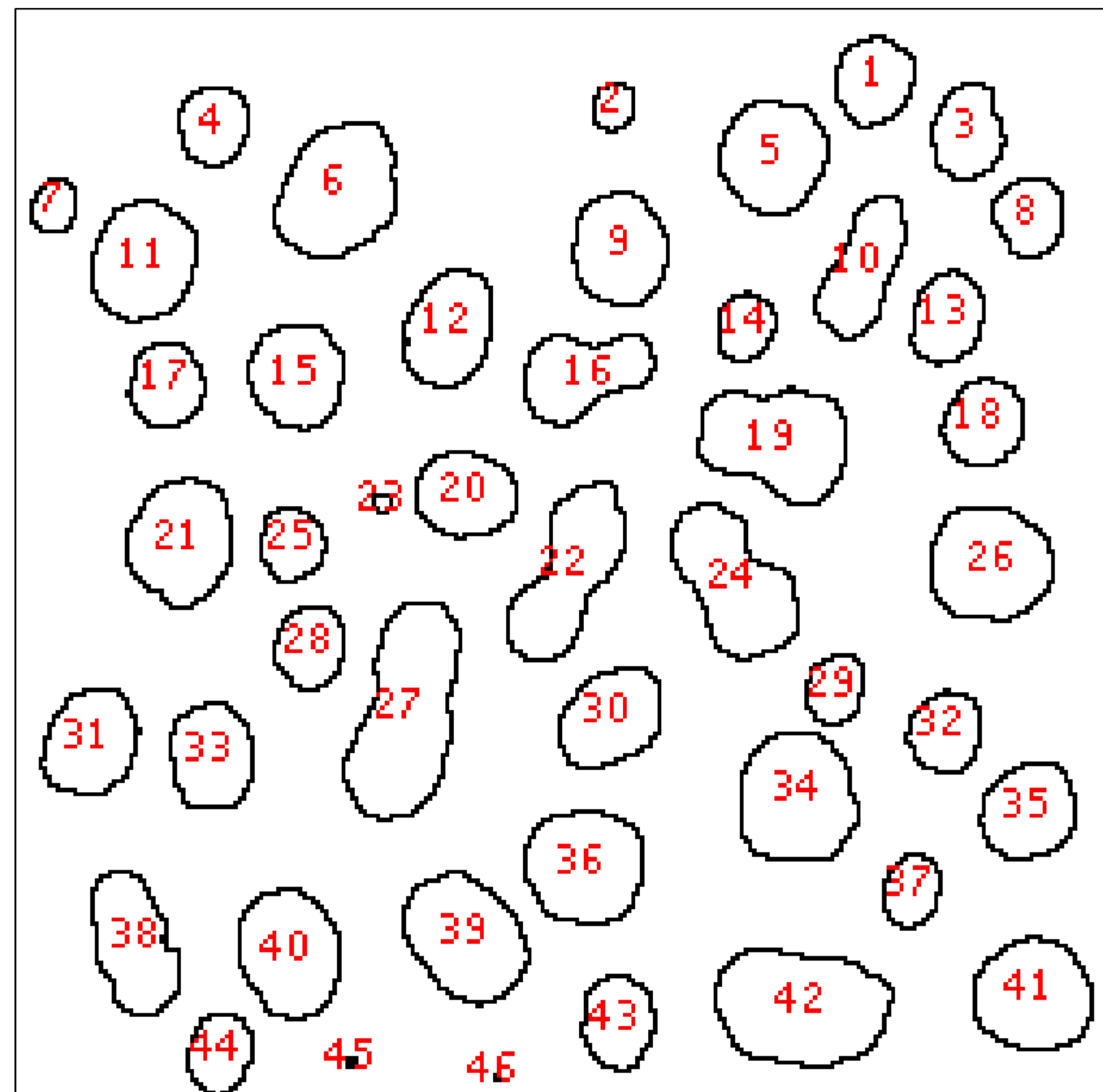
0	1	1	1	0
0	0	1	0	0
0	0	1	0	0
0	1	0	1	0
1	0	0	0	1

"Binary" image

What is “Image Segmentation”?

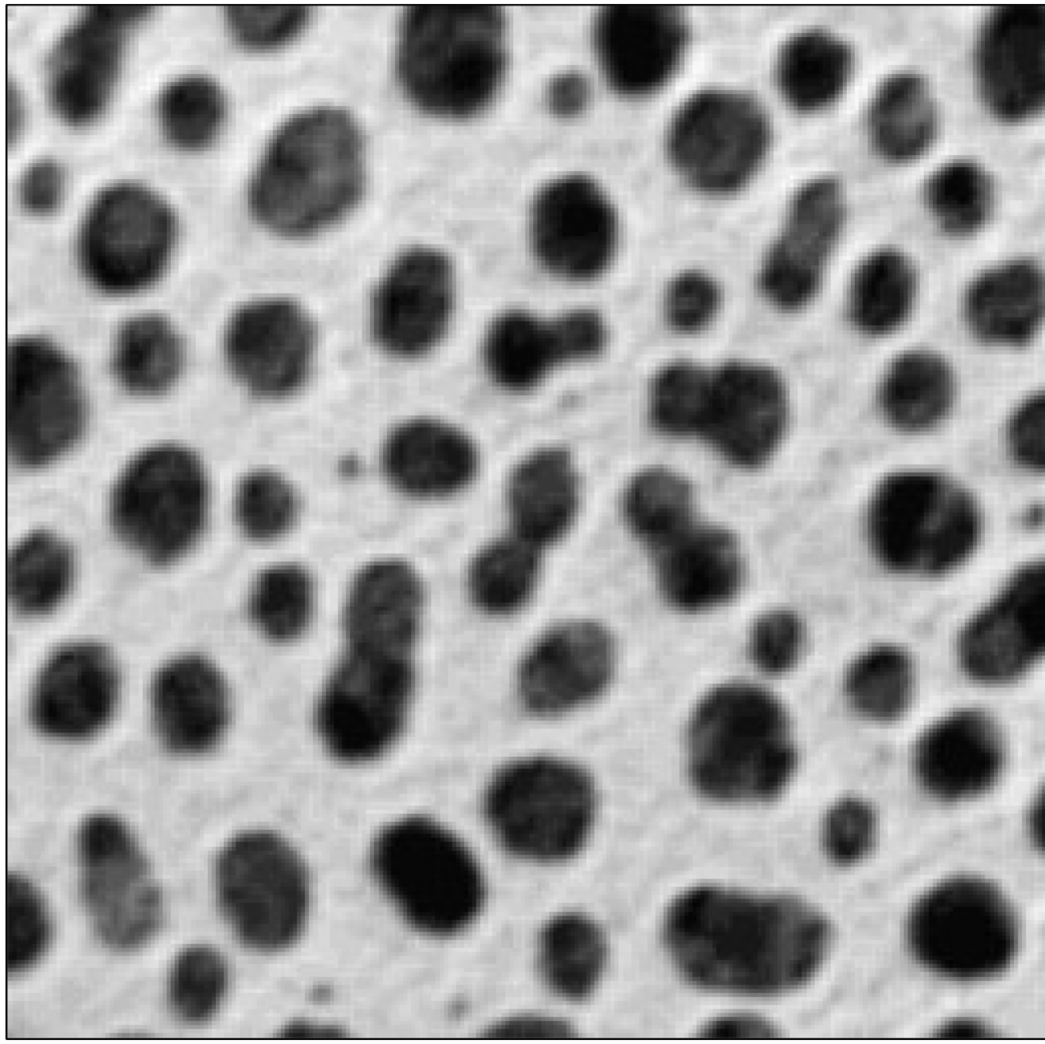


“Scalar Intensity” image

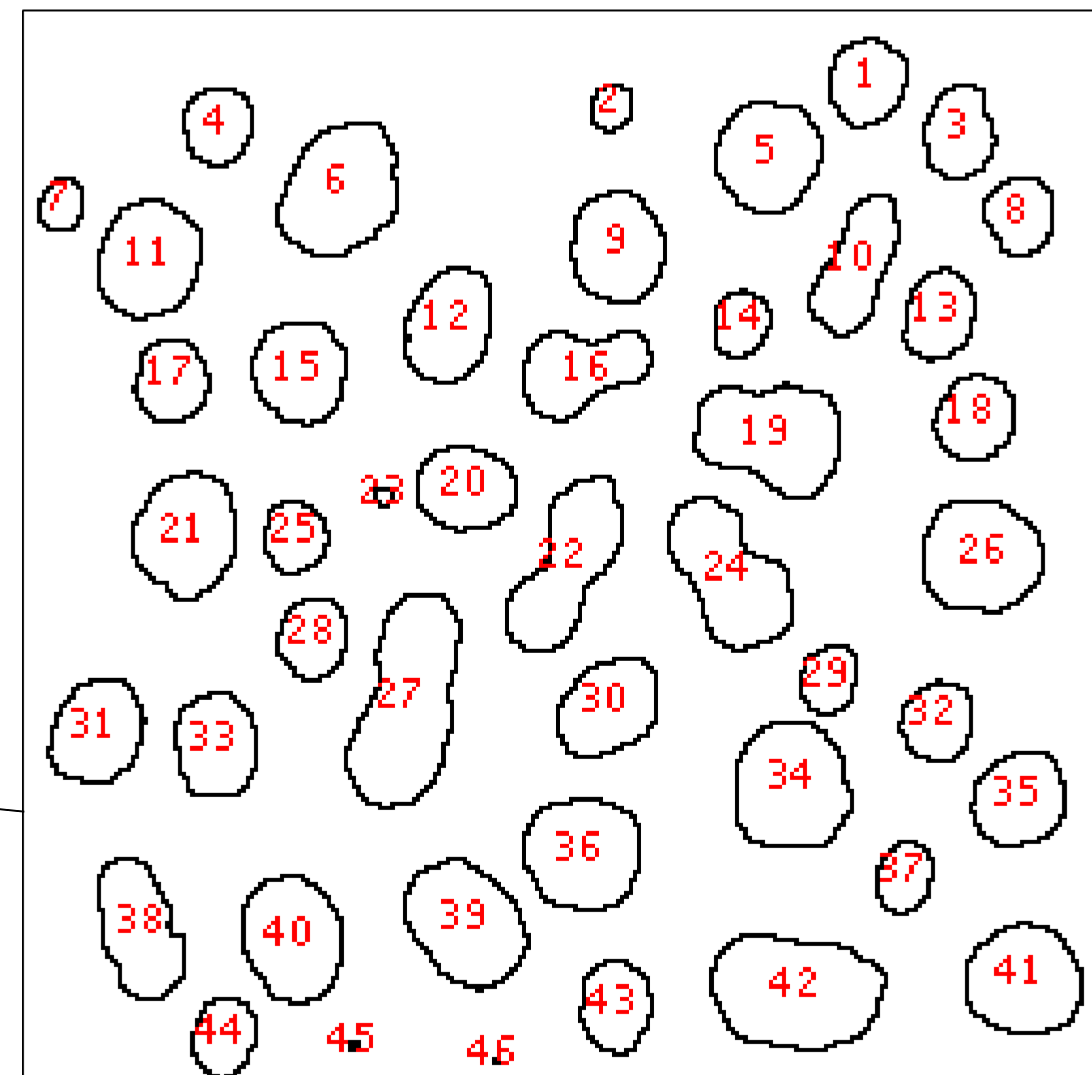


Labeled objects

What is "Image Segmentation"?

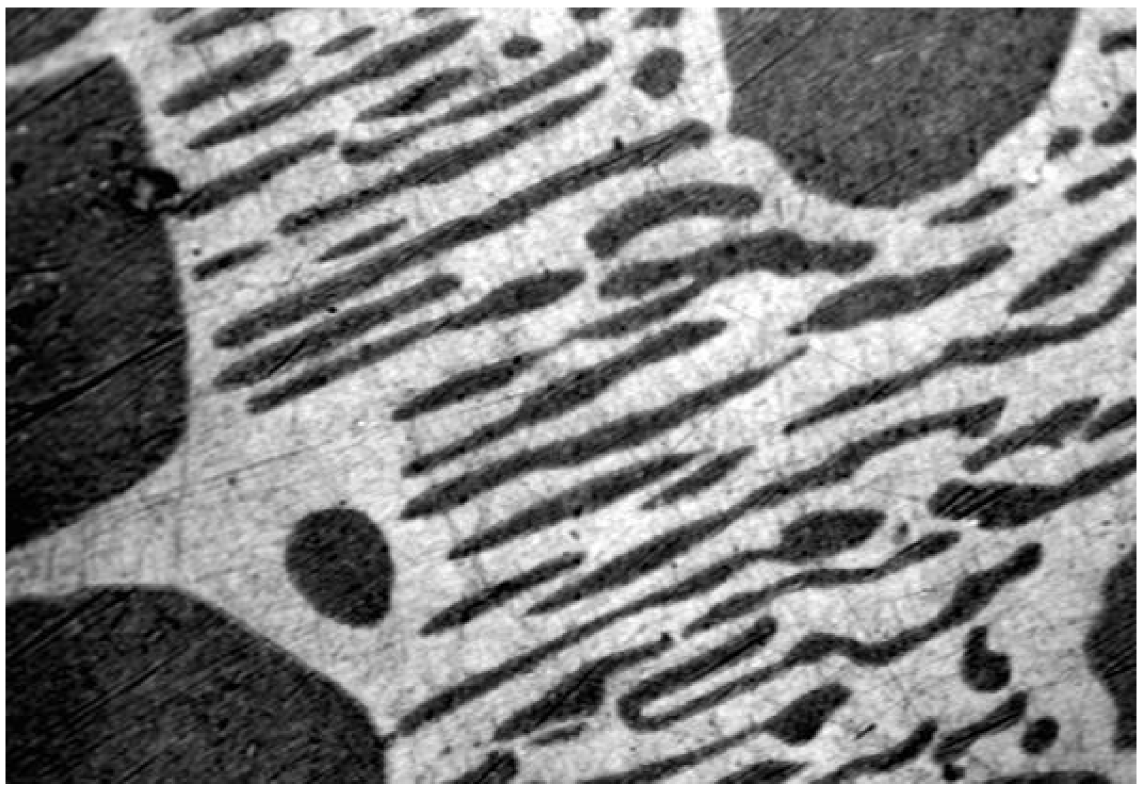


High Information Content
65536 pixels, 0-255 value



Lower Information Content, but easier to interpret biological meaning…
45 "objects" with properties: size, shape, intensity etc.

"Thresholding" (Intensity Histogram Split)



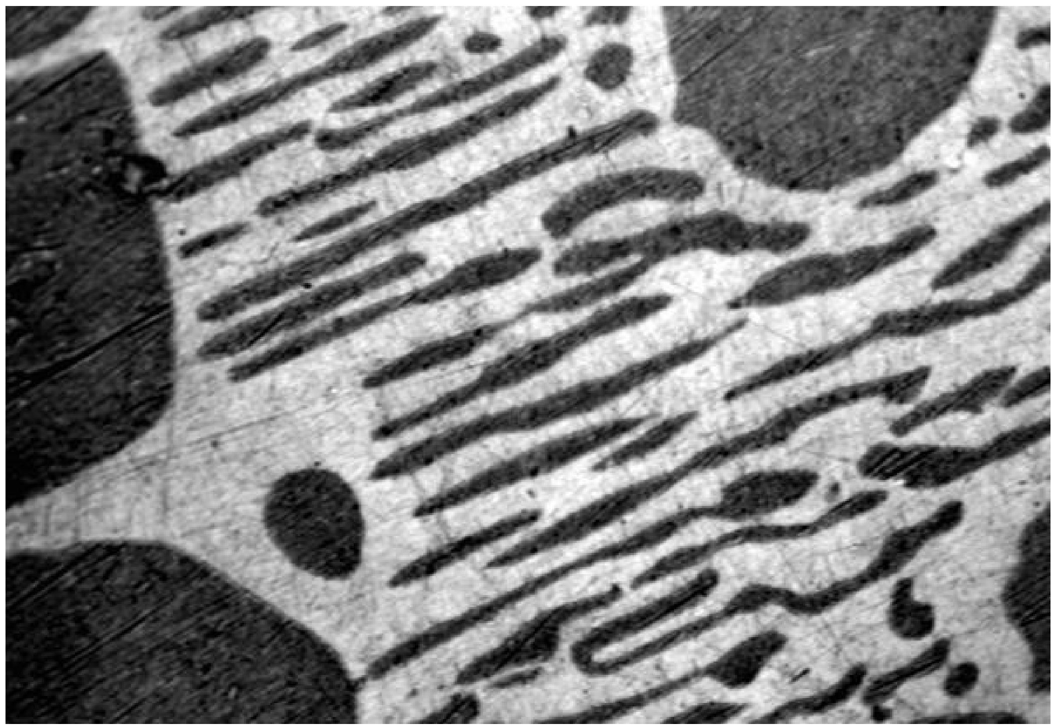
Clear difference between foreground and background?

Image not very noisy?

Choose an intermediate grey value = "threshold"

Determines foreground and background.

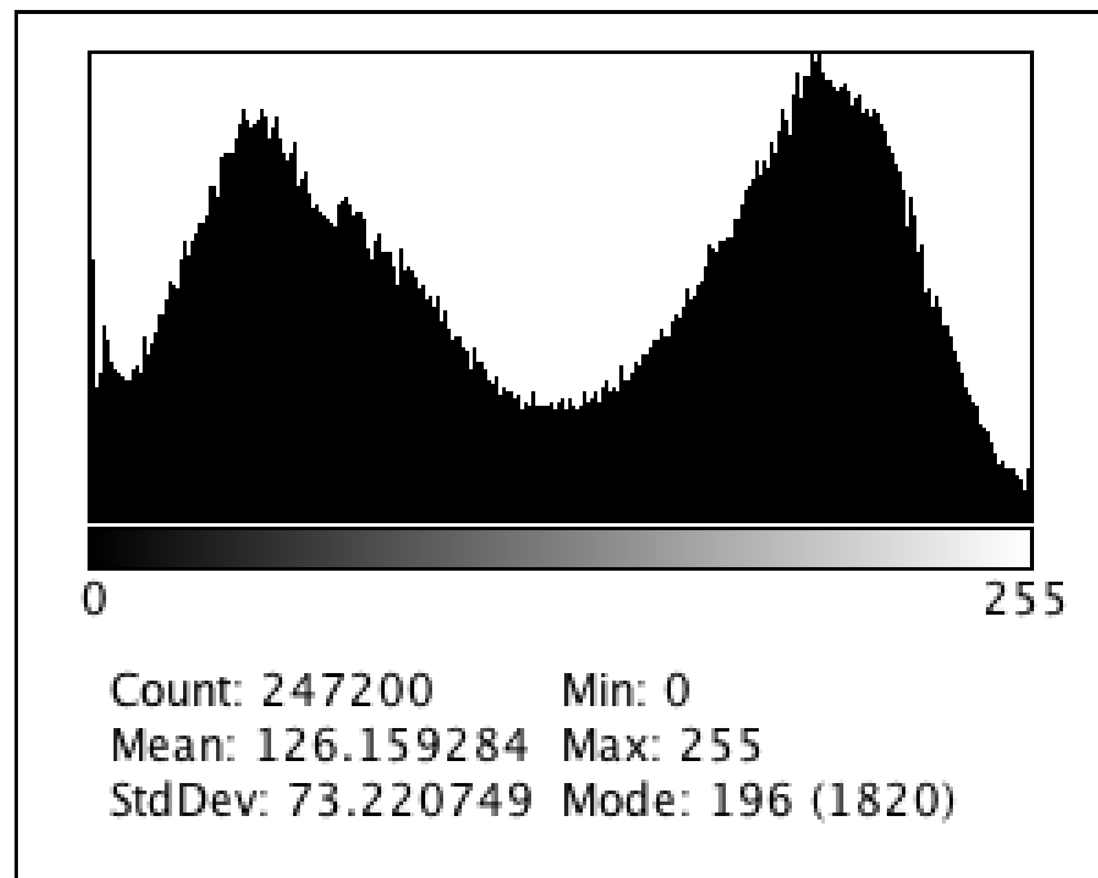
"Thresholding" (Intensity Histogram Split)



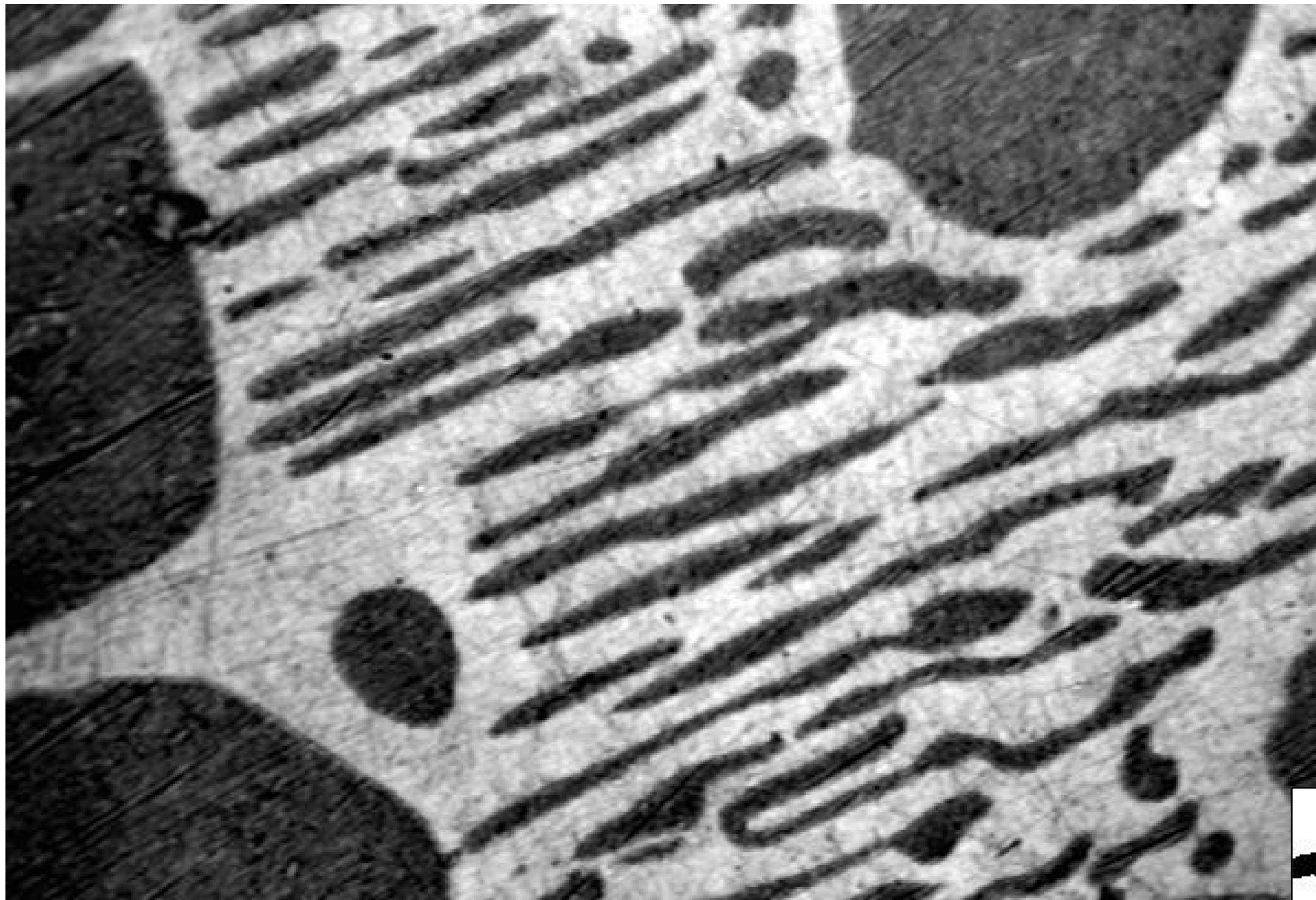
How to choose the grey level for thresholding?

Look at pixel intensity histogram of whole image…

Is there an obvious place?



“Thresholding” (Intensity Histogram Split)



“Greyscale” image

Create foreground background using: auto threshold → otsu

Practical Session 2c

Simple Image Segmentation

(1) File - Open Samples – Blobs (inverse)

(1) Thesholds

✓ Image - Adjust - Auto Threshold and Auto Local Threshold

✓ Many more methods, and “local” method

Deconvolution

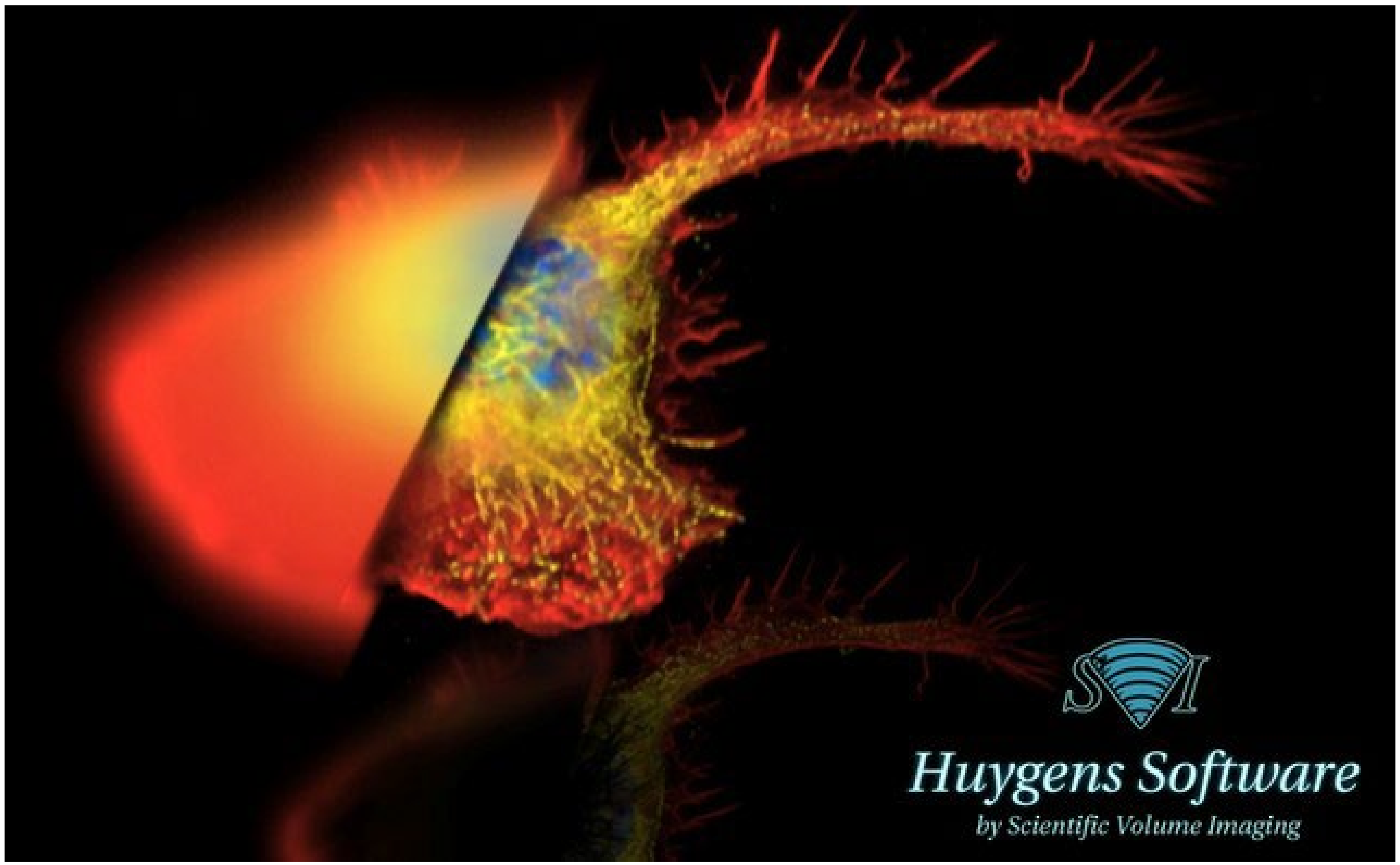
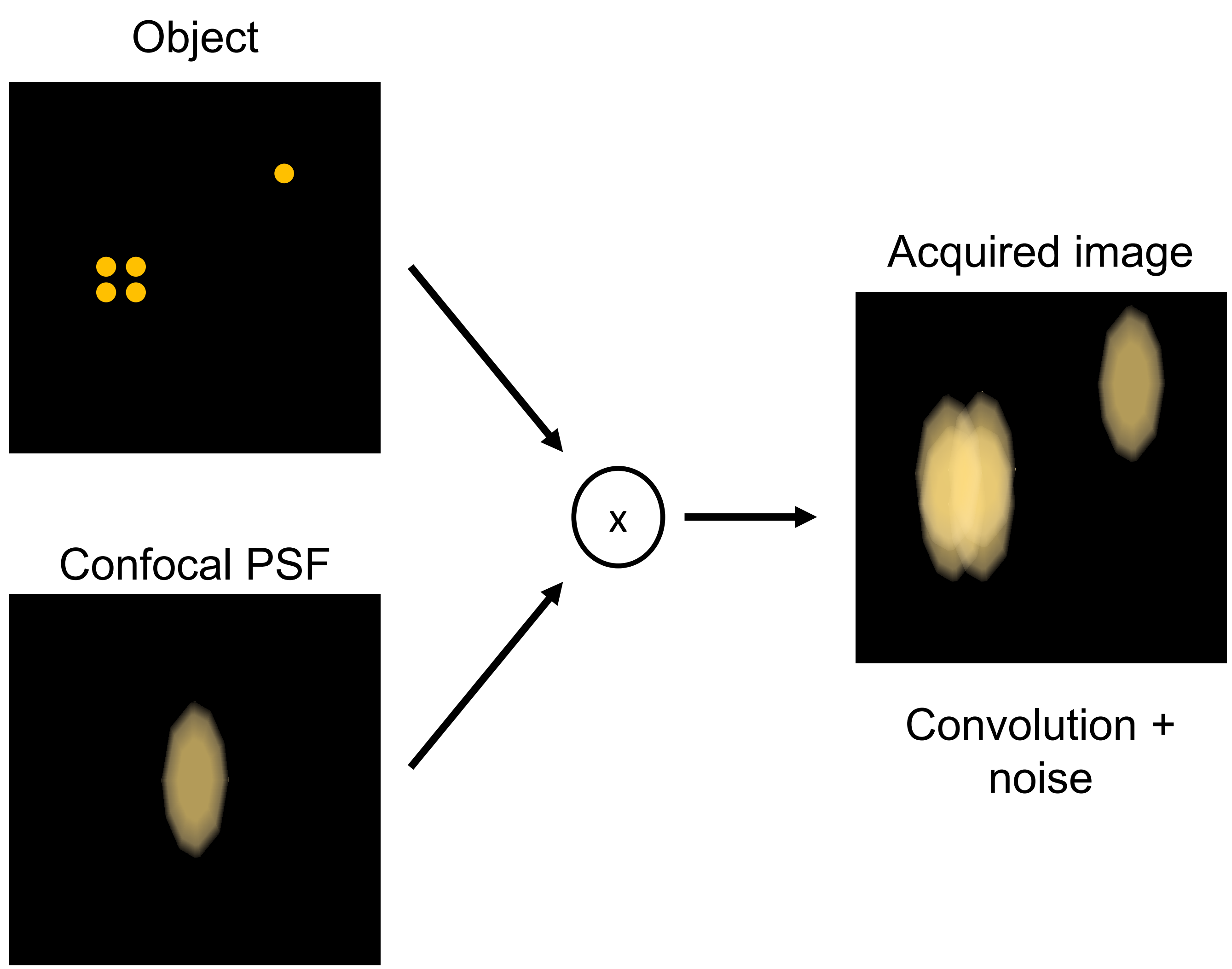
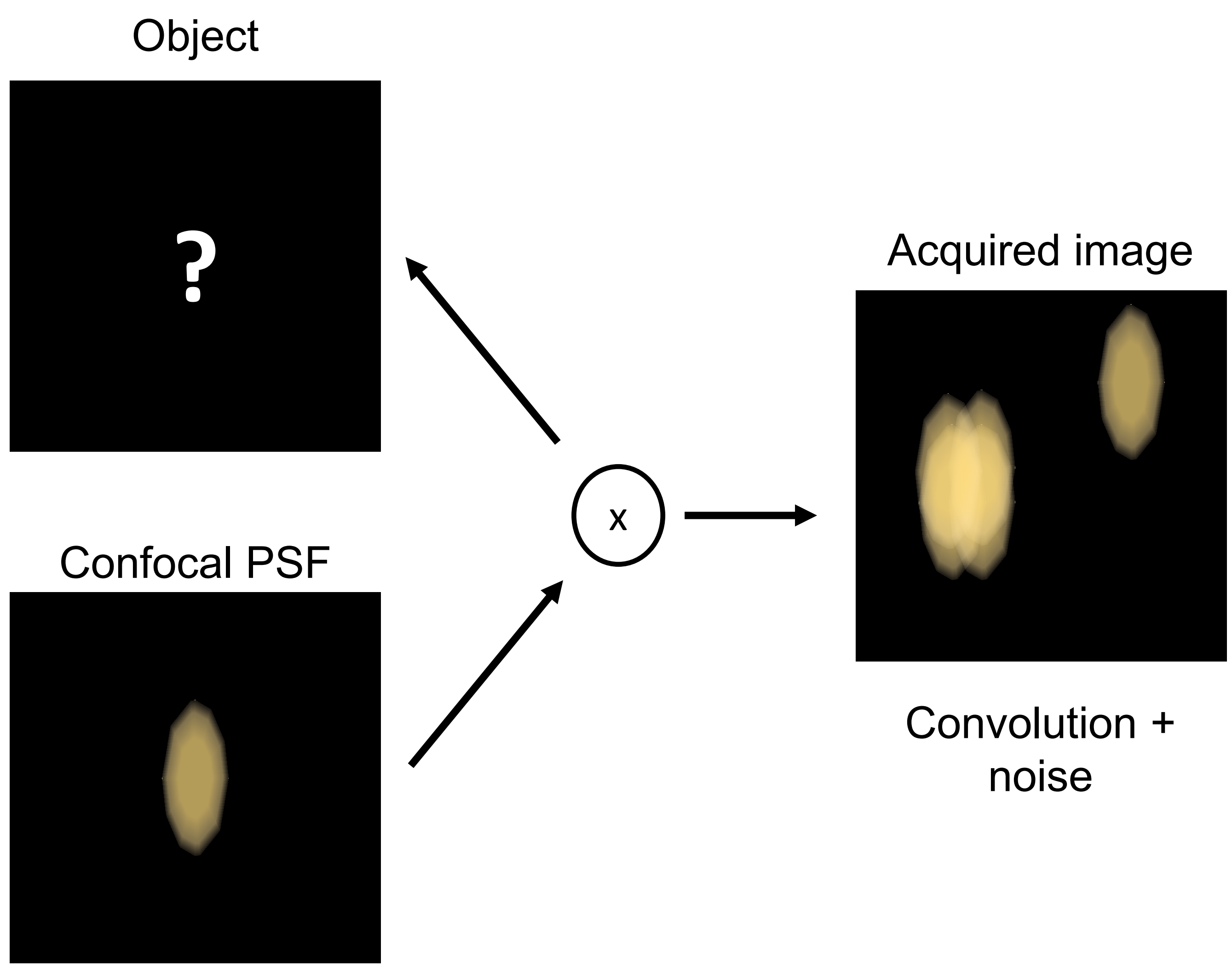


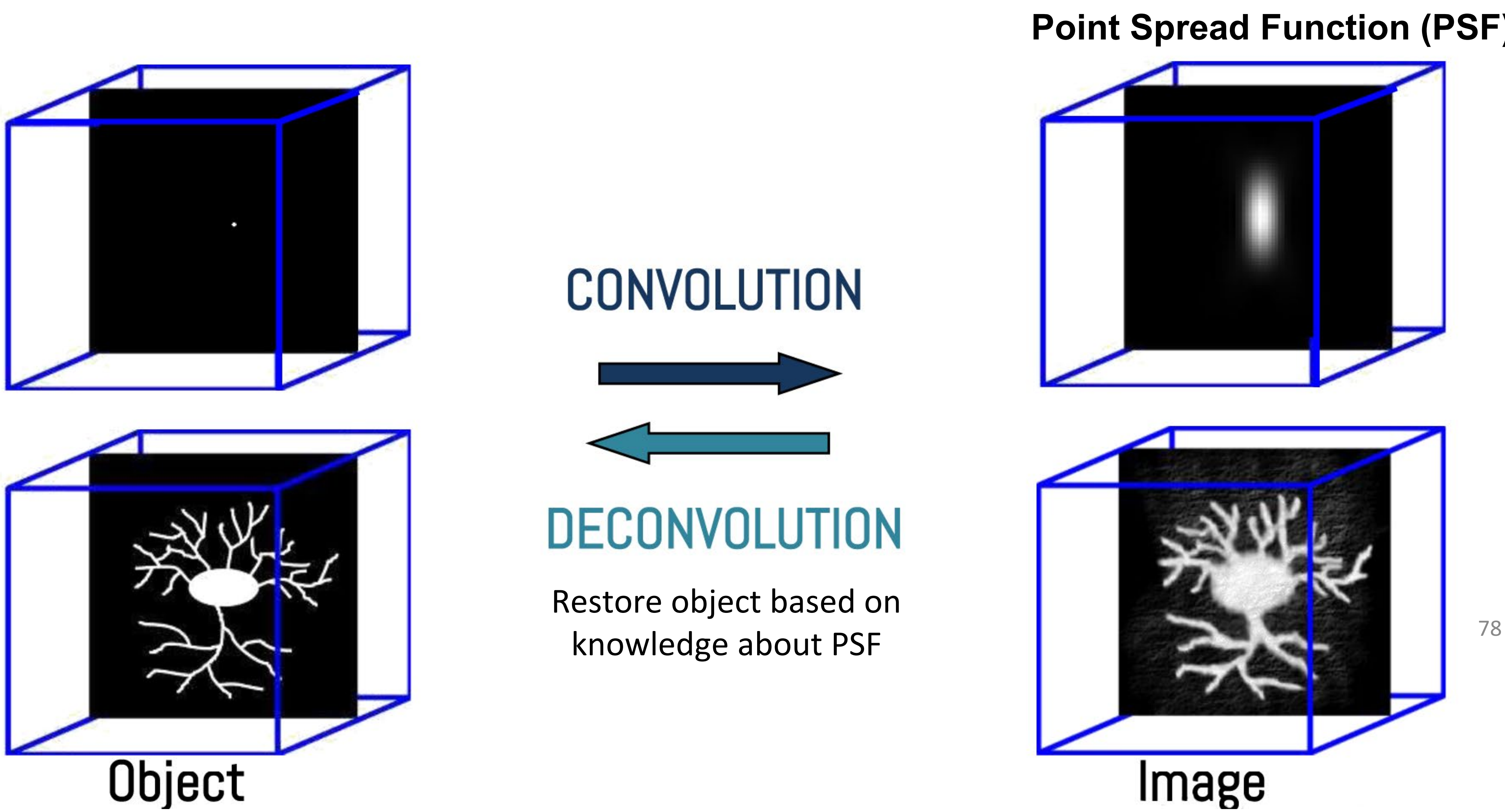
Image formation – convolution of signal



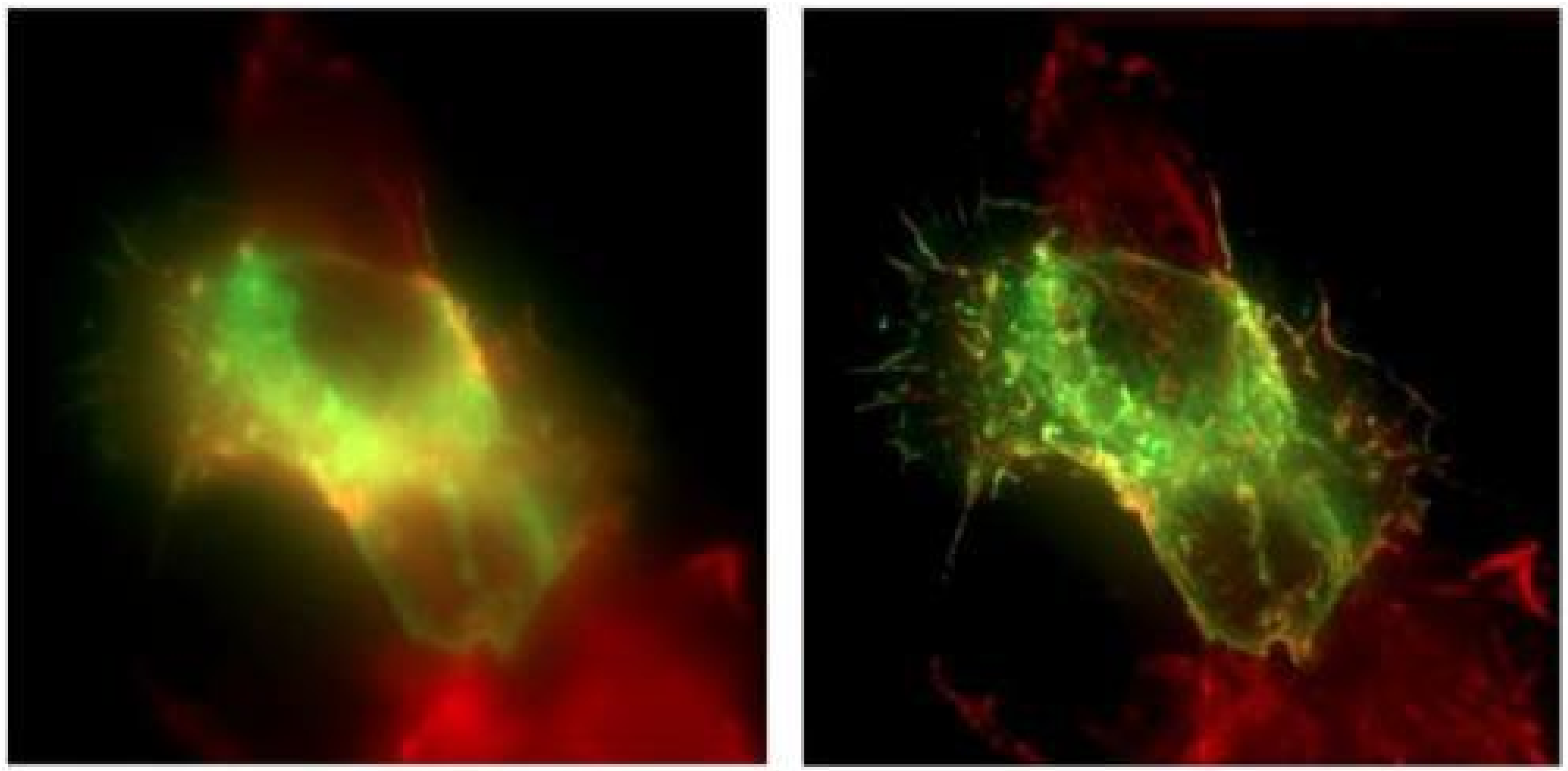
Deconvolution



Convolution←→Deconvol



Deconvolution of WF image



Deconvolution reveals structure hidden caused by imaging acquisition

Topics:

- ImageJ / FIJI
- Basic properties of digital images (sampling, formats, size, depth,)
- Image enhancement (how to make your image look better)
  - histograms, LUT, contrast enhancement
  - background subtraction
  - filter masks (smoothing and sharpening)
- Working with image stacks
- Colocalization
- Image segmentation by thresholding
- Deconvolution